

Computational Work Extraction: The Complexity of Catalysts

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Abstract

Thermodynamics has been repeatedly reshaped by improving how one models the capabilities of agents that extract work. For an isolated quantum system, the maximal extractable work, the *ergotropy*, assumes that the agent can apply any unitary. However, achieving it can be computationally intractable. In this work we introduce and study *computational ergotropy*, restricting extraction to *polynomially-sized* uniform unitary circuits acting on the system alone.

We prove *maximal separations*: n -qubit systems can have $\Theta(n)$ ergotropy, while every efficient process extracts negligible work, even for Hamiltonians consisting of single-qubit terms. We establish an *unconditional* existential separation and give an explicit construction in the *random oracle model*. Assuming the existence of quantum-secure pseudorandom functions, this separation extends to the *plain model*.

This work uncovers an important connection between ergotropy and the *complexity of catalytic computation*—computation where auxiliary qubits must be finally restored to their initial state. Relative to a random oracle, we establish relational and decision problems that: (i) can be solved efficiently with λ catalysts; but (ii) cannot be solved by any algorithm with $c\lambda$ catalysts, for any $c < 1$. We show this by proving query lower bounds for *quantum-space bounded* algorithms.

As a consequence, for computational ergotropy, catalysts prove to be surprisingly powerful—there is a family of Hamiltonians and states for which catalysts enable efficient extraction of the full $\Theta(n)$ ergotropy, while every efficient non-catalytic process extracts negligible work. Furthermore, catalysts also allow us to introduce and instantiate the notion of *pseudoergotropy*—analogous to pseudorandomness. On the other hand, we show catalysts do not change (information-theoretic) ergotropy.

Finally, our work also sheds light on the classical aspect of the problem. First, most of our constructions rely on classical states and Hamiltonians and therefore imply analogous results for *classical ergotropy*. Second, we show that certain *proof of quantumness* protocols can be used to generically *separate classical and quantum* catalytic ergotropy.

Overall, these results point towards a theory of thermodynamics where computational complexity plays a fundamental role.

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1 Introduction

Thermodynamics is the theory of how energy can be transformed, and at its heart lies the distinction between energy and useful energy. Energy is conserved, yet only part of it can be harnessed. Some of it can be extracted as work, while the rest cannot. This raises a basic question, namely *how much work can be extracted from a given system?* Its answer has direct consequences for quantum technologies. It determines how much energy a quantum battery can deliver, how much work a quantum engine can produce in each cycle, and what it costs to power quantum devices as they grow in size [CGQ⁺24, MAD22]. Remarkable experimental advances have brought these questions within reach, including heat engines whose working medium is a single trapped ion [RDT⁺16], quantum batteries charged collectively in organic microcavities and superconducting circuits [QMG⁺22, HQS⁺22], and direct measurements of the extractable work stored in individual quantum systems [NWW⁺24].

Ergotropy. How much work a system yields is determined not by its state alone but also by the process used to extract it. The answer changes, for instance, depending on whether the system may exchange heat with a reservoir or interact with other systems along the way. The cleanest setting is an isolated system taken through a cyclic process, that is, one that returns its Hamiltonian H to its initial form. Quantum mechanics then dictates that every such process acts on the state ρ of the system as a unitary, $\rho \mapsto U\rho U^\dagger$, and the work it extracts is the resulting decrease in energy. Remarkably, this minimal setting already reproduces the structure of thermodynamics! States from which no unitary can extract work are called *passive*, and those that remain passive even when arbitrarily many copies are processed jointly are called *completely passive*. Completely passive states turn out to precisely be thermal states [PW78, Len78]. The Kelvin–Planck form of the second law, which forbids any cyclic process from extracting work from a system in thermal equilibrium, thus follows from unitary work extraction alone. For a general state ρ on an n -qubit Hilbert space, the maximal work that unitaries can extract is its *ergotropy* [ABN04], defined as:

$$\text{erg}_{\rho,H} := \max_{U \in \mathcal{U}(n)} \text{tr} \left[H \left(\rho - U\rho U^\dagger \right) \right], \quad (1)$$

where $\mathcal{U}(n)$ is the group of unitaries acting on the Hilbert space of n qubits.

Efficiency. While the unitary setting already produces a rich thermodynamic structure, it fails to capture various important aspects of the physical world. For instance, since the maximum in Eq. (1) ranges over every unitary on the system, this implicitly assumes optimal unitaries can be physically realised. An optimal unitary rotates the state into the eigenbasis of H and rearranges its populations so that the largest occupy the lowest energies. For a pure state it maps the state to the ground state, which amounts to running a preparation of the state in reverse. In any physical realisation, whether by control fields, a sequence of quenches or a circuit of elementary gates, the cost of a unitary grows with its complexity, and almost all unitaries on n qubits require a number of elementary operations exponential in n [Kni95]. The complexity of preparing quantum states and implementing unitaries has become a subject of its own [Aar16, BEM⁺25]. It reveals that the two tasks differ sharply. Every quantum state can be prepared by an efficient algorithm with a single query to a suitable classical oracle [Ros24], whereas some unitaries cannot be implemented in this way even approximately [AK07, LMW24]. Work extraction fits neither framework. It asks neither for a particular final state nor for a particular unitary, since any process that lowers the energy extracts work and many processes may come close to the optimum. Therefore, the overarching question in this work is the following:

*How does our understanding of work extraction from an isolated system change
when extraction is limited by computational efficiency?*

To make progress towards obtaining concrete answers in this direction, one could consider the simplest setting captured by ergotropy.

Question 1. *How does one define efficiency in the context of ergotropy? Does it result in a meaningfully different notion from information-theoretic ergotropy?*

Computational Ergotropy. We first develop a framework that allows formally posing and rigorously studying such questions. A prerequisite to this, is a definition of efficiency, and in particular, efficient work extraction. Such a notion is not well-defined for a single state, since for any fixed system the optimal unitary is some fixed circuit, however large. Efficiency is meaningful only asymptotically, and we therefore consider families of states and Hamiltonians of growing size. The definition must also specify what the extraction process has access to. We give it polynomially many copies of the state and an efficient description of the Hamiltonian, from which it produces a circuit that is then applied to the system. The *computational ergotropy* of a family is the largest amount of work that efficient processes of this kind extract, asymptotically in the size of the system.

With this definition in place, our first result is that the computational ergotropy can differ from the ergotropy as much as possible. There are pure states, the ideal batteries of the information-theoretic theory, whose ergotropy is essentially all of their energy, yet from which no efficient process extracts more than a negligible amount of work. This happens even though the Hamiltonian is local and consists of single-qubit terms, so the entire difficulty lies in the state. Since states obtained by an existence argument may themselves be impossible to prepare, we then show that the gap persists relative to a random oracle, the least structured oracle there is, and in the plain model for explicit, efficiently preparable states under standard cryptographic assumptions.

One could justifiably ask whether the separation between computational ergotropy and ergotropy really tells us something fundamental about nature? Or if it is an artefact of considering work extraction using a very restricted class of operations? For instance, in various places, *catalysts* have been used to show surprising strengthening of operations one can apply, without changing the underlying physical justification for choosing the permitted operation.

Catalysts. Registers that are borrowed and returned in their original state, called *catalysts*, are recurring theme in quantum information and computation [LBS21, LBWN24a]. In entanglement theory, an entangled catalyst enables transformations that are impossible under local operations and classical communication [JP99, Tur07, Kli07], while a catalyst that is returned only approximately makes essentially any transformation possible [vdH03]. In complexity theory, a memory that must be restored to its initial contents still increases the computational power [BCK⁺14, CLMP25, CGM⁺25], a phenomenon that has recently been extended to quantum computation [BFM⁺25]¹. In quantum thermodynamics, the catalysts change the laws themselves. Allowing them relaxes the thermo-majorization criterion for state transitions [HO13] to a weaker family of second laws [BHN⁺15], and this family collapses to a single law once the catalyst is returned exactly but only marginally, that is, once it is allowed to remain correlated with the system [Mī8].

Some catalysts are too permissive. The choice of which borrowed registers to allow is a substantive part of the model, and for our purpose a register that may remain correlated with the system is too permissive (See bottom row of Figure 1). A unitary acting on the system alone preserves its spectrum, whereas such a register lifts that constraint entirely and makes every state of at least the same entropy reachable to arbitrary accuracy [BEG⁺19, Wil21]. The extractable work would then be governed by entropy alone, and the question we are asking would change character before any computational restriction enters. Physically, such a register looks untouched when examined on its own, yet the joint state has changed. The work

¹In catalytic space the borrowed memory starts in arbitrary contents rather than in the all-zero state, which is what makes restoring it a genuine constraint.

extracted can be traced to the correlations built along the way, and creating correlation is known to carry a thermodynamic cost [HPLH⁺15].

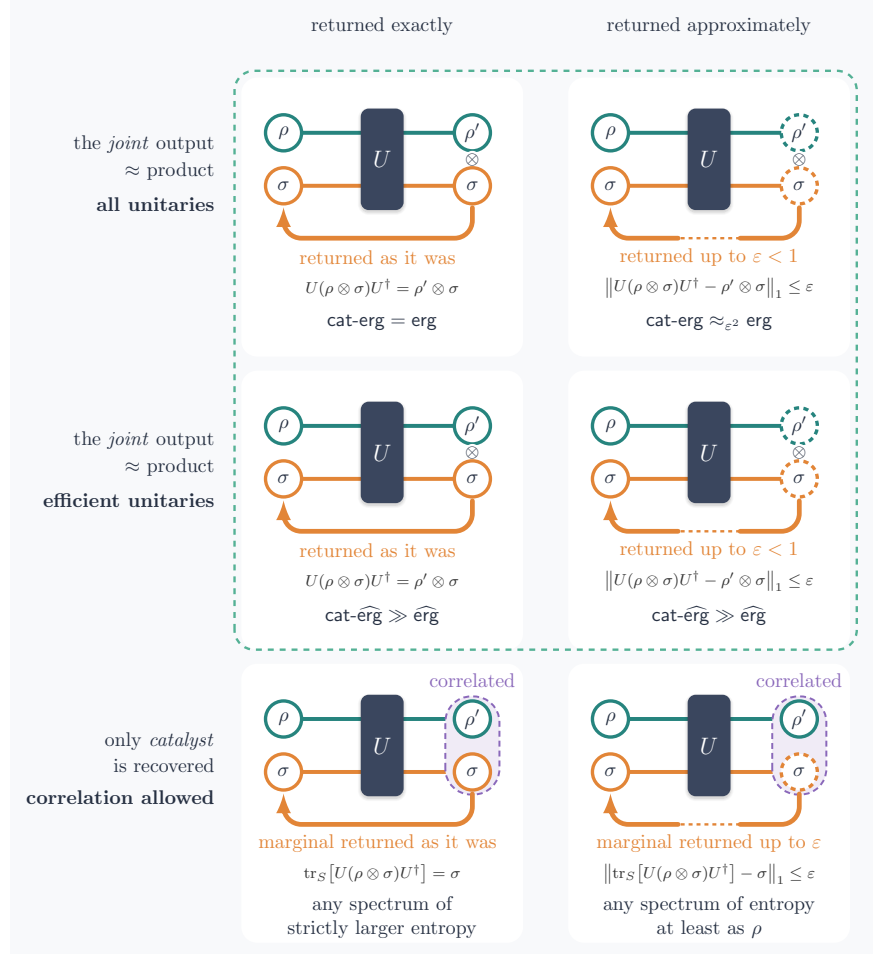


Figure 1: *The role of catalysts in information-theoretic and computational work extraction.* The system and catalyst initially have states ρ and σ . The left column requires exact restoration, while the right column allows a trace-norm error of at most ε . In the upper two rows, the final joint state must equal or approximate a product containing the original catalyst state; these rows allow arbitrary unitaries and efficient circuits, respectively. The dashed outline encloses the settings studied in this work. (Nearly) exact product restoration leaves (information-theoretic) ergotropy unchanged, yet can increase computational ergotropy. The bottom row is an instance of a catalyst that is *too permissive*. Here, restoration only of the catalyst marginal, permitting correlations with the system and changes in the system’s spectrum.

Question 2. *How does the notion of ergotropy change when one allows the use of catalysts? What happens if one also restricts to efficient operations?*

Understanding Catalytic Computation. To answer this question, we had to first define and study the *complexity* of the appropriate *catalytic model of computation*. While similar models have been considered in the literature, none of those quite matched our requirements here. We begin with the information-theoretic setting and investigate whether using catalyst qubits, permits one to apply operations beyond what is possible by unitary evolution of the system. We show that even if one is required to return a fixed catalyst to its original state, there is no advantage to using catalysts. In fact, even in the

most lenient setting—i.e. even if the joint system (after the operation) is brought ε close (in trace distance, say) to being a state where the catalyst is restored to being a tensor product—one does not gain much.

We then turn our attention to considering the *complexity of catalysts in the efficient computation setting*. Here, we allow the model to access catalysts initialised to the all-zero state, in addition to the input state, on which it can apply a unitary. We require this unitary to be produced efficiently by an efficient classical machine. The catalytic constraint is that the catalyst must be restored to its initial state exactly. In this *efficient setting*, we show that there is a problem that one can solve with catalysts but cannot solve without catalysts. In fact, we show something stronger—with insufficient quantum space (i.e. even if the catalyst is arbitrary correlated and not required to be restored in any way), the problem cannot be solved. These results are also unconditional, in the random oracle model. More fundamentally, this establishes exponential query-complexity separations for space bounded unitary quantum algorithms, which may be of independent interest. For random functions on λ -bit strings, the problem admits a classical reversible algorithm using three queries and λ auxiliary qubits returned exactly to zero. For every fixed $0 < c < 1$, quantum unitary algorithms with at most $c\lambda$ auxiliary qubits require $2^{\Omega(\lambda)}$ queries for constant success probability, even when these qubits need not be restored. The lower bound imposes no restriction on the number of oracle-independent gates.

Catalytic Ergotropy. Using our understanding of the complexity of catalytic computation, we study catalytic ergotropy. Starting with (information-theoretic) catalytic ergotropy, we show that catalysts do not help—even under a very lenient requirement on catalytic restoration. We then define computational catalytic ergotropy (i.e. the setting where efficiency is enforced). We show that there is a family of states and Hamiltonians for which one can obtain near-maximal separation between *computational ergotropy* and *computational catalytic ergotropy*. As above, the results are stronger—in the sense that with $c\lambda$ quantum space (for $c < 1$), no non-negligible work can be extracted efficiently while with λ catalysts, catalytic ergotropy is maximal.

Catalytic work-potential. We define catalytic work-potential as the difference between the work one can efficiently extract with catalysts and without. We prove a sufficient criterion (as a consequence of the preceding analysis) that allows one to identify whether a Hamiltonian has (maximal) catalytic work potential. Informally, it says that any efficiently describable Hamiltonian whose ground state is easy to prepare, can be shown to have maximal catalytic work potential.

Figure 1 summarizes the contrast between information-theoretic and computational catalytic work extraction across the different catalyst restoration requirements. These results show that in the presence of catalysts, the computational theory changes dramatically while the information-theoretic theory remains unchanged. It may be worth noting that this is different from showing one quantity in the information-theoretic setting is larger than the analogous quantity in the computational setting—here, the comparison is being made between objects of the same theory. The relation between these objects is what has changed. This suggests that computational considerations may result in fundamental new insights in our understanding of nature. One way of probing this, is to ask if something concrete becomes possible using catalysts that was not possible without it.

Pseudoergotropy. It turns out that without catalysts, it is unclear if even simple efficient operations such as applying a keyed permutation are possible. On the other hand, we show that using catalysts, one can construct a family of keyed states and Hamiltonians that exhibit *pseudoergotropy*. Intuitively, this shows that there are two such families of keyed states and Hamiltonians where (i) without the key, they are indistinguishable (computationally), (ii) the first family has low catalytic ergotropy, even when the key

is given and (iii) the second family has high catalytic ergotropy, when the key is given. These properties are akin to what one expects from *pseudorandomness*—without the key, the strings are indistinguishable but given the key, the first string has low entropy while the second has high entropy. Here, the property being captured is “extractable work” (or useful energy), instead of entropy.

Classical ergotropy. Unlike purely quantum phenomena, such as entanglement [ABV23], ergotropy remains well defined and meaningful even when one restricts to classical reversible operations. Furthermore, computational bounds apply to the classical world as well. It is therefore perhaps more natural to first ask questions about efficiency and catalysts in the classical setting. Indeed, when the state and the Hamiltonian are diagonal in the computational basis, classical reversible operations permute the populations among the energy levels. For a quantum unitary U , the transition probabilities $B_{ij} = |U_{ij}|^2$ form a doubly stochastic matrix, hence a convex combination of permutation matrices by the Birkhoff–von Neumann theorem. The final mean energy is therefore a weighted average of the energies obtained by those permutations. Some permutation extracts at least as much work as U , so information-theoretically, classical and quantum ergotropy coincide [ABN04].

Question 3. *How does efficiency affect ergotropy in the classical setting? Are there separations between quantum and classical ergotropy that are absent information-theoretically but emerge in the bounded setting?*

Computational Classical Ergotropy. Besides the existential separation between ergotropy and catalytic ergotropy, all of our aforementioned results were established using *classical states* and Hamiltonians diagonal in the computational basis. Furthermore, the efficient processes we use to achieve maximal work extraction, are nothing but permutations. Therefore, our preceding results directly translate to analogous classical statements.

Classical-Quantum Ergotropy Separation. We show that there is a family of states and (extensive) Hamiltonians such that all of the following hold (under cryptographic assumptions):

- (i) information-theoretically, quantum ergotropy and classical ergotropy for this family is identical.
- (ii) Classical catalytic ergotropy is negligible.
- (iii) Quantum catalytic ergotropy is constant.

This is another feature of thermodynamics that seems to only appear in the computational theory—and it is again aided by catalysts.

1.1 Main results

We now state our results a bit more formally. See Figure 2 for a schematic sketch of our main separations.

Computational Ergotropy. We start with our separations between computational ergotropy and (information-theoretic) ergotropy.

Existential Separation. We consider families of states $\rho = \{\rho_\lambda\}_\lambda$ and Hamiltonians $H = \{H_\lambda\}_\lambda$ on $n(\lambda)$ -qubits, where members are parametrised by λ . We use $\text{erg}_{\rho, H}$ to denote the (information-theoretic) ergotropy and $\widehat{\text{erg}}_{\rho, H}$ to denote the computational ergotropy. Informally, one may treat both these as functions of λ . We show that there is an unconditional separation between these two notions.

Theorem 1 (Unconditional existential separation (informal version of Theorem 42)). *There is a family of states and Hamiltonians ρ, H such that $\text{poly}(n) = (1 - o(1))n$. Then, there exists a family of pure states ρ and a negligible function negl such that*

$$\text{erg}_{\rho, H} \geq_\infty \text{poly}, \quad \text{and} \quad \widehat{\text{erg}}_{\rho, H} \leq_\infty \text{negl}.$$

To make comparison between functions, let's say $a, b : \mathbb{N} \rightarrow \mathbb{N}$ we use $a \leq_\infty b$ to mean that $a(\lambda) \geq b(\lambda)$ for all large λ .

The theorem establishes unconditionally that the work stored in a quantum state can be almost entirely inaccessible to computationally bounded agents. The proof relies on Haar-random states, concentration inequalities and a counting argument.

Constructive separation. For constructive separations, it helps to consider *distributions* $\mathcal{D}$ over families of states and Hamiltonians ρ, H . We often write $(\rho, H, O) \leftarrow \mathcal{D}$ to denote that (ρ, H, O) is sampled from $\mathcal{D}$, where O is an oracle (a random oracle in our case). For such distributions, we use $\text{erg}_{\mathcal{D}}$ and $\widehat{\text{erg}}_{\mathcal{D}}$ to denote ergotropy and its computational variant.

We show that for a Hamming weight Hamiltonian and state of the form

$$\rho_\lambda := \frac{1}{2^{n(\lambda)}} \sum_{r \in \{0,1\}^{n(\lambda)}} |f_\lambda(r)\rangle \langle f_\lambda(r)|$$

where f is a length doubling function encoded in the oracle, the following holds.

Theorem 2 (Separation in the random oracle model (informal version of Theorem 52)). *Let $\text{poly}(\lambda) = (1/2 - o(1))n(\lambda)$. In the random oracle model, it holds that*

$$\text{erg}_{\mathcal{D}} \geq_\infty \text{poly}, \quad \text{and} \quad \nexists \text{ non-negl s.t. } \widehat{\text{erg}}_{\mathcal{D}} \geq_\infty \text{non-negl}.$$

where non-negl is any non-negligible function.

Due to maximisation over efficient algorithms in the definition of $\widehat{\text{erg}}$, it is not exactly correct to say $\widehat{\text{erg}} \leq_\infty \text{negl}$ here but we will often say in words that $\widehat{\text{erg}}$ is negligible if no non-negl work can be extracted.

Returning to the construction, the intuition here is that to a computationally bounded algorithm, the state looks maximally mixed, even given access to the random oracle. However, an unbounded algorithm can essentially invert the function and return to a state of the form $\sum_r |r, 0\rangle \langle r, 0|$.

While this is not true in general, in our construction, we can instantiate the random oracle using a (post-quantum) pseudorandom function to obtain the following.

Theorem 3 (Separating $\text{erg}_{\mathcal{D}}$ from $\widehat{\text{erg}}_{\mathcal{D}}$ in the plain model (informal version of Theorem 56)). *Suppose pseudorandom functions exist. Then, there is an efficiently samplable distribution $\mathcal{D}$ over a family of states and Hamiltonians (see Definition 55) such that*

$$\nexists \text{ non-negl s.t. } \widehat{\text{erg}}_{\mathcal{D}} \geq_\infty \text{non-negl}, \text{ while} \\ \text{erg}_{\mathcal{D}} \geq_\infty \frac{n}{2}$$

where non-negl is any non-negligible function.

Understanding Catalytic Computation We start with the information theoretic setting.

Catalytic Computation (unbounded/information theoretic). We denote by $\text{cat-}\mathcal{U}(n, \ell, \varepsilon)$ unitaries that act on an n -dimensional IO register and an ℓ -dimensional catalyst register such that the resulting state is ε close in trace distance to a product state. We call these *approximate catalytic unitaries*. Let us denote by $C_U^\eta(\cdot)$ the channel implemented by such unitaries U where the catalyst is $|\eta\rangle$ and one only keeps the IO register (the catalyst register is traced out). Then, we show that all such channels are essentially just unitaries acting on the IO register alone.

Theorem 4. Let $0 \leq \varepsilon < 1$, and let $U \in \text{cat-}\mathcal{U}(n, \ell, \varepsilon)$. For every pure catalyst state $\eta = |\eta\rangle\langle\eta| \in \text{Den}(\ell)$, there exists a unitary $V_\eta \in \mathcal{U}(n)$ such that

$$\max_{\rho \in \text{Den}(n)} \text{TD}(C_U^\eta(\rho), C_{V_\eta}(\rho)) \leq \frac{3\varepsilon^2}{2}. \quad (2)$$

Efficient Catalytic Computation. Since nothing changes substantially when catalysts are introduced in the information theoretic setting, we now turn to the computationally bounded setting. To consider the efficient catalytic setting, we define $\text{cat-}\mathcal{U}_{|\eta\rangle}(n, \ell)$ just as we defined $\text{cat-}\mathcal{U}(n, \ell, \varepsilon)$ but with the added restriction that the catalyst $|\eta\rangle$ must be restored exactly.

With this, we can describe the class $(n, \ell)\text{-}\widehat{\text{catGens}}$: A collection of efficient classical algorithms $\mathcal{A}$ that on input 1^λ , produce a description of a unitary $U_\lambda \in \text{cat-}\mathcal{U}_{|\eta\rangle}(n, \ell)$ and a projector Π_λ that acts on the IO register. We define corresponding class of decision problems (i.e. decision problems that algorithms in $(n, \ell)\text{-}\widehat{\text{catGens}}$ can solve) to be $(n, \ell)\text{-}\widehat{\text{cat}}$.

Theorem 5 (Decision separation between catalysts (informal version of Theorem 84)). Let $n, \ell, \ell' : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomials $n(\lambda) = 2\lambda$, $\ell'(\lambda) = \lambda$ and $\ell(\lambda) = c \cdot \lambda$ for $c < 1$. In the random oracle model, it holds that

$$(n, \ell)\text{-}\widehat{\text{cat}} \subsetneq (n, \ell')\text{-}\widehat{\text{cat}}.$$

This shows that there is a decision problem that can be solved with λ catalysts but not with $c \cdot \lambda$ for any $c < 1$. In effect, even though catalysts do not affect the information theoretic picture, the computational complexity of catalysts changes when one restricts to efficient computations.

Catalytic Ergotropy Building on our complexity theoretic results, we now look at how catalysts affect ergotropy. When discussing ergotropy, we restrict to exact catalytic restoration in our explanations. Robustness to arbitrary errors is handled by showing quantum space lower bounds.

Let $\text{cat-erg}_{\rho, H}$ denote the work one can extract from the state ρ using catalytic unitaries that restore the catalyst exactly. Then, it is immediate from the preceding discussion that

$$\text{erg}_{\rho, H} = \text{cat-erg}_{\rho, H}$$

for all states and Hamiltonians.

To define the analogous quantity in the computational setting, we denote the work extracted by an efficient algorithm $\mathcal{A}$, using ℓ catalysts, from a state ρ and Hamiltonian H as

$$\mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)} := \mathbb{E} \left[\text{tr} \left[H \otimes \mathbb{I}_{2^\ell} \left(\rho \otimes |0^\ell\rangle\langle 0^\ell| - U \left(\rho \otimes |0^\ell\rangle\langle 0^\ell| \right) U^\dagger \right) \right] : \right. \\ \left. \begin{array}{l} \text{desc}(U) \leftarrow \mathcal{A}^{O_{\rho, H}, \text{desc}}(1^\lambda), \\ U \in \text{cat-}\mathcal{U}_{|0^\ell\rangle}(n, \ell) \end{array} \right]$$

where the number of catalysts and the system size is now a function of λ . Intuitively, this is computing the work one can extract using the unitary specified by $\mathcal{A}$ using ℓ -many catalysts—where the catalyst is restored exactly.

We can now define *computational ergotropy in the catalytic setting* that we denote by $\text{cat-}\widehat{\text{erg}}$. We say $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \geq_\infty \text{lb}$ if there is some efficient algorithm $\mathcal{A}$ such that $\mathbb{E}_{(\rho, H) \leftarrow \mathcal{D}} \mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)} \geq_\infty \text{lb}$. We show that $\widehat{\text{erg}}$ can be maximally separated from $\widehat{\text{cat-erg}}$.

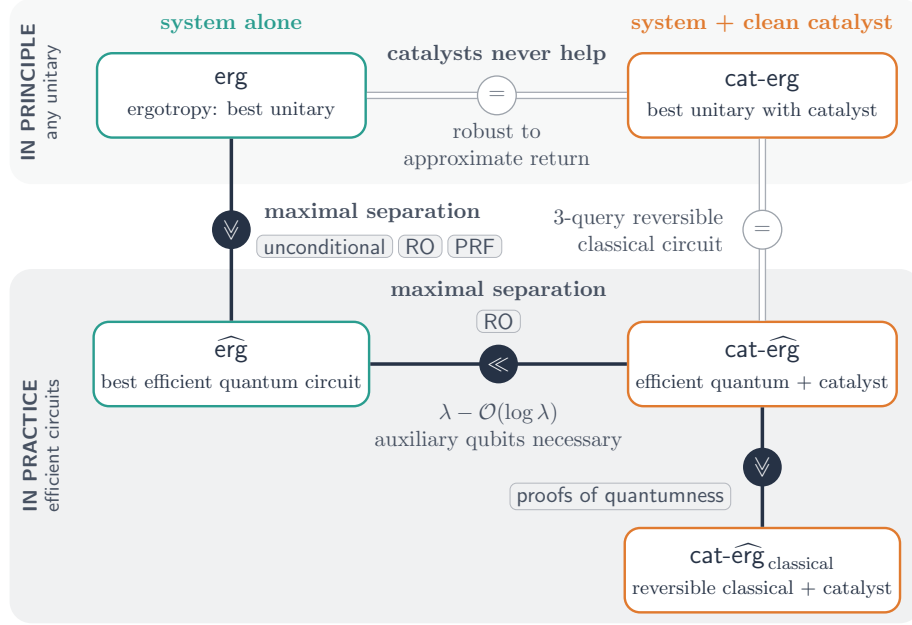


Figure 2: Overview of the separation results.

Theorem 6 (Separation between computational ergotropy and catalytic computational ergotropy (informal version of Theorem 92)). *There is a distribution $\mathcal{D}$ over the family of states, Hamiltonians and the random oracle, $\rho, \mathbf{H}, \mathcal{O}$ for which, in the random oracle model, it holds that*

$$\nexists \text{ non-negl } s.t. \widehat{\text{erg}}_{\mathcal{D}} \geq_{\infty} \text{non-negl, while} \\ \text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \geq_{\infty} \frac{n}{2}$$

where non-negl is any non-negligible function.

Catalytic Work-Potential We now give a sufficient condition to conclude that an efficient extensive Hamiltonian has catalytic work potential, i.e. Δ is not negligible (recall Δ is the difference between efficiently extractable work with and without catalysts).

Theorem 7 (Criterion for catalytic-work-potential (informal version of Theorem 94)). *Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomial $n(\lambda) = 2\lambda$. Let $\mathbf{H} = \{H_{\lambda}\}_{\lambda}$ be a family of efficient extensive Hamiltonians acting on $n(\lambda)$ qubits. Then, $\mathbf{H}$ has catalytic work potential if there is a polynomial $\ell : \mathbb{N} \rightarrow \mathbb{N}$ and an algorithm $\mathcal{A} \in (n, \ell)\text{-cat}$ such that $\text{tr}[H_{\lambda}\sigma_{\lambda}] = \text{non-negl}(\lambda)$ where*

$$\sigma_{\lambda} := \mathbb{E}_{\text{desc}(\Pi_{\lambda}, U_{\lambda}) \leftarrow \mathcal{A}(1^{\lambda})} U \left(\mathbb{I}_{2^{\lambda}} \otimes |0\rangle\langle 0|^{\lambda} \otimes |0\rangle\langle 0|^{\ell(\lambda)} \right) U^{\dagger}.$$

Pseudo-ergotropy Here, it helps to introduce keyed distributions. Just as earlier we had distributions over families of states and Hamiltonians, now we also include families of keys. We say a keyed distribution $\mathcal{D}$ over states and Hamiltonians has (c, d) -catalytic-pseudoergotropy if there is another keyed distribution $\mathcal{D}'$ such that $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \leq_{\infty} c$, $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}'} \geq_{\infty} d$ (assume the extraction algorithms see the key) and without

the key, the states produced by the two distributions are computationally indistinguishable. Here, we require both. Here, it helps to introduce keyed distributions. Just as earlier we had distributions over families of states and Hamiltonians, now we also include families of keys. We say a keyed distribution $\mathcal{D}$ over states and Hamiltonians has (c, d) -catalytic-pseudoergotropy if there is another keyed distribution $\mathcal{D}'$ such that $\widehat{\text{cat-erg}}_{\mathcal{D}} \leq_{\infty} c$, $\widehat{\text{cat-erg}}_{\mathcal{D}'} \geq_{\infty} d$ and without the key, the states produced by the two distributions are computationally indistinguishable.

Theorem 8 (Pseudoergotropy can be realised (informal version of Theorem 99)). *Suppose quantum secure pseudorandom permutations exist. Then, for $d(\lambda) = 4\lambda$, there is some polynomial $\ell : \mathbb{N} \rightarrow \mathbb{N}$ and an efficiently sampleable keyed distribution $\mathcal{D}$ that has $(0, d)$ -catalytic-pseudoergotropy using ℓ catalysts.*

The number of catalysts depends on the details of the pseudorandom permutation used. This seems to be a concrete application of catalysts in the computational setting. It is unclear whether one can realise any reasonable notion of pseudoergotropy without using catalysts.

Classical-Quantum Ergotropy Separation The fact that our separations work for classical ergotropy is a direct consequence of our constructions that we detail later. Here we present our separation between quantum and classical catalytic ergotropy, denoted by $\widehat{\text{cat-erg}}_{\mathcal{C}_{\mathcal{D}}}(\ell)$ and $\widehat{\text{cat-erg}}_{\mathcal{Q}_{\mathcal{D}}}(\ell)$ respectively. Deviating slightly from our previous definitions, here we allow negligible error in catalytic restoration. This is needed since we need to be able to run pseudo-deterministic circuits catalytically.

Theorem 9 (Classical vs quantum catalytic ergotropy separation (informal version of Theorem 111)). *Suppose factoring is hard for efficient classical machines. Then, in the common-reference string model, there is a distribution $\mathcal{D}$ over families of states, extensive Hamiltonians and common-reference strings, such that*

$$\nexists \text{ non-negl } \text{ s.t. } \widehat{\text{cat-erg}}_{\mathcal{C}_{\mathcal{D}}}(\ell) \geq_{\infty} \text{ non-negl} \quad (3)$$

$$\exists \text{ negl}' \text{ s.t. } \widehat{\text{cat-erg}}_{\mathcal{Q}_{\mathcal{D}}}(\ell) \geq_{\infty} 1 - \text{negl}' \quad (4)$$

where ℓ is some polynomial, non-negl is a non-negligible function while negl' is some negligible function.

This result can be instantiated with any Proof of Quantumness (PoQ) protocol that satisfies certain properties—we require it to be a 2-message, public-coin protocol whose soundness and completeness can be made almost perfect.

We end by emphasising that all our separations are relative to local Hamiltonians.

1.2 Prior Work

Recent work has looked at the problem of qubit erasure under computational constraints. [MKH⁺25] uses complexity entropies to bound the work cost of erasure with a bounded number of computational gates. Subsequently, [ZZP26] obtains optimal many-copy erasure protocols when quantum learning and state preparation are efficient, and proves cryptographic obstructions to optimal erasure and work extraction. [ZZP26] also discusses encrypted batteries with restored ancillary registers, but their extraction converts purity into work using a thermal reservoir. Thus, these processes permit heat exchange with a reservoir and changes to the system's spectrum. In contrast, our work concerns ergotropy, the work extracted under unitary transformations from isolated systems.

Exact catalysts returned uncorrelated with the system are already known to leave unrestricted ergotropy unchanged [LBWN24b]. We show that approximate restoration permits only a quadratic increase in extractable work: if a joint unitary maps every pure system state tensored with a fixed pure catalyst to within trace distance ε of a product state whose catalyst factor is the initial catalyst state, then allowing

such unitaries increases the maximum extractable work by at most $\varepsilon^2 \|H\|_\infty$. Munson et al. [MKH⁺25] also discuss how pure ancillas could reduce erasure complexity. Relative to a random oracle, we show that auxiliary qubits initialized to zero and returned exactly and uncorrelated enable efficient extraction of the full $\Theta(n)$ ergotropy from systems whose computational ergotropy is negligible.

Classical catalytic space [BCK⁺14] and its quantum analogue [BFM⁺25] study computation with borrowed memory whose unknown initial contents must be restored. Our catalytic model instead assumes that the catalyst is in some known fixed state, uncorrelated with the system. In fact, for our algorithms, we assume that the auxiliary qubits are in fact initialized to zero. Hao, Huang, and Liu [HHL26] show that attaining optimal query complexity for short-answer problems can require memory exponential in the answer length. Subsequently, [GJX26] exhibits a total Boolean function whose quantum query advantage disappears when workspace is restricted. On the other hand, we establish an exponential separation within unitary computation using only a constant-factor increase in total space, in the random oracle model. This separation makes reusable workspace essential for efficient work extraction despite its leaving information-theoretic ergotropy unchanged.

Recent work studies how computational restrictions affect entanglement [ABF⁺24, LREJ25], magic [GLG⁺24], nonlocality [GBGR24], quantum entropies [AHRA26], and deep thermalization [CCGGT25]. In particular, [ABV23] constructs computationally indistinguishable ensembles with low entanglement cost and high distillable entanglement, respectively, under efficient local operations and classical communication. Our construction uses their reduced states to conceal an extensive difference in ergotropy between fully separable mixed states. For a public, key-independent Hamiltonian consisting of single-qubit terms, the maximally mixed reference has zero ergotropy, whereas the other ensemble has extensive ergotropy on average over the key. The permutation key enables classical reversible extraction of the full ergotropy with exactly restored auxiliary qubits. Without it, efficient quantum protocols satisfying the same restoration requirement extract negligible expected work.

The same state ensembles, equipped with a zero Hamiltonian, have zero and $\Theta(n)$ free-energy excess above equilibrium, respectively, while remaining computationally indistinguishable given polynomially many copies. Assuming quantum-secure pseudorandom permutations, this gives a counterexample to the conjectured nonexistence of pseudo-nonequilibrium states [WT26] (See Remark 104).

2 Technical Overview

We briefly explain the key ideas and constructions we use to establish the main results.

2.1 Separating ergotropy from computational ergotropy

For the unconditional existential separation, we consider the Hamiltonian $H_n = \sum_{i=1}^n X_i$ and a Haar-random pure state. Its energy concentrates near zero, while its ergotropy is nearly n because the ground energy is $-n$. Every fixed unitary preserves the Haar distribution. Concentration is strong enough to take a union bound over all circuits with $2^{n/4}$ gates, simultaneously bounding their extracted work by $2n 2^{-n/3}$. Thus, learning copies cannot help an algorithm choose a successful polynomial-size circuit (see Lemma 39 and Theorem 42).

We first prove a constructive separation in the random-oracle model. Let $n = n(\lambda)$ and let $f_\lambda : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a uniformly random function. For our state-Hamiltonian pair, we have

$$\rho_\lambda = \frac{1}{2^n} \sum_{r \in \{0, 1\}^n} |f_\lambda(r)\rangle \langle f_\lambda(r)|, \quad H_\lambda = \mathbb{I}_{2^n} \otimes H_{\text{hw}(n)}. \quad (5)$$

Sampling r uniformly and making only one oracle query prepares ρ_λ , which has a rank of at most 2^n , so its

entire support can be moved into the 2^n -dimensional ground space of H_λ . Its ergotropy therefore equals its initial energy and has expectation $n/2$ over f_λ .

Nevertheless, ρ_λ is computationally indistinguishable from the maximally mixed state, even when the distinguisher can query f_λ . The proof compares $f_\lambda(r)$ with an independent uniform string by reprogramming the oracle at the hidden argument r and applying the one-way-to-hiding lemma (see Lemma 47). Since the maximally mixed state is invariant under unitaries, non-negligible work extraction would distinguish the ensembles by estimating their energies. The reduction also generates any copies used to choose the extraction circuit through forward oracle queries. This establishes the random-oracle separation (see Theorem 52).

For the plain-model separation, we replace the random function by a quantum-secure PRF whose key is unavailable to the extraction algorithm. PRF security transfers the indistinguishability argument, giving efficiently preparable states with extensive ergotropy but negligible efficient work extraction, leading to a maximal separation (see Theorem 56).

Next, we move on to the catalytic case. Exact restoration of a fixed pure catalyst for every system state leaves ergotropy unchanged. We show that this is robust: even if we allow for the final product state of the system and register to be ε -close to the ideal state (with exactly restored catalyst), we are able to show that ergotropy does not change other than by a factor of roughly ε^2 . The difficulty in the approximate restoration case, is that we must identify one unitary for all inputs. Suppose every pure output is within trace distance $\varepsilon < 1$ of a product containing the original catalyst state $|\eta\rangle$. Projecting onto $|\eta\rangle$ defines a system map L whose singular values lie in $[\sqrt{1 - \varepsilon^2}, 1]$. Its polar decomposition gives a unitary V with $\|L - V\|_\infty \leq \varepsilon^2$. The orthogonal catalyst component has weight at most ε^2 , and its cross terms vanish under the partial trace. Thus the reduced channel is within trace distance $3\varepsilon^2/2$ of $V(\cdot)V^\dagger$ on every input. Additional work is at most this quadratic error times the spectral width of H (see Lemma 64).

2.2 Query separations for catalytic computation

We prove both relational and decision separations. Let $f, g : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ be independent random functions. For the relational version, we consider the mapping $|x, y\rangle \mapsto |x, y \oplus f(g(x))\rangle$. It is easy to see that with $\ell(\lambda) = \lambda$ catalyst qubits and given access to standard random oracles for f and g , only three queries suffice to implement

$$|x, y, 0^\lambda\rangle \mapsto |x, y \oplus f(g(x)), 0^\lambda\rangle.$$

Our main technical contribution is to prove that no quantum algorithm making only polynomially many queries to oracles for f and g can solve this problem with constant probability using $\ell(\lambda) = \lfloor c\lambda \rfloor$ auxiliary qubits, for any fixed $0 < c < 1$. For this, we consider the inverse problem of

$$|x, f(g(x)), 0^\ell\rangle \mapsto |x, 0^\lambda, 0^\ell\rangle.$$

In fact, our lower bound proves something much more general: we do not require the $\ell(\lambda)$ auxiliary qubits to be exactly restored. Thus, this result is a query complexity separation for space bound quantum algorithms. We establish this by quantitatively relating the success probability of the relational computation to the probability of recovering x from $g(x)$, where g is a uniformly random function (see Lemma 76).

The challenge is to rule out quantum circuits that distribute intermediate information across the system and auxiliary registers; one cannot simply assume that they must store $g(x)$ in a separate register. We overcome this by quantitatively comparing erasure success with random-function inversion. Constant erasure success would give an inverter that recovers x from $g(x)$ with probability at least $2^{-\ell}/\text{poly}(\lambda)$. Although this probability can be exponentially small, for $\ell = c\lambda$ it exceeds the $\text{poly}(\lambda)2^{-\lambda}$ success permitted by polynomially many quantum queries.

The decisive technical point is to obtain a loss determined only by the auxiliary register's dimension 2^ℓ . We show that successful erasure must involve components that query f at $g(x)$: the component avoiding this address contributes at most $2^{\ell-\lambda}$ to average success. Such queries provide the starting point for an inverter. Given $g(x)$, it initializes the other registers uniformly at random and reverses a suitable circuit prefix to recover x . Randomizing these registers appears to incur a factor $2^{\lambda+\ell}$, but averaging over the uniform value $f(g(x))$ cancels the factor 2^λ associated with the second register. Apart from polynomial factors in the query counts, the reduction therefore loses only 2^ℓ . Comparing its inversion probability with the quantum query bound for random-function inversion limits average erasure success to a polynomial in λ times $2^{-(1-c)\lambda}$. This is negligible, even without requiring the erasing circuit to restore its auxiliary register.

For the decision separation, we consider the problem of distinguishing $(x, f(g(x)))$ from (x, r) , where x, r are independent uniform strings. A distinguisher need not erase the second register or preserve x . Its accepting subspace can therefore have large dimension, so the dimension bound used for relational computation does not establish the decision lower bound.

We instead bound the difference between the two average acceptance probabilities. As in the relational argument, the non-oracle gates and accepting projector are fixed independently of f, g . Conditional on g, x and the other values of f , the operator representing the component that never queries f at $g(x)$ is fixed, while both $f(g(x))$ and r are uniform. This component consequently gives the same average acceptance probability in both cases, even if that probability is large.

Distinguishing advantage must therefore involve components that query f at $g(x)$, including their interference with the component avoiding this address. We control this interference through the amplitudes of these components. The same reversed-prefix construction then relates the squared distinguishing advantage to inversion success, with a loss of 2^ℓ and polynomial factors in the query counts. The random-function inversion bound consequently makes the advantage negligible for polynomially many queries and $\ell(\lambda) = \lfloor c\lambda \rfloor$, for any fixed $0 < c < 1$. No restoration of the auxiliary register is required (see Lemma 83).

2.3 Catalytic Work Extraction and Catalytic Work Potential

We turn the query separations into a computational work-extraction separation by assigning energy to the second register. Once again, consider

$$\rho_\lambda = \frac{1}{2^\lambda} \sum_{x \in \{0,1\}^\lambda} |x, f(g(x))\rangle \langle x, f(g(x))|, \quad H_\lambda = \mathbb{I}_{2^\lambda} \otimes H_{\text{hw}(\lambda)}. \quad (6)$$

With λ auxiliary qubits, the three-query circuit maps each $(x, f(g(x)))$ to $(x, 0^\lambda)$ and restores every auxiliary qubit exactly to zero. Clearing the second register therefore moves the entire support into the ground space, decreasing the average energy. Hence catalytic computational ergotropy is near-maximal.

The lower bound for the decision problem immediately proves that lowering the energy is nearly impossible for any quantum algorithm. Indeed, the states ρ_λ and $\mathbb{I}/2^{2\lambda}$ have the same expected initial energy, while every system unitary leaves the latter unchanged. Consequently, non-negligible expected extraction from ρ_λ would produce a non-negligible difference in their final energies. Measuring a uniformly chosen term of H_λ would detect this difference, contradicting the decision lower bound. Thus polynomially many queries give negligible expected extraction without auxiliary qubits, whereas the exactly restored work-space enables full extraction. Here the extraction circuit's non-oracle gates are fixed independently of f, g , as required by the query lower bound (see Section 6).

The same connection allows us to define a notion of catalytic work potential that works for broad classes Hamiltonians. For any public, oracle-independent local H_λ of polynomial operator norm, ρ_λ has expected energy $\bar{E}_\lambda = \text{tr}(H_\lambda)/2^{2\lambda}$, and the decision bound again excludes non-negligible extraction without

auxiliary qubits. The catalytic circuit instead makes $(\mathbb{I}_{2^\lambda}/2^\lambda) \otimes |0^\lambda\rangle\langle 0^\lambda|$ efficiently reachable. If an exactly catalytic circuit can transform this state into one of energy $E_\lambda < \bar{E}_\lambda$, their composition extracts $\bar{E}_\lambda - E_\lambda$ expected work. A non-negligible gap therefore gives catalytic work potential.

2.4 Pseudoergotropy

While for most pseudo-resources instantiating the low resource state using PRS works, it does not suffice for pseudoergotropy, since the Haar random state is a low-ergotropy state.

We leverage the ability to use keyed cryptographic primitives in the catalytic setting (see Section 5) to give an explicit distribution $\mathcal{D}$ with $(0, \mathcal{O}(n))$ -catalytic pseudoergotropy. Taking inspiration from [ABV23], we define contracting and expanding functions $h : \{0, 1\}^m \rightarrow \{0, 1\}^{m/2}$ and $g : \{0, 1\}^{m/2} \rightarrow \{0, 1\}^m$, by truncating the output and padding the input to the PRP respectively, and $f : \{0, 1\}^m \rightarrow \{0, 1\}^m$ which applies the PRP as is. One can show that $\text{expanding} \circ \text{contracting}(x)$ is computationally indistinguishable from applying the PRP on x , when the key is not known (see Lemma 27, Lemma 100). Using this, one can construct the states $\sigma = \sum_{x \in \{0, 1\}^{m(\lambda)}} |f(x)\rangle\langle f(x)|$ (maximally mixed), and state $\rho = \sum_{x \in \{0, 1\}^{m(\lambda)}} |g(h(x))\rangle\langle g(h(x))|$ (relatively less mixed), that are computationally indistinguishable in the absence of the key k (see Definition 98). With access to the key k one can invert the function g , bringing the last $m(\lambda) - n(\lambda)$ -qubits of ρ back to 0 (see Lemma 103). For the padded Hamming weight Hamiltonian, one can therefore catalytically extract work from the state ρ while no work can be extracted from the state σ .

2.5 Separating classical and quantum ergotropy

Our classical-quantum separation uses classical states and classical Hamiltonians: the advantage comes from the computational power of the extraction operations, rather than from coherence in the initial state. We start from an amplified, pseudo-deterministic public-coin proof of quantumness. Given a challenge c , a quantum prover produces its unique accepting response with overwhelming probability, whereas any efficient classical prover succeeds only with negligible probability. We take c to be the common reference string and encode the verifier's computation into a 4-local classical Hamiltonian H_c . The Hamiltonian essentially penalises inconsistent computation steps and rejection, thus its zero-energy basis states encode valid accepting transcripts. An accepting response can therefore be efficiently recovered from any such ground state (see Claim 110).

For the initial state ρ_c , we use the valid computation transcript obtained by running the verifier on the all-zero input. This state is efficiently preparable and violates no computation constraints: its only possible penalty is for rejection. Its expected energy is consequently $1 - \text{negl}(\lambda)$, since otherwise the all-zero response would violate classical soundness. The important feature here is that every nonzero energy of H_c is at least one. Thus, a classical extractor's expected work is bounded by the probability that its output is a ground state. Any non-negligible work extraction would therefore produce an accepting response with non-negligible probability, contradicting the soundness of the proof of quantumness.

For quantum extraction, finding an accepting response is not enough: the computation must also return its auxiliary workspace. This is where the pseudo-deterministic property is used. A compute-copy-uncompute transformation turns the quantum prover into a catalytic computation with negligible restoration error (see Claim 109). The extractor first undoes the initial transcript computation, then generates the accepting response and constructs its valid verifier transcript, uncomputing the auxiliary workspace. The ideal output is a ground state and the negligible construction error contributes only negligible expected energy as $\|H_c\|_\infty$ is polynomially bounded. As a result, polynomially many catalyst qubits suffice for quantum extraction of $1 - \text{negl}(\lambda)$ work, while efficient classical extraction yields no non-negligible work (Section 8.3).

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3 Preliminaries and Notation

We introduce the relevant notation and the requisite preliminaries.

3.1 Notations

Quantum Notation

We start by specifying standard quantum notation.

Notation 10. We adopt the following notations throughout this article.

- Symbols for integer valued parameters:
 - λ : Security parameter,
 - n : Number of qubits in the system,
 - ℓ : Number of catalytic qubits,
 - q : Number of queries (typical; depends on the context).
- Registers are denoted by math sans-serif symbols (e.g. IO, catalyst) and the corresponding Hilbert spaces are denoted by $\mathcal{H}$ and the register in the subscript (e.g. $\mathcal{H}_{\text{IO}}, \mathcal{H}_{\text{catalyst}}$).
- $\text{Den}(n)$: The set of density operators on $(\mathbb{C}^2)^{\otimes n}$.
- $\text{Herm}(n)$: The set of Hermitian operators on $(\mathbb{C}^2)^{\otimes n}$.
- $\text{Proj}(n)$: The set of all projectors on n qubits, i.e. $\Pi \in \text{Herm}(n)$ satisfying $\Pi^2 = \Pi$.
- $\text{Perm}(n)$: The set of all permutations over n -bit strings.
- Hamiltonians are denoted by capital H .
Random oracles (and permutations) are generally denoted by O, O' .
For the big-O notation, we use $\mathcal{O}(\cdot)$.
- $\mathcal{U}(n)$: The unitary group on $(\mathbb{C}^2)^{\otimes n}$.
- $\text{hw}(s)$: The Hamming weight of $s \in \{0, 1\}^*$.
- $\text{TD}(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1$: The trace distance between the states ρ and σ . For pure states, $\text{TD}(|a\rangle\langle a|, |b\rangle\langle b|) = \sqrt{1 - |\langle a | b \rangle|^2}$.
- $\text{negl}(\lambda)$: A function that grows slower than $\frac{1}{p(\lambda)}$ for every polynomial p , asymptotically.
- By *canonical circuit description*, we mean the collection of gates and wires (qubits) the gate acts on, sequenced by the order in which the gate is present in the circuit.
- $\text{desc}(U)$: A canonical description of a circuit implementing the unitary U , corresponding to a fixed constant sized universal gate set G .
- The pure states $|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|$ are abbreviated as ψ, ϕ when no confusion arises.
- X, Y, Z

- These are used to represent the Pauli matrices, and by X_i we denote the X gate acting on the i -th qubit.
- These may also be used to denote sets and registers. This should be clear from the context.
- PPT: Probabilistic Poly Time Turing Machine
- QPT: Quantum Poly-time Turing Machine (see Definition 24)
- For a set S , by $s \leftarrow S$ we mean s is sampled uniformly from S .
- For $H = \sum_i \lambda_i |i\rangle \langle i| \in \text{Herm}$, $\|H\| = \|H\|_\infty := \max_i \lambda_i$ and $\|H\|_1 := \sum_i |\lambda_i|$.
- By $[n]$ we denote the set $\{1, 2, \dots, n\}$.
- $\binom{n}{k} := \frac{n!}{k!(n-k)!}$.

Hamiltonians

Most physically relevant Hamiltonians happen to be local in the following sense.

Definition 11 (k -local Hamiltonian). A Hamiltonian $H \in \text{Herm}(n)$ is a k -local Hamiltonian if it can be decomposed as follows:

$$H = \sum_{i=1}^m H^{(i)}$$

where each $H^{(i)}$ is a Hamiltonian on n -qubits that acts non-trivially only on k -qubits, i.e. each $H^{(i)}$ corresponds to a subset $S_i \subseteq [n]$ (see Notation 10) such that $|S_i| \leq k$, such that

$$H^{(i)} = h^{(i)} \otimes \mathbb{I}_{[n] \setminus S_i}$$

for some operator $h^{(i)} \in \text{Herm}(|S_i|)$.

Remark 12. We say a family of k -local Hamiltonians $H_n \in \text{Herm}(n)$ is simply a *local Hamiltonian* if k does not depend on n .

An instructive instance of a local Hamiltonian is the Hamming-Weight Hamiltonian that we use at various points in our exposition. Not only does it serve as a simple running example throughout this article but also models a physically relevant system—*independent spins in a fixed magnetic field*.

Definition 13 (Hamming-Weight Hamiltonian). Let n denote the size of the system. We define the Hamming weight Hamiltonian over n -qubits as follows,

$$H_{\text{hw}(n)} = \sum_{i=1}^n \frac{(\mathbb{I} - Z_i)}{2} = \sum_{s \in \{0,1\}^n} \text{hw}(s) |s\rangle \langle s|$$

where hw is as in Notation 10.

3.2 Ergotropy

The main quantity of interest in this work, is ergotropy. Here, we recall how it is conventionally defined, before introducing its computational variant.

Consider an n -qubit quantum system in a state ρ with Hamiltonian H . The extractable work is a process-dependent quantity. The work extractable from any quantum system can be defined in multiple ways depending upon the quantum operations used for work extraction. Ergotropy captures the notion of maximum work extractable from a quantum system under *unitary* operation. Formally, it is defined as follows.

Definition 14 (Ergotropy). Given an n -qubit state $\rho \in \text{Den}(n)$ and its associated Hamiltonian $H \in \text{Herm}(n)$, we define *ergotropy* as

$$\text{erg}_{\rho, H} := \max_{U \in \mathcal{U}(n)} \text{tr} \left[H \left(\rho - U \rho U^\dagger \right) \right] = \text{tr} [H \rho] - \min_{U \in \mathcal{U}(n)} \text{tr} \left[H U \rho U^\dagger \right].$$

To establish an unconditional separation between the computational notion of ergotropy (that we introduce later), and the notion above, it is helpful to consider so-called passive states.

A state is said to be *passive* with respect to a Hamiltonian if no work can be extracted from it through unitary transformations.

Definition 15 (Passive state). An n -qubit quantum state $\rho \in \text{Den}(n)$ is passive with respect to $H \in \text{Herm}(n)$ if $\text{erg}_{\rho, H} = 0$.

There is a well known characterisation of passive states that will be helpful. One direction is immediate—the other was established by [Len78].

Theorem 16 (Characterisation of passive states [Len78]). Let $H \in \text{Herm}(n)$ have a spectral decomposition $H = \sum_i e_i |e_i\rangle \langle e_i|$ with $e_1 \leq e_2 \leq \dots$, and let $\rho \in \text{Den}(n)$. The state ρ is passive (see Definition 15 above) with respect to H iff ρ and H are simultaneously diagonalisable in some orthonormal basis $\{|v_i\rangle\}_i$, such that $\rho = \sum_i \lambda_i |v_i\rangle \langle v_i|$ and $H = \sum_i |v_i\rangle \langle v_i|$, where the eigenvalues are anti-ordered, i.e.

$$e_i < e_j \implies \lambda_i \geq \lambda_j.$$

We end by stating the following fact (that can also be seen as a simple application of the theorem).

Fact 17 (Ergotropy of pure states). For every n -qubit pure state $|\psi\rangle$ and every $H \in \text{Herm}(n)$,

$$\text{erg}_{\psi, H} = \langle \psi | H | \psi \rangle - E_{\min}(H),$$

where $\psi = |\psi\rangle \langle \psi|$ (see Notation 10) and $E_{\min}(H)$ is the minimum eigenvalue of the Hamiltonian H .

3.3 Haar measure

We use the Haar measure to establish an existential separation between ergotropy and its computational variant. To this end, we recall the relevant definition and the key results.

Definition 18 (Haar measure). The *Haar measure on the unitary group* $\mathcal{U}(n)$ is the unique probability measure μ that is both left and right invariant over the group $\mathcal{U}(n)$, i.e. for all integrable functions f and for all $V \in \mathcal{U}(n)$, we have

$$\int_{\mathcal{U}(n)} f(U) d\mu(U) = \int_{\mathcal{U}(n)} f(UV) d\mu(U) = \int_{\mathcal{U}(n)} f(VU) d\mu(U).$$

By *sampling an n -qubit pure state from the Haar measure*, we mean sampling a unitary from the Haar measure on the unitary group $\mathcal{U}(n)$, and applying it on $|0\rangle^{\otimes n}$.

Lemma 19 (Levy’s lemma [Led01]). Consider the set $\mathbb{S}^{2d-1} := \{v \in \mathbb{C}^d : \|v\|_2 = 1\}$. Let $f : \mathbb{S}^{2d-1} \rightarrow \mathbb{R}$ be a function satisfying the Lipschitz condition $|f(v) - f(w)| \leq L \|v - w\|_2$. For all $\varepsilon \geq 0$, we have the probability bound:

$$\Pr_{|\phi\rangle \sim \mu} \left[\left| f(\phi) - \mathbb{E}_{|\psi\rangle \sim \mu} [f(\psi)] \right| \geq \varepsilon \right] \leq 2 \exp \left(-\frac{2d\varepsilon^2}{9\pi^3 L^2} \right).$$

Corollary 20 (Levy’s lemma for expectation value of observables). Let H be a Hermitian operator on $\mathbb{C}^d$ and μ_H be a Haar measure on $\mathbb{C}^d$. For all $\varepsilon \geq 0$, we have

$$\text{Prob}_{|\psi\rangle \sim \mu} \left(\left| \langle \psi | H | \psi \rangle - \frac{\text{tr}(H)}{d} \right| \geq \varepsilon \right) \leq 2 \exp \left(-\frac{d\varepsilon^2}{18\pi^3 \|H\|_\infty^2} \right).$$

This is a straightforward consequence of Levy’s lemma [Led01]. For a pedagogical exposition, see [Mel24].

3.4 Oracles

Many of our results are in the random oracle model. We review and setup the notation for algorithms having oracle access, and state the O2H lemma.

Oracles Access The oracle O corresponding to a function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ is defined as $O |x\rangle_X |y\rangle_Y = |x\rangle_X |y \oplus f(x)\rangle_Y$ where X and Y denote registers with Hilbert space $\mathcal{H}_X = (\mathbb{C}^2)^{\otimes n}$ and $\mathcal{H}_Y = (\mathbb{C}^2)^{\otimes m}$ respectively.

Definition 21 (Quantum Oracle Algorithm). A q -query quantum oracle algorithm $\mathcal{A}^O$ that acts on an input register I and produces the output in register O , is a quantum circuit that has at most q -many O gates.

Remark 22. When no confusion arises, we use $\mathcal{A}^f$ instead of $\mathcal{A}^O$ to refer to an algorithm $\mathcal{A}$ having oracle access to f .

The O2H Lemma In the random oracle model, the algorithms are given access to a (uniformly) random function $O : \{0, 1\}^n \rightarrow \{0, 1\}^m$ such that, for each input $x \in \{0, 1\}^n$, the output $O(x)$ is independently and uniformly distributed in $\{0, 1\}^m$. In cryptographic contexts, one often considers two oracles O, O' that are drawn from some joint distribution and differ only on a small domain. Classically, the only way to distinguish these oracles is to query them on this domain. The quantum analogue of this statement is the so-called One-way-to-Hiding lemma—an abstraction of the well-known BBBV theorem [BBBV97]. The lemma also allows the algorithm to take as input, a string that is arbitrarily correlated with the oracles.

Lemma 23 (O2H Lemma (rephrased) [Unr13, AHU18]). Let $S \subseteq \{0, 1\}^n$ be a random subset, $O, O' : \{0, 1\}^n \rightarrow \{0, 1\}^m$ be random functions satisfying $\forall x \notin S, O(x) = O'(x)$. Let z be a random string in $\{0, 1\}^m$. Note that S, O, O' and z may be drawn from a joint distribution. Let $\mathcal{A}$ be a quantum oracle algorithm making at most q queries. Let $\mathcal{B}^O$ be a quantum oracle algorithm with oracle access to O , which upon input z does the following:

- Picks $i \leftarrow \{1, 2, \dots, q\}$
- Runs $\mathcal{A}^O(z)$ until just before the i -th query, measures all query input registers in the computational basis, outputs the set T of measurement outcomes.

Define

$$\begin{aligned} P_{\text{left}} &:= \Pr [1 \leftarrow \mathcal{A}^O(z)], \\ P_{\text{right}} &:= \Pr [1 \leftarrow \mathcal{A}^{O'}(z)], \\ P_{\text{guess}} &:= \Pr [S \cap T \neq \emptyset : T \leftarrow \mathcal{B}^O(z)]. \end{aligned}$$

Then, it holds that

$$\frac{|P_{\text{left}} - P_{\text{right}}|}{2q} \leq \sqrt{P_{\text{guess}}}.$$

We end this discussion by considering families of oracles and QPT machines.

QPT machines with oracle access. Let $n, m : \mathbb{N} \rightarrow \mathbb{N}$ be functions. Let $f_\lambda : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{m(\lambda)}$. The family of oracles $\mathcal{O} := \{O_\lambda\}_\lambda$ corresponding to the family of functions $\{f_\lambda\}_\lambda$, is defined as $O_\lambda |x\rangle_X |y\rangle_Y = |x\rangle_X |y \oplus f_\lambda(x)\rangle_Y$ where X and Y denote registers with Hilbert space $\mathcal{H}_X = (\mathbb{C}^2)^{\otimes n(\lambda)}$ and $\mathcal{H}_Y = (\mathbb{C}^2)^{\otimes m(\lambda)}$ respectively.

An algorithm with oracle access to $\mathcal{O}$ is denoted by $\mathcal{A}^{\mathcal{O}}$. By $\mathcal{A}^{\mathcal{O}}$ we mean that the algorithm can use O_λ (for any choice of λ) as it would any other gate.

Definition 24 (QPT with oracle access). A *quantum polynomial time Turing machine* (QPT) is specified by a probabilistic polynomial time Turing machine (PPT) that on input 1^λ , produces a canonical description (see Notation 10) of a quantum circuit C_λ that takes an $a(\lambda)$ -qubit input and produces a $b(\lambda)$ -qubit output.

On input $|\psi\rangle_{A(\lambda)}$, QPT($1^\lambda, |\psi\rangle_{A(\lambda)}$) runs PPT(1^λ) to obtain C_λ and returns $C_\lambda(|\psi\rangle_{A(\lambda)})$ where $A(\lambda)$ is an $a(\lambda)$ -qubit input register.

Let $\mathcal{O} := \{O_{\lambda'}\}_{\lambda'}$ be as described above. Then, a *quantum polynomial time Turing machine with oracle access* to $\mathcal{O}$, QPT $^{\mathcal{O}}$ is specified by a probabilistic polynomial time machine PPT that on input 1^λ , produces a canonical description of a quantum circuit C_λ —that can use any of $\{O_{\lambda'}\}_{\lambda'}$ as gates—and acts on an $a(\lambda)$ qubit input and produces a $b(\lambda)$ qubit output.

On input $|\psi\rangle_{A(\lambda)}$, QPT $^{\mathcal{O}}$ ($1^\lambda, |\psi\rangle_{A(\lambda)}$) runs PPT $^{\mathcal{O}}$ (1^λ) to obtain C_λ and returns $C_\lambda(|\psi\rangle_{A(\lambda)})$ where $A(\lambda)$ is an $a(\lambda)$ -qubit input register.

Random Oracle Model. As alluded above, a random function $f : \{0, 1\}^* \rightarrow \{0, 1\}$ is one that, on each input, outputs an independent uniformly sampled bit. In the *Random Oracle Model*, each party/algorithm, is given oracle access to f .

It is clear that f outputting a bit is not a limitation—it can be used to produce arbitrary length outputs (e.g. by using $f(i|\cdot)$ where i encodes the i -th bit of the desired output).

In our constructions, for instance, we consider families of random functions of the form $f_\lambda : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{m(\lambda)}$ and access to this is understood in the sense described in Definition 24 above.

3.5 Cryptography

For our first constructive separation in the plain model, we rely on pseudorandom functions.

Definition 25 (PRF). A Pseudorandom Function Family (PRF) is defined by a classical deterministic polynomial-time algorithm as follows:

- $F(k, x)$ takes as input a key $k \in \{0, 1\}^n$ and an input $x \in \{0, 1\}^n$. It returns $y \in \{0, 1\}^n$.

We note that here we chose the key and input lengths to be equal (and implicitly both are also equal to the security parameter λ) for the sake of minimising the number of parameters. We require the following property.

- **Security:** For any polynomial-time quantum algorithm $\mathcal{A}$ with binary output, it holds that

$$\left| \Pr_{k \leftarrow \{0,1\}^n} \left[1 \leftarrow \mathcal{A}^{F(k, \cdot)}(1^n) \right] - \Pr_{f \leftarrow \mathcal{F}_n} \left[1 \leftarrow \mathcal{A}^f(1^n) \right] \right| = \text{negl}(n).$$

Here, $\mathcal{F}_n$ denotes the set of all possible functions mapping $\{0, 1\}^n$ to $\{0, 1\}^n$. The superscript notation $\mathcal{A}^{F(k, \cdot)}$ means that the (quantum) algorithm $\mathcal{A}$ is allowed to make arbitrary (quantum) queries to evaluate the functions $F(k, \cdot)$.

In the construction of pseudo-ergotropy (see Section 7) we use pseudorandom permutations. This in turn, also gives us other plain-model separations (e.g. between ergotropy and its computational variant, see Theorem 56). The definition below and the following result have been taken verbatim from [ABV23].

Definition 26 (PRP). A *Pseudorandom Permutation Family (PRP)* is defined by a pair of classical deterministic polynomial time algorithms as follows.

- $F(k, x)$ takes as input a key $k \in \{0, 1\}^n$ and an input $x \in \{0, 1\}^n$. It returns $y \in \{0, 1\}^n$.
- $\text{Flnv}(k, y)$ takes as input a key $k \in \{0, 1\}^n$ and an input $y \in \{0, 1\}^n$. It returns $x \in \{0, 1\}^n$.

We note that here we chose the key and input lengths to be equal (and implicitly both are also equal to the security parameter λ) for the sake of minimising the number of parameters. We require the following properties.

1. **Correctness:** For all n , for all $k \in \{0, 1\}^n$, and for all $x, y \in \{0, 1\}^n$, it holds that $F(k, x) = y$ if and only if $\text{Flnv}(k, y) = x$
2. **Pseudorandomness:** For any polynomial time quantum algorithm $\mathcal{A}$ with binary output it holds that

$$\left| \Pr_{k \leftarrow \{0,1\}^n} \left[1 \leftarrow \mathcal{A}^{F(k, \cdot), \text{Flnv}(k, \cdot)}(1^n) \right] - \Pr_{\pi \leftarrow S_{\{0,1\}^n}} \left[1 \leftarrow \mathcal{A}^{\pi, \pi^{-1}}(1^n) \right] \right| = \text{negl}(n).$$

Here, the superscript notation $\mathcal{A}^{F(k, \cdot), \text{Flnv}(k, \cdot)}$ means that the (quantum) algorithm $\mathcal{A}$ is allowed to make arbitrary (quantum) queries to evaluate the functions $F(k, \cdot)$ and $\text{Flnv}(k, \cdot)$.

Lemma 27 (Permutation $\approx$ Expansion $\circ$ Contraction; [ABV23]). *Consider a PRP family (F, Flnv) and let $n, m \in \mathbb{N}$ be s.t. $m > n$. We consider $k_f, k_g, k_h \in \{0, 1\}^m$, and define the following functions: $f : \{0, 1\}^m \rightarrow \{0, 1\}^m$ as $f(x) = F(k_f, x)$, $g : \{0, 1\}^n \rightarrow \{0, 1\}^m$ as $g(x) = F(k_g, x \| 0^{m-n})$, and $h : \{0, 1\}^m \rightarrow \{0, 1\}^n$ as $h(x) = F(k_h, x)_{|n}$ (i.e. the n -bit prefix of the outcome). Then for any polynomial function $m = m(n)$ and for any polynomial-time quantum algorithm $\mathcal{A}$ with binary output it holds that*

$$\left| \Pr_{k_g, k_h} \left[1 \leftarrow \mathcal{A}^{g(h(\cdot))}(1^n) \right] - \Pr_{k_f} \left[1 \leftarrow \mathcal{A}^{f(\cdot)}(1^n) \right] \right| = \text{negl}(n).$$

For completeness, we include the proof in Section A.

4 Computational work extraction

Recall that Ergotropy (see Definition 14) is defined to be the maximum amount of work that can be extracted from a state (given the corresponding Hamiltonian) using a *unitary* operation. However, there are no restrictions on which unitary one uses—in this sense, it is an information theoretic quantity. Clearly,

unitary operations that extract the maximum work, may not be efficiently realisable. Therefore, it is physically more realistic to revise the notion of work extraction to account for computational boundedness. In this section, we define such a computational notion of work extraction—computational ergotropy (see Definition 33)—and show that indeed, this notion is distinct from Definition 14. More concretely, we show an unconditional existential separation between ergotropy and computational ergotropy, followed by a concrete separation in the random oracle model. Section 4.1 introduces computational ergotropy while Section 4.2 establishes the separations.

There are a few challenges in defining a computational notion of ergotropy. We start by first looking at why one must consider families of states as opposed to studying a single state (and their corresponding Hamiltonians).

4.1 Defining computational ergotropy $\widehat{\text{erg}}$

Ergotropy (as in Definition 14) considers a given state-Hamiltonian pair. However, the notion of computational efficiency is only defined asymptotically—one cannot say whether a circuit with 500 gates is efficient or not in isolation. Computational efficiency is therefore defined in terms of, for instance, the number of gates in a circuit as a function of the input size. Typically, when one uses the word “efficient” one thinks of quantities such as number of gates, being polynomial in the input size.

Thus, to restrict to computationally efficient unitaries in ergotropy, one must first reformulate ergotropy in the asymptotic setting. In the following discussion, we start by parametrising the system using a security parameter $\lambda \in \mathbb{N}$ and replace single instances of state-Hamiltonian pairs, with families $\{\rho_\lambda\}$ and $\{H_\lambda\}$. Since we are no longer in the information theoretic setting, one needs to be more careful about defining how these objects are accessed by an algorithm.

Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be a function that denotes the size (number of qubits) of the quantum system. For all $\lambda \geq 1$, let $\rho = \{\rho_\lambda\}_\lambda$ and $H = \{H_\lambda\}_\lambda$ denote a family of states and Hamiltonians, such that $\rho_\lambda \in \text{Den}(n(\lambda))$, $H_\lambda \in \text{Herm}(n(\lambda))$.

Definition 28 (State generation oracle). The *state generation oracle* O_ρ , upon being queried at 1^λ , outputs the state $n(\lambda)$ -qubit state $\rho_\lambda \in \rho$, i.e.

$$O_\rho \left(|1^\lambda\rangle \langle 1^\lambda| \right) = \rho_\lambda.$$

Definition 29 (Efficiently describable Hamiltonian). A family of Hamiltonians H is *efficiently describable* if there is a PPT machine (see Notation 10) $H\text{-desc}$ that on input 1^λ outputs a canonical circuit description of H_λ . We say H is *extensive* if $\|H_\lambda\|_\infty \in \mathcal{O}(n(\lambda))$.

Example 30. A k -local Hamiltonian (see Definition 11) is efficiently describable (implicit in the proof of Lemma 50 that appears later).

In this work, we restrict ourselves to studying only normalised Hamiltonians. This is because by using arbitrary Hamiltonians, one can scale the separation between information theoretic ergotropy and computational ergotropy (see Section 4.2) arbitrarily.

Definition 31 (Work extracted by an algorithm). Let ρ, H be as above, and let $\mathcal{A}$ be a quantum oracle algorithm. We define the work extracted by the algorithm $\mathcal{A}$ from the state ρ_λ with respect to the Hamiltonian H_λ as

$$\mathcal{W}_{\rho, H, \mathcal{A}}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_\lambda \left(\rho_\lambda - U_\lambda \rho_\lambda U_\lambda^\dagger \right) \right] : \begin{array}{l} \text{desc}(U_\lambda) \leftarrow \mathcal{A}^{O_\rho, H\text{-desc}}(1^\lambda), \\ U_\lambda \in \mathcal{U}(n(\lambda)) \end{array} \right].$$

Since ergotropy captures the maximum work that can be extracted from a state with respect to a Hamiltonian through a unitary operation, one can define computational ergotropy as the maximum of $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}$ over all algorithms $\mathcal{A}$ of a particular computational class. Since $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}$ is a function, $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}} : \mathbb{N} \rightarrow \mathbb{R}$, we need a notion of comparison of these functions to meaningfully define the maximum of $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}$ over all algorithms $\mathcal{A}$.

Notation 32 (Function Comparison). Let $f, g : \mathbb{N} \rightarrow \mathbb{R}$ be functions. By $f \leq_{\infty} g$, we mean that there is a $\lambda_0 \in \mathbb{N}$ such that for all $\lambda \geq \lambda_0$, it holds that $f(\lambda) \leq g(\lambda)$. We say $f =_{\infty} g$ if $f \leq_{\infty} g$ and $f \geq_{\infty} g$ simultaneously.

With all these notions in place, we can finally formally state what we mean by computational ergotropy.

Definition 33 (Bounds on Computational Ergotropy ($\mathcal{C}$ -erg, $\widehat{\text{erg}}$)). Let $\mathcal{C}$ be a class of Turing machines (e.g. QPT, PPT), $\text{ub}, \text{lb} : \mathbb{N} \rightarrow \mathbb{R}$ be functions, and $\rho, \mathbf{H}$ be families of states and Hamiltonians as above. We say that the *computational ergotropy for the class $\mathcal{C}$* , denoted by $\mathcal{C}\text{-erg}_{\rho, \mathbf{H}}$, is (lb, ub) bounded if

$$\forall \mathcal{A} \in \mathcal{C}, \text{ it holds that } \mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}} \leq_{\infty} \text{ub}, \text{ and} \quad (7)$$

$$\exists \mathcal{A} \in \mathcal{C} \text{ s.t. } \mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}} \geq_{\infty} \text{lb}. \quad (8)$$

When $\mathcal{C}$ is taken to be the class of QPT (or PPT) algorithms (see Definition 24), we say *computational ergotropy*, denoted by $\widehat{\text{erg}}_{\rho, \mathbf{H}}$, is (lb, ub) bounded.

In subsequent discussions, we give upper bounds for computational ergotropy. To express them concisely, we overload Notation 32 as follows.

Notation 34. We use $\mathcal{C}\text{-erg}_{\rho, \mathbf{H}} \leq_{\infty} \text{ub}$ to denote $\mathcal{C}\text{-erg}$ is upper bounded by ub (i.e. Equation (7) holds) and $\mathcal{C}\text{-erg}_{\rho, \mathbf{H}} \geq_{\infty} \text{lb}$ to denote $\mathcal{C}\text{-erg}$ is lower bounded by lb (i.e. Equation (8)).

While we defined $\mathcal{C}\text{-erg}$ in terms of upper and lower bounds, in the following, we explain how one might intuitively think about $\mathcal{C}\text{-erg}$ when these bounds coincide.

Remark 35 (Matching bounds to specify $\mathcal{C}\text{-erg}$). For cases, such as when ρ is just the maximally mixed state, it is intuitively clear that $\mathcal{C}\text{-erg}$ must be 0. Formally, this corresponds to cases where lb and ub match.

1. We say $\mathcal{C}\text{-erg}_{\rho, \mathbf{H}}$ is represented by a function $b : \mathbb{N} \rightarrow \mathbb{R}$ if $\mathcal{C}\text{-erg}$ is (b, b) bounded.
2. Note that if $\mathcal{C}\text{-erg}$ is represented both by functions b and b' then $b =_{\infty} b'$.
3. When $\mathcal{C}\text{-erg}$ is represented by a function b , then there is an algorithm $\mathcal{A}^* \in \mathcal{C}$ such that

$$\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}^*} =_{\infty} b.$$

4. Another way of viewing $\mathcal{A}^*$ is as the best algorithm for extracting work, asymptotically, i.e.

$$\mathcal{A}^* \in \arg \max_{\mathcal{A} \in \mathcal{C}} \mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}$$

where we say an algorithm $\mathcal{A}' \in \arg \max_{\mathcal{A} \in \mathcal{C}} \mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}$ if for all $\mathcal{A} \in \mathcal{C}$, $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}} \leq_{\infty} \mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}'}$.

To study the separation between ergotropy and its computational variant, it also helps to define ergotropy for families of states and Hamiltonians.

Notation 36. Let $\rho = \{\rho_{\lambda}\}_{\lambda}$ be a family of states, and $\mathbf{H} = \{H_{\lambda}\}_{\lambda}$ be a family of Hamiltonians. By $\text{erg}_{\rho, \mathbf{H}}(\lambda)$ we denote the quantity $\text{erg}_{\rho_{\lambda}, H_{\lambda}}$, the ergotropy for each state-Hamiltonian pair in the family.

With $\text{erg}_{\rho, \mathbf{H}}(\lambda)$ now defined, we show that by taking $\mathcal{C}$ to be the class of unbounded computations, $\mathcal{C}\text{-erg}$ becomes essentially equivalent to $\text{erg}_{\rho, \mathbf{H}}$.

Remark 37. Let $\epsilon : \mathbb{N} \rightarrow \mathbb{R}$ be any function such that with $\epsilon(\lambda) > 0$ for all $\lambda \in \mathbb{N}$. Then, we have the following.

1. There exists a (unbounded) quantum Turing machine $\mathcal{A}$ such that

$$\text{erg}_{\rho, H}(\lambda) - \mathcal{W}_{\rho, H, \mathcal{A}}(\lambda) \geq \epsilon(\lambda) \quad \text{for all } \lambda \in \mathbb{N}.$$

2. It holds that $\mathcal{C}$ -erg is $(\text{erg}_{\rho, H} - \epsilon, \text{erg}_{\rho, H})$ bounded, when $\mathcal{C}$ is the class of all quantum Turing machines that halt on all inputs.

The first point in Remark 37 is immediate: by querying the state generation oracle O_ρ (see Definition 28) exponentially many times, one can do state tomography, and obtain the description of the state up to arbitrary precision. With the description of the state and Hamiltonian, one can describe a unitary that maps this state to a passive state (as characterised in Theorem 16) therefore extracting maximal work.

The second point in Remark 37 is an immediate consequence of the first point: For each λ , $\text{erg}_{\rho, H}(\lambda)$ is already the information theoretically maximal amount of work one can extract—so even with unbounded computation, one cannot exceed this value; and point 1 shows that there is an algorithm $\mathcal{A}$ that extracts at least $\text{erg}_{\rho, H}(\lambda) - \epsilon(\lambda)$ amount of energy.

4.2 Unconditional existential separation between erg and $\widehat{\text{erg}}$

Given that $\widehat{\text{erg}}_{\rho, H}$ is most $\text{erg}_{\rho, H}$, (i.e. $\text{erg}_{\rho, H} \geq_\infty \widehat{\text{erg}}_{\rho, H}$ using Notation 34 and Remark 37 above), one can ask the following natural question.

What is the maximal separation possible $\text{erg}_{\rho, H}$ and $\widehat{\text{erg}}_{\rho, H}$?

We show a near-maximal separation between ergotropy and its computational variant is possible. Section 4.2.1 establishes an unconditional existential separation and Section 4.4.1 gives an explicit construction in the random oracle model (see last paragraph in Section 3.4). We note that our proof techniques generalise to most physically relevant extensive Hamiltonians (see Definition 29).

4.2.1 Unconditional Existential Separation

Here, we show that there exists a family of states and Hamiltonians (ρ, H) such that $\widehat{\text{erg}}_{\rho, H} \leq_\infty \text{negl}$ for some negligible function, while $\text{erg}_{\rho, H} \geq_\infty \text{poly} - \text{negl}'$ for $\text{poly}(n) = n/2$ and some other negligible function negl' .

This result is established using a counting argument. To this end, we first upper bound the number of circuits, in terms of the number of gates and wires it uses.

For Lemmas 38 and 39 below, let G be a (finite) set of universal gates consisting of one and two-qubit gates. We assume desc (see Notation 10) is defined relative to G .

Lemma 38 (Circuit Count). *Let $p : \mathbb{N} \rightarrow \mathbb{N}$ be a function. The number of unitary operations that can be implemented on n -qubits with at most $p(n)$ gates from G is at most $\mathcal{O}(n^{p(n)})$.*

Proof sketch. A circuit of size at most $p(n)$ gates, is a sequence of $p(n)$ slots, where for each slot, we choose 2 out of the n -qubits the gate acts on ($\mathcal{O}(n^2)$), and specific gate in G ($|G| = \mathcal{O}(1)$). Therefore, there are at most $\mathcal{O}(n^2 \cdot |G|^{p(n)}) = \mathcal{O}(n^{p(n)})$ circuits. Some of these circuits might correspond to the same unitary, therefore, $|\mathcal{U}_p(n)| \leq \mathcal{O}(n^{p(n)})$. $\square$

In Lemma 39 and Theorem 42 we consider the n -qubit Hamiltonian $H_n = \sum_{i=1}^n X_i$ (see Notation 10). For the lemma, it would also be useful to consider a variant of ergotropy where the unitary operations are restricted to having at most p many gates. We define

$$\text{erg}_{\rho_n, H_n}^{\mathcal{U}_p(n)} := \text{tr}[H_n \rho_n] - \min_{U_n \in \mathcal{U}_p(n)} \text{tr}[H_n U_n \rho_n U_n^\dagger]$$

where $U \in \mathcal{U}_p(n)$ iff $\text{desc}(U)$ has at most $p(n)$ gates.

The following lemma shows that there exists a state for which erg is essentially maximal while $\text{erg}^{\mathcal{U}_p}$ is negligible. We first show that for Haar random pure states, erg is maximal with high probability. Then using the unitary invariance of Haar measures and union bound, we argue that, by maximising only over unitaries in $\mathcal{U}_p$, one can't extract non-negligible amount of work from a Haar random pure states.

Lemma 39. *Let $p : \mathbb{N} \rightarrow \mathbb{N}$ be the function $p(n) = 2^{n/4}$. For the n -qubit Hamiltonian H_n and any choice of $\delta(n) = 2^{-n/3}$, there exists $n_0 = n_0(\delta, p, G)$ such that for all $n \geq n_0$ there exists an n -qubit pure state $|\psi_n\rangle$ satisfying*

$$\begin{aligned} \text{erg}_{\psi_n, H_n} &\geq (1 - \delta(n)) \cdot n, \text{ and} \\ \text{erg}_{\psi_n, H_n}^{\mathcal{U}_p(n)} &< 2\delta(n) \cdot n. \end{aligned}$$

Proof. We break this proof into the following two claims.

Claim 40. For a Haar random state over n -qubits (Definition 18), the information theoretic ergotropy is at least $n(1 - \delta)$ with high probability, i.e.

$$\Pr_{\phi_n \sim \mu} [\text{erg}_{\psi_n, H_n} < (1 - \delta(n)) n] \leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right).$$

Proof. Let ϕ_n be an n -qubit Haar random pure state, i.e. $\phi_n \sim \mu$. Then for any $\varepsilon > 0$, it holds that,

$$\Pr_{\phi_n \sim \mu} [|\langle \phi_n | H_n | \phi_n \rangle| \geq \delta(n) \cdot n] \leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right). \quad (9)$$

This follows from directly from Corollary 20, since for the Hamiltonian H_n , $\text{tr}[H_n] = 0$ and $-E_{\min}(H_n) = n = E_{\max}(H_n) = \|H_n\|_\infty$.

From Fact 17 we know that the ergotropy of a pure state ϕ with respect to the Hamiltonian H can be written as,

$$\text{erg}_{\phi, H} = \langle \phi | H | \phi \rangle - E_{\min}(H).$$

Combining this with Equation (9), we get,

$$\begin{aligned} \Pr_{\phi_n \sim \mu} [\text{erg}_{\psi_n, H_n} < (1 - \delta(n)) n] &= \Pr_{\phi_n \sim \mu} [\langle \phi_n | H_n | \phi_n \rangle + n < (1 - \delta(n)) n] \\ &= \Pr_{\phi_n \sim \mu} [\langle \phi_n | H_n | \phi_n \rangle < -\delta(n) \cdot n] \\ &\leq \Pr_{\phi_n \sim \mu} [|\langle \phi_n | H_n | \phi_n \rangle| \geq \delta(n) \cdot n] \\ &\leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right). \end{aligned}$$

□

Claim 41 ($\text{erg}_{\phi'_n, H_n}^{\mathcal{U}_p(n)} < 2\delta(n) \cdot n$ w.h.p.). For a Haar random state over n -qubits, the ergotropy in the setting where the circuit size is at most $p(n)$, is upper bounded by $2\delta(n) \cdot n$, with high probability, i.e. there exists a negligible function negl_2 such that

$$\Pr_{\phi'_n \sim \mu} \left[\text{erg}_{\phi'_n, H_n}^{\mathcal{U}_p(n)} > 2\delta(n) \cdot n \right] \leq \text{negl}_2(n).$$

Proof. From Equation (9) we know that

$$\Pr_{\phi'_n \sim \mu} \left[|\langle \phi'_n | H_n | \phi'_n \rangle| \geq \delta(n) \cdot n \right] \leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right).$$

Since Haar unitaries are left and right unitary invariant, for any fixed n -qubit unitary V , the following holds,

$$\begin{aligned} & \Pr_{\phi'_n \sim \mu} \left[|\langle \psi | V^\dagger H V | \psi \rangle| \geq \delta(n) \cdot n \right] \leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right) \\ \implies & \Pr_{\phi'_n \sim \mu} \left[\exists V \in \mathcal{U}_p(n) \text{ s.t. } |\langle \psi | V^\dagger H V | \psi \rangle| \geq \delta(n) \cdot n \right] \leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right) \cdot |\mathcal{U}_p(n)| \quad \text{through union bound} \\ \implies & \Pr_{\phi'_n \sim \mu} \left[\exists V \in \mathcal{U}_p(n) \text{ s.t. } |\langle \psi | V^\dagger H V | \psi \rangle| \geq \delta(n) \cdot n \right] \leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right) \cdot \mathcal{O}(n^{p(n)}) \\ & = \mathcal{O} \left(n^{p(n)} / 2^{2^n \delta^2(n)} \right) \quad \text{for } p(n) = 2^{n/4} \text{ and} \\ & = \text{negl}(n) \quad \delta^2(n) = 2^{-n/3}. \end{aligned}$$

We get the third inequality because the size of $\mathcal{U}_p(n) \leq \mathcal{O}(n^{p(n)})$ (from Lemma 38).

We now bound the ergotropy optimised over the set $\mathcal{U}_p(n)$ as,

$$\begin{aligned} \text{erg}_{\phi'_n, H_n}^{\mathcal{U}_p(n)} &= \langle \phi'_n | H_n | \phi'_n \rangle - \min_{V_n \in \mathcal{U}_p(n)} \langle \phi'_n | V_n H_n V_n^\dagger | \phi'_n \rangle \\ &\Downarrow \\ \Pr_{\phi'_n \sim \mu} \left[\text{erg}_{\phi'_n, H_n}^{\mathcal{U}_p(n)} > 2\delta(n) \cdot n \right] &\leq 1 - \left(1 - \Pr_{\phi'_n \sim \mu} \left[|\langle \phi'_n | H_n | \phi'_n \rangle| \geq \delta(n) \cdot n \right] \right) \\ &\quad \times \left(1 - \Pr_{\phi'_n \sim \mu} \left[\exists V \in \mathcal{U}_p \text{ s.t. } |\langle \phi'_n | V_n H_n V_n^\dagger | \phi'_n \rangle| \geq \delta(n) \cdot n \right] \right) \\ &\Downarrow \\ \Pr_{\phi'_n \sim \mu} \left[\text{erg}_{\phi'_n, H_n}^{\mathcal{U}_p(n)} > 2\delta(n) \cdot n \right] &\leq \Pr_{\phi'_n \sim \mu} \left[|\langle \phi'_n | H_n | \phi'_n \rangle| \geq \delta(n) \cdot n \right] \\ &\quad + \Pr_{\phi'_n \sim \mu} \left[\exists V \in \mathcal{U}_p \text{ s.t. } |\langle \phi'_n | V_n H_n V_n^\dagger | \phi'_n \rangle| \geq \delta(n) \cdot n \right] \\ &\leq 2 \exp \left(-\frac{2^n \delta^2(n)}{18\pi^3} \right) + \mathcal{O} \left(2^n / 2^{2^n \delta^2(n)} \right) = \text{negl}(n). \end{aligned}$$

□

Putting together Claims 40 and 41 we get the following:

1. For a $1 - \text{negl}_1(n)$ fraction of states, $\text{erg}_{\phi_n, H_n} \geq (1 - \delta(n)) \cdot n$, for some negligible function negl_1

2. For a $1 - \text{negl}_2(n)$ fraction of states, $\text{erg}_{\phi'_n, H_n}^{\mathcal{U}_p(n)} \leq 2\delta(n) \cdot n$ where negl_2 is as in Claim 41

Therefore there exists at least one n -qubit pure state $|\psi_n\rangle$ for which $\text{erg}_{\psi_n, H_n} \geq (1 - \delta(n)) \cdot n$ and $\text{erg}_{\psi_n, H_n}^{\mathcal{U}_p} \leq 2\delta(n) \cdot n$. $\square$

It is immediate that the proof generalises to any extensive Hamiltonian (i.e. Hamiltonians with polynomial operator norm). With this proof in place, we are now ready to state the main result of this section.

Theorem 42 (Unconditional Existential Separation). *Let $\text{poly}(n) := (1 - o(1))n$ and $\mathbf{H} = \{H_n\}_n$. Then, there exists a family of pure states $\rho = \{|\psi_n\rangle\langle\psi_n|\}_n$ and a negligible function negl such that*

$$\begin{aligned} \text{erg}_{\rho, \mathbf{H}} &\geq_{\infty} \text{poly} \\ \widehat{\text{erg}}_{\rho, \mathbf{H}} &\leq_{\infty} \text{negl} \end{aligned}$$

where we are using Notations 32 and 34.

Proof. From Lemma 39 we know that there exists a family of states $\{\psi_n\}_n$ such that $\text{erg}_{\rho, \mathbf{H}}(n) \geq (1 - \delta(n))n$ and $\text{erg}_{\psi_n, H_n}^{\mathcal{U}_p(n)} < 2\delta(n) \cdot n$ for $p \in \mathcal{O}(2^{n/4})$. Since the output of any QPT algorithm is at most polynomially sized, for each QPT algorithm $\mathcal{A}$, there exists an $n_{\mathcal{A}} \in \mathbb{N}$ such that for all $n \geq n_{\mathcal{A}}$, the circuit whose description is output by $\mathcal{A}$ is in $\mathcal{U}_p(n)$. Therefore, $\widehat{\text{erg}}_{\rho, \mathbf{H}} \leq_{\infty} \text{erg}_{\rho, \mathbf{H}}^{\mathcal{U}_p} \leq_{\infty} \text{negl}$ (using Notations 32 and 34) where $\text{erg}_{\rho, \mathbf{H}}^{\mathcal{U}_p}(n) := \text{erg}_{\psi_n, H_n}^{\mathcal{U}_p(n)}$. Therefore, we have $\widehat{\text{erg}}_{\rho, \mathbf{H}} \leq_{\infty} \text{negl}$ while $\text{erg}_{\rho, \mathbf{H}}(n) \geq (1 - \mathcal{O}(2^{-n/2})) \cdot n = (1 - o(1)) \cdot n$ for all $n \in \mathbb{N}$. $\square$

4.3 Defining computational ergotropy for distributions $\widehat{\text{erg}}_{\mathcal{D}}$

For the explicit separation, we first introduce ergotropy for distributions.

Let $\mathcal{D}$ denote a probability distribution over families of states $(\rho_i)_i$ and families of Hamiltonian $(\mathbf{H}_i)_i$. We use $\rho, \mathbf{H} \leftarrow \mathcal{D}$ to denote the result of sampling a pair of state and Hamiltonian families from $\mathcal{D}$.

Definition 43 (Bounds on Computational Ergotropy for a distribution $\mathcal{D}$ ($\mathcal{C}\text{-erg}_{\mathcal{D}}, \widehat{\text{erg}}_{\mathcal{D}}$)). Let $\mathcal{C}$ be a class of Turing machines (e.g. QPT, PPT), $\text{ub}, \text{lb} : \mathbb{N} \rightarrow \mathbb{R}$ be functions, and $\mathcal{D}$ be a distribution over families of states and Hamiltonians (as described above). We say that the *computational ergotropy for the class $\mathcal{C}$ and distribution $\mathcal{D}$* , denoted by $\mathcal{C}\text{-erg}_{\mathcal{D}}$, is (lb, ub) bounded if

$$\begin{aligned} \forall \mathcal{A} \in \mathcal{C}, \text{ it holds that } \mathbb{E}_{\rho, \mathbf{H} \leftarrow \mathcal{D}} [\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}] &\leq_{\infty} \text{ub}, \text{ and} \\ \exists \mathcal{A} \in \mathcal{C} \text{ s.t. } \mathbb{E}_{\rho, \mathbf{H} \leftarrow \mathcal{D}} [\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}] &\geq_{\infty} \text{lb}. \end{aligned}$$

When $\mathcal{C}$ is taken to be the class of PPT (or QPT) algorithms, we say computational ergotropy (for the distribution $\mathcal{D}$), denoted by $\widehat{\text{erg}}_{\mathcal{D}}$, is (lb, ub) bounded.

It would be helpful to make statements of the form “ $\mathcal{C}\text{-erg}$ is negligible”. However, stating $\mathcal{C}\text{-erg} \leq_{\infty} \text{negl}$ for some negligible function turns out to be too strong in most cases—it requires that for every algorithm $\mathcal{A} \in \mathcal{C}$, the work extracted is bounded by the same negligible function negl . Therefore, we instead use the following to captures the same intuition that no algorithm can extract non-negligible work.

Notation 44. By $\mathcal{C}\text{-erg}$ is *negligible* we denote the following property: there is no non-negligible function non-negl for which $\mathcal{C}\text{-erg} \geq_{\infty} \text{non-negl}$.

The perspective of a distribution over the family of states and Hamiltonians is particularly helpful in the random oracle model. One can define families of states and Hamiltonians that depends on the random oracle.

Remark 45 (Ergotropy in the random oracle model). Consider the random oracle model where $\mathcal{O}$ is the random oracle.

1. The probability distribution $\mathcal{D}$ is over the family of states, Hamiltonians ρ, H and the oracle $\mathcal{O}$.
2. For a particular family of states, Hamiltonians and oracle, $(\rho, H, \mathcal{O}) \leftarrow \mathcal{D}$, we define

$$\mathcal{W}_{\rho, H, \mathcal{A}^{\mathcal{O}}}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_{\lambda} \left(\rho_{\lambda} - U_{\lambda} \rho_{\lambda} U_{\lambda}^{\dagger} \right) \right] : \begin{array}{l} \text{desc}(U_{\lambda}) \leftarrow \mathcal{A}^{\mathcal{O}, H, \text{desc}, \mathcal{O}}(1^{\lambda}) \\ U_{\lambda} \in \mathcal{U}(n(\lambda)) \end{array} \right]$$

where $\mathbb{E}$ is for a fixed $\rho, H, \mathcal{O}$.

3. We say $\mathcal{C}\text{-erg}_{\mathcal{D}}$ is (lb, ub) bounded if

$$\begin{aligned} \forall \mathcal{A} \in \mathcal{C}, \text{ it holds that } & \mathbb{E}_{\rho, H, \mathcal{O} \leftarrow \mathcal{D}} [\mathcal{W}_{\rho, H, \mathcal{A}^{\mathcal{O}}}] \leq_{\infty} \text{ub}, \text{ and} \\ \exists \mathcal{A} \in \mathcal{C} \text{ s.t. } & \mathbb{E}_{\rho, H, \mathcal{O} \leftarrow \mathcal{D}} [\mathcal{W}_{\rho, H, \mathcal{A}^{\mathcal{O}}}] \geq_{\infty} \text{lb}. \end{aligned}$$

4.4 Constructive separation between $\text{erg}_{\mathcal{D}}$ from $\widehat{\text{erg}}_{\mathcal{D}}$

In Theorem 42 above, we showed the *existence* of states and Hamiltonians for which ergotropy is near-maximally separated from its computational variant. We now leverage computational hardness assumptions to *explicitly* specify states and Hamiltonians that exhibit *near-maximal* separation.

Here, we first work in the random oracle model and subsequently show how to instantiate the oracle using a pseudorandom function to obtain a separation in the plain model.

The separation in the random oracle model is established through a reprogramming technique and arguing that the output of a length doubling random oracle (on a hidden input) is indistinguishable from a uniformly random string.

4.4.1 Constructive Separation in the random oracle model

We start by specifying the states and Hamiltonians that we use for the separation. These can be prepared efficiently.

Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be a polynomial which describes the size of the system. By $\mathcal{F}$ we denote the set of functions

$$\mathcal{F} := \left\{ g : g = \{g_{\lambda}\}_{\lambda} \text{ where } g_{\lambda} \text{ is from } \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{2n(\lambda)} \right\}.$$

Let $\mathcal{O}_f = \{O_{\lambda}\}_{\lambda}$ be the oracle, corresponding to a *length doubling function* $f = \{f_{\lambda}\}_{\lambda}$.

Definition 46. We define the distribution $\mathcal{D}$ over $(\rho, H, \mathcal{O})$ in the random oracle model as follows.

- The length doubling function $f = \{f_{\lambda}\}_{\lambda}$ is uniformly sampled from the set $\mathcal{F}$.
- For each fixed $\mathcal{O}_f$ (the oracle corresponding to f) we define the families of states $\rho^{\mathcal{O}_f} = \{\rho_{\lambda}\}_{\lambda}$ and $\sigma^{\mathcal{O}_f} = \{\sigma_{\lambda}\}_{\lambda}$ over $2n(\lambda)$ -qubits where

$$\rho_{\lambda} := \frac{1}{2^{n(\lambda)}} \sum_{r \in \{0, 1\}^{n(\lambda)}} |f_{\lambda}(r)\rangle \langle f_{\lambda}(r)| \quad (10)$$

and

$$\sigma_\lambda := \frac{1}{2^{2n(\lambda)}} \sum_{u_\lambda \in \{0,1\}^{2n(\lambda)}} |u_\lambda\rangle \langle u_\lambda|. \quad (11)$$

Clearly, σ_λ is the maximally mixed state (and independent of $\mathbf{O}_f$).

- $\mathbf{H} := \{H_\lambda\}_\lambda$ is independent of $\mathbf{O}$ and is defined as

$$H_\lambda := \mathbb{I} \otimes H_{\text{hw}(n(\lambda))} \quad (12)$$

acting on $2n(\lambda)$ qubits.

Using the O2H lemma, we show that no efficient procedure can distinguish whether it holds ρ or σ , even given access to the oracle $\mathbf{O}$.

Lemma 47. *Let $\mathbf{O}$ be the length doubling random oracle, and $q_1, q_2 : \mathbb{N} \rightarrow \mathbb{N}$ be polynomials. For every q_1 -query quantum algorithm $\mathcal{B}$ with oracle access to $\mathbf{O}$, there exists a q_2 -query quantum algorithm $\mathcal{A}$ with oracle access to O_λ corresponding to the function f_λ such that,*

$$\begin{aligned} & \left| \Pr_{\mathcal{D}} [\mathcal{B}^{\mathbf{O}}(1^\lambda, \rho_\lambda) = 1] - \Pr_{\mathcal{D}} [\mathcal{B}^{\mathbf{O}}(1^\lambda, \sigma_\lambda) = 1] \right| \\ &= \left| \Pr_{\mathcal{D}} [\mathcal{A}^{O_\lambda}(1^\lambda, \rho_\lambda) = 1] - \Pr_{\mathcal{D}} [\mathcal{A}^{O_\lambda}(1^\lambda, \sigma_\lambda) = 1] \right| \end{aligned} \quad (13)$$

$$\leq \text{negl}(\lambda). \quad (14)$$

Proof. We first prove that there exists an algorithm $\mathcal{A}$ such that the output distributions of $\mathcal{B}^{\mathbf{O}}(1^\lambda, \tau_\lambda)$ and $\mathcal{A}^{O_\lambda}(1^\lambda, \tau_\lambda)$ are identical for any τ_λ that depends only on O_λ . Observe that each f_λ is sampled uniformly and independently from $\mathcal{F}_\lambda$.

Now, for every algorithm $\mathcal{B}^{\mathbf{O}}$, one can construct such an algorithm $\mathcal{A}^{O_\lambda}$ as follows: $\mathcal{A}$ runs $\mathcal{B}$ identically except when $\mathcal{B}$ makes queries. When $\mathcal{B}$ makes a query to $\mathbf{O}$ at $\{0, 1\}^{n(\lambda)}$, then $\mathcal{A}$ uses O_λ , and when $\mathcal{B}$ makes any other query, it simulates a random oracle independently. This establishes Equation (13).

To establish Equation (14), we use a hybrid argument and the One-Way to Hiding (O2H) Lemma. We define a sequence of hybrid experiments $\text{Exp}_0, \text{Exp}_1, \text{Exp}_2$ as follows

- Exp_0 : The challenger samples $O_\lambda \leftarrow \mathcal{F}_\lambda$ and $r \leftarrow \{0, 1\}^{n(\lambda)}$. It outputs $\mathcal{A}^{O_\lambda}(O_\lambda(r))$.
- Exp_1 : The challenger samples $O_\lambda \leftarrow \mathcal{F}_\lambda$, $r \leftarrow \{0, 1\}^{n(\lambda)}$ and $u \leftarrow \{0, 1\}^{2n(\lambda)}$. It defines O'_λ such that $O'_\lambda(r) = u$ and $O'_\lambda(x) = O_\lambda(x)$ for all $x \neq r$. It outputs $\mathcal{A}^{O'_\lambda}(u)$.
- Exp_2 : The challenger samples $O_\lambda \leftarrow \mathcal{F}_\lambda$, $r \leftarrow \{0, 1\}^{n(\lambda)}$ and $u \leftarrow \{0, 1\}^{2n(\lambda)}$. It outputs $\mathcal{A}^{O_\lambda}(u)$.

Now, evidently Exp_0 and Exp_1 are identically distributed (we just relabelled some variables).

Claim 48. $\Pr_{O_\lambda, r} [1 \leftarrow \text{Exp}_0] = \Pr_{O_\lambda, r, u} [1 \leftarrow \text{Exp}_1]$.

Furthermore, Exp_1 and Exp_2 have negligibly close output distributions from the O2H lemma (see Lemma 23).

Claim 49. $|\Pr_{O_\lambda, r, u} [1 \leftarrow \text{Exp}_1] - \Pr_{O_\lambda, u} [1 \leftarrow \text{Exp}_2]| \leq \text{negl}(n)$.

We defer the proofs of these claims to Section A in the Appendix, as these are standard arguments. Now using the hybrid argument, we conclude $|\Pr_{O_\lambda, r} [1 \leftarrow \text{Exp}_0] - \Pr_{O_\lambda, u} [1 \leftarrow \text{Exp}_2]| \leq \text{negl}(n)$ proving the theorem statement. $\square$

The following allows us to show that when two states are computationally indistinguishable, then the average energy one can efficiently extract is essentially the same.

Lemma 50 (Indistinguishability of states under efficiently preparable observables). *Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be a polynomial describing the size of the system and $\tau_1 = \{\tau_{1,\lambda}\}_\lambda$ and $\tau_2 = \{\tau_{2,\lambda}\}_\lambda$ be families of states, such that the $n(\lambda)$ -qubit state $\tau_{1,\lambda}$ is computationally indistinguishable from $\tau_{2,\lambda}$ when given a single copy of the state. For any fixed $k \in \mathbb{N}$ and any family of extensive k -local Hamiltonians $\mathbf{H} = \{H_\lambda\}_\lambda$ (see Definition 11), there exists a negligible function negl and $\lambda_0 \in \mathbb{N}$, such that*

$$|\text{tr}[H_\lambda \tau_{1,\lambda}] - \text{tr}[H_\lambda \tau_{2,\lambda}]| \leq \text{negl}(\lambda) \quad (15)$$

for some $\lambda > \lambda_0$. Furthermore, if U_λ is produced by a QPT algorithm on input 1^λ , then

$$\left| \text{tr}[H_\lambda U_\lambda \tau_{1,\lambda} U_\lambda^\dagger] - \text{tr}[H_\lambda U_\lambda \tau_{2,\lambda} U_\lambda^\dagger] \right| \leq \text{negl}(\lambda). \quad (16)$$

Proof. Since every $H_\lambda \in \text{Herm}(n(\lambda))$ is a k -local Hamiltonian, it can be written as,

$$H_\lambda = \sum_{i=1}^m h_\lambda^{(i)} \otimes \mathbb{I}_{[n(\lambda)] \setminus S_i} \quad (17)$$

where $S_{i,\lambda} \subseteq [n(\lambda)]$, such that $|S_{i,\lambda}| \leq k$ and $h_{i,\lambda} \in \text{Herm}(|S_i|)$ (see Definition 11). Since $k \in \mathbb{N}$ is a constant, the total number of subsets of $[n(\lambda)]$ of size at most k is $\binom{n(\lambda)+k}{k} \in \mathcal{O}(n(\lambda)^k)$ (see Notation 10) therefore, $m \in \mathcal{O}(n(\lambda)^k) = \text{poly}(\lambda)$ for some polynomial function poly .

Furthermore, each $h^{(i)}$ can be written as a linear combination of projectors as follows:

$$h_\lambda^{(i)} = \sum_{j=1}^{2^k} e_\lambda^{(ij)} \Pi_\lambda^{(ij)}.$$

This follows because $h_\lambda^{(i)} \in \text{Herm}(|S_i|)$ is an operator on a Hilbert space of dimension at most 2^k . Therefore, its spectral decomposition has at most 2^k elements. Combining this with Equation (17) and using the bound on m , we get that

$$H_\lambda = \sum_{i=1}^{\text{poly}(\lambda)} \sum_{j=1}^{2^k} e_\lambda^{(ij)} \Pi_\lambda^{(ij)} \quad \text{where } \Pi_\lambda'^{(ij)} := \Pi_\lambda^{(ij)} \otimes \mathbb{I}_{[n(\lambda)] \setminus S_i}$$

Therefore, every Hamiltonian $H_\lambda \in \mathbf{H}$ can be written as the linear combination of $\mathcal{O}(\text{poly}(\lambda))$ projectors. Replacing H_λ with its linear combination of projectors, we get

$$\begin{aligned} |\text{tr}[H_\lambda \tau_{1,\lambda}] - \text{tr}[H_\lambda \tau_{2,\lambda}]| &= \left| \text{tr} \left[\sum_{i=1}^{\text{poly}(\lambda)} \sum_{j=1}^{2^k} e_\lambda^{(ij)} \Pi_\lambda'^{(ij)} \tau_{1,\lambda} \right] - \text{tr} \left[\sum_{i=1}^{\text{poly}(\lambda)} \sum_{j=1}^{2^k} e_\lambda^{(ij)} \Pi_\lambda'^{(ij)} \tau_{2,\lambda} \right] \right| \\ &\leq \sum_{i=1}^{\text{poly}(\lambda)} \sum_{j=1}^{2^k} \underbrace{\left| e_\lambda^{(ij)} \right| \cdot \left| \text{tr}[\Pi_\lambda'^{(ij)} \tau_{1,\lambda}] - \text{tr}[\Pi_\lambda'^{(ij)} \tau_{2,\lambda}] \right|}_{:= \alpha_{ij}(\lambda)} \end{aligned} \quad \text{using } \triangle\text{-inequality.}$$

For contradiction, suppose the left-hand side, i.e. $|\text{tr}[H_\lambda \tau_{1,\lambda}] - \text{tr}[H_\lambda \tau_{2,\lambda}]| = \text{non-negl}(\lambda)$. Then, from the pigeon-hole principle one can conclude that there is some index $i^*(\lambda), j^*(\lambda)$ such that $\alpha_{i^*(\lambda), j^*(\lambda)}(\lambda) \geq \text{non-negl}(\lambda) / (2^k \text{poly}(\lambda))$. This means that

$$\left| \text{tr}[\Pi_\lambda'^{(ij)} \tau_{1,\lambda}] - \text{tr}[\Pi_\lambda'^{(ij)} \tau_{2,\lambda}] \right| \geq \text{non-negl}'(\lambda) \quad (18)$$

(as the $e_\lambda \cdot |\text{something bounded by } 2|$ is non-negligible so e_λ is non-negligible). Now consider the following distinguisher $\mathcal{B}$:

- on input $(\tau_{b,\lambda}, 1^\lambda)$, it samples $i' \leftarrow [\text{poly}(\lambda)]$, $j' \leftarrow [2^k]$ (see Notation 10)
- measures $\tau_{b,\lambda}$ using $\Pi_\lambda^{(i'j')}$ and outputs 1 if the outcome corresponds to $\Pi_\lambda^{(i'j')}$ and 0 otherwise.

The distinguishing probability of $\mathcal{B}$ is at least

$$\begin{aligned} \frac{1}{2^k \cdot \text{poly}(\lambda)} \sum_{ij} \left| \text{tr}[\tau_{1,\lambda} \Pi_\lambda^{(ij)}] - \text{tr}[\tau_{2,\lambda} \Pi_\lambda^{(ij)}] \right| &\geq \frac{1}{2^k \cdot \text{poly}(\lambda)} \left| \text{tr}[\tau_{1,\lambda} \Pi_\lambda^{(i^*(\lambda)j^*(\lambda))}] - \text{tr}[\tau_{2,\lambda} \Pi_\lambda^{(i^*(\lambda)j^*(\lambda))}] \right| \\ &= \frac{1}{2^k \cdot \text{poly}(\lambda)} \cdot \text{non-negl}'(\lambda) \quad \text{using eq. (18)} \\ &= \text{non-negl}''(\lambda). \end{aligned}$$

However, this contradicts the indistinguishability of $\tau_{1,\lambda}$ and $\tau_{2,\lambda}$. Thus,

$$|\text{tr}[H_\lambda \tau_{1,\lambda}] - \text{tr}[H_\lambda \tau_{2,\lambda}]| \leq \text{negl}(\lambda)$$

which proves Equation (15).

Given Equation (15) holds, establishing Equation (16) is rather straightforward: if no efficient algorithm can distinguish between $\tau_{1,\lambda}$ and $\tau_{2,\lambda}$, then no efficient algorithm can distinguish between $U_\lambda \tau_{1,\lambda} U_\lambda^\dagger$ and $U_\lambda \tau_{2,\lambda} U_\lambda^\dagger$ as well, where $\text{desc}(U_\lambda)$ is produced by an efficient algorithm. $\square$

Remark 51. Consider the cases where one is working relative to oracles $\mathcal{O}$, i.e. where the distinguishing algorithms have access to an oracle $\mathcal{O}$, and where τ s can depend on $\mathcal{O}$. Given that τ_1 is computationally indistinguishable from τ_2 , one can verify that the results in Lemma 50 continue to hold.

We are now ready to give the explicit families of states and Hamiltonians for which we obtain maximal separation.

Theorem 52 below, is one of our main results which shows that for the family states and extensive Hamiltonians we described in Definition 46, $\text{erg}_{\mathcal{D}}$ is at least $n/2$ asymptotically while $\widehat{\text{erg}}_{\mathcal{D}}$ is negligible.

Theorem 52 (Separating $\text{erg}_{\mathcal{D}}$ from $\widehat{\text{erg}}_{\mathcal{D}}$ in the random oracle model). *In the random oracle model, for the family of states, Hamiltonians and the oracle, $\rho, \mathbf{H}, \mathcal{O}$ sampled from distribution $\mathcal{D}$ as defined above (see Definition 46), it holds that*

$$\begin{aligned} \nexists \text{ non-negl s.t. } \widehat{\text{erg}}_{\mathcal{D}} \geq_\infty \text{ non-negl, while} \\ \text{erg}_{\mathcal{D}} \geq_\infty \frac{n}{2} \end{aligned}$$

where non-negl is any non-negligible function.

Proof. We show the first inequality in Claim 53 and the second in Claim 54.

Claim 53. In the random oracle model, there does not exist a non-negligible function non-negl such that $\widehat{\text{erg}}_{\mathcal{D}} \geq_\infty \text{ non-negl}$.

Proof. For every QPT algorithm $\mathcal{B}^{\mathcal{O}, O_\rho, \text{H-desc}}(1^\lambda)$ that outputs the description of a unitary, one can do the following analysis to upper bound $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{B}}$. We proceed in two steps. First we show that one can extract a unitary U from $\mathcal{B}$ without giving access to O_ρ . Then, from the fact that ρ and σ are computationally indistinguishable, we can use Remark 51 and prove that the associated energies are the same, up to negligible factors.

More precisely, one can construct a simulator $\mathcal{D}^{\mathcal{O}}$, which upon receiving the input state $\tau_\lambda \in \{\rho_\lambda, \sigma_\lambda\}$, $\mathcal{D}^{\mathcal{O}}(1^\lambda)$ simulates $\mathcal{B}^{\mathcal{O}, O_\rho, \text{H-desc}}(1^\lambda)$ to obtain unitary U_λ . $\mathcal{D}^{\mathcal{O}}$ does the following:

1. $\mathcal{D}$ runs $\mathcal{B}$ exactly, when $\mathcal{B}$ does not make any queries to the oracles.
2. When $\mathcal{B}$ queries the oracle $\mathcal{O}$, $\mathcal{D}$ also queries the oracle $\mathcal{O}$.
3. When $\mathcal{B}$ queries the oracle O_ρ , $\mathcal{D}$ simulates the output of O_ρ by randomly sampling the bit string $r_1 \leftarrow \{0, 1\}^{n(\lambda)}$, querying the oracle $\mathcal{O}$ and returning the state $|f_\lambda(r_1)\rangle \langle f_\lambda(r_1)|$. Since r_1 is sampled uniformly and independently, the state so prepared is identical to the state ρ_λ .
4. When $\mathcal{B}$ queries the oracle H-desc, $\mathcal{D}$ can simulate H-desc exactly, because it is independent of the oracle $\mathcal{O}$. Therefore, $\mathcal{D}$ need not make any additional queries to $\mathcal{O}$.

From Lemma 47 we know that ρ_λ and σ_λ are computationally indistinguishable. Using Lemma 50 and Remark 51 this means that for any unitary U_λ obtained by $\mathcal{D}^{\mathcal{O}}(1^\lambda)$ (which is a QPT algorithm), there exists negligible functions $\text{negl}_1, \text{negl}_2$ such that

$$\begin{aligned} |\text{tr}[H_\lambda \rho_\lambda] - \text{tr}[H_\lambda \sigma_\lambda]| &\leq \text{negl}_1(\lambda), \text{ and} \\ \left| \text{tr}[H_\lambda U_\lambda \rho_\lambda U_\lambda^\dagger] - \text{tr}[H_\lambda U_\lambda \sigma_\lambda U_\lambda^\dagger] \right| &\leq \text{negl}_2(\lambda). \end{aligned}$$

Using triangle inequality, we have

$$\begin{aligned} \left| \left(\text{tr}[H_\lambda \rho_\lambda] - \text{tr}[H_\lambda U_\lambda \rho_\lambda U_\lambda^\dagger] \right) - \underbrace{\left(\text{tr}[H_\lambda \sigma_\lambda] - \text{tr}[H_\lambda U_\lambda \sigma_\lambda U_\lambda^\dagger] \right)}_{0 \text{ since } \sigma_\lambda \text{ is invariant under unitary operations}} \right| &\leq \text{negl}_1(\lambda) + \text{negl}_2(\lambda) \\ \implies \left(\text{tr}[H_\lambda \rho_\lambda] - \text{tr}[H_\lambda U_\lambda \rho_\lambda U_\lambda^\dagger] \right) &\leq \text{negl}_1(\lambda) + \text{negl}_2(\lambda) \\ \implies \mathcal{W}_{\rho, \mathbf{H}, \mathcal{B}}(\lambda) &\leq \text{negl}_1(\lambda) + \text{negl}_2(\lambda) \\ &= \text{negl}_{\mathcal{B}}(\lambda). \end{aligned}$$

Since for all $\mathcal{B} \in \text{QPT}$, we can construct a simulator $\mathcal{D}$ which only uses the random oracle $\mathcal{O}$, there exists some negligible function $\text{negl}_{\mathcal{B}}$ such that, $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{B}}(\lambda) \leq \text{negl}_{\mathcal{B}}(\lambda)$, it follows that there does not exist a non-negligible function non-negl such that $\widehat{\text{erg}}_{\mathcal{D}} \geq_\infty \text{non-negl}$. $\square$

Claim 54. It holds that $\text{erg}_{\mathcal{D}}(\lambda) = \frac{n(\lambda)}{2}$.

Proof. For the family of states and Hamiltonians $\rho, \mathbf{H}$ it holds that

$$\text{tr}[H_\lambda \rho_\lambda] = \frac{n(\lambda)}{2}.$$

This is because, each of the last $n(\lambda)$ -qubits of ρ_λ corresponds to 0 or 1 in the computational basis with equal probability and since H_λ is the hamming weight Hamiltonian on the last $n(\lambda)$ -qubits, the expected hamming weight of the last $n(\lambda)$ -qubits is $\frac{n(\lambda)}{2}$.

A query unbounded algorithm $\mathcal{A}^{O, O_\rho, H\text{-desc}}(1^\lambda)$ can query the random oracle at all points in $\{0, 1\}^{n(\lambda)}$ and find the complete truth table of the random oracle O_λ . With this information, it can output the description of a unitary U_λ , which maps $|f_\lambda(r)\rangle \mapsto |r^*, 0^{n(\lambda)}\rangle$, where r^* is the lexicographically smallest inverse of $f_\lambda(r)$, i.e. $r^* \in \{0, 1\}^{n(\lambda)}$ is the smallest value such that $f_\lambda(r^*) = f_\lambda(r)$.

For such a unitary U_λ , it holds that

$$\text{tr} [H_\lambda U_\lambda \rho_\lambda U_\lambda^\dagger] = 0.$$

This is because after applying the unitary U_λ on the state ρ_λ , the resulting state would always have $0^{n(\lambda)}$ in the last $n(\lambda)$ -qubits and therefore, the hamming weight over the last $n(\lambda)$ -qubits is 0.

Therefore, for using this explicit unbounded algorithm $\mathcal{A}$, we are able to construct a unitary U_λ such that,

$$\text{tr} [H_\lambda \rho_\lambda] - \text{tr} [H_\lambda U_\lambda \rho_\lambda U_\lambda^\dagger] = \frac{n(\lambda)}{2}.$$

From this, we conclude that $\text{erg}_\mathcal{D} \geq \infty \frac{n(\lambda)}{2}$. □

Combining Claims 53 and 54 we have the proof for the theorem. □

4.4.2 Constructive Separation in the plain model

In Theorem 56 below, we lift the separation in the random oracle model (see Theorem 52) to the plain model by instantiating the random oracle with a pseudorandom function (PRF) whose key is hidden.

Let $F_\lambda : \{0, 1\}^{k(\lambda)} \times \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{2n(\lambda)}$ be an expanding pseudorandom function (see Definition 25).

Definition 55. Let $n, \kappa : \mathbb{N} \rightarrow \mathbb{N}$ be polynomials denoting the size of the quantum system and size of the key respectively. We define a distribution $\mathcal{D}$ over families of states $(\rho, \mathbf{H})$ as follows.

- The family of keys $\mathbf{k} = \{k_\lambda\}_\lambda$ where $k_\lambda \leftarrow \{0, 1\}^{\kappa(\lambda)}$.
- The family of states $\rho = \{\rho_\lambda\}_\lambda$ acting on $2n(\lambda)$ -qubits where

$$\rho_\lambda := \frac{1}{c} \sum_{r \in \{0, 1\}^{n(\lambda)}} |F_\lambda(k_\lambda, r)\rangle \langle F_\lambda(k_\lambda, r)|$$

where c is chosen to ensure $\text{tr}(\rho_\lambda) = 1$.

- The family of Hamiltonians $\mathbf{H} = \{H_\lambda\}_\lambda$ (independent of $\mathbf{k}$) where H_λ is the padded Hamming weight Hamiltonian (see Equation (12)).

Intuitively, we have PRF $F_\lambda(k, \cdot)$ play the role of the random oracle O_λ . While this cannot be done in general, it is straightforward to see that the proof of Theorem 52 can be extended to work for a PRF.

Theorem 56 (Separating $\text{erg}_\mathcal{D}$ from $\widehat{\text{erg}}_\mathcal{D}$ in the plain model). *Assuming the existence of quantum secure PRFs, for the family of states, Hamiltonians, $\rho, \mathbf{H}$ sampled from $\mathcal{D}$ as defined above (see Definition 55), it holds that*

$$\nexists \text{ non-negl s.t. } \widehat{\text{erg}}_\mathcal{D} \geq \infty \text{ non-negl, while } \text{erg}_\mathcal{D} \geq \infty \frac{n}{2}$$

where non-negl is any non-negligible function.

Proof sketch. To establish $\text{erg}_{\mathcal{D}} \geq_{\infty} \frac{n}{2}$, observe that an unbounded algorithm $\mathcal{A}$ can query the state generation oracle O_{ρ} exponentially many times, to enumerate all possible values $F_{\lambda}(r)$ assumes. With this information, $\mathcal{A}$ can output the description of a unitary that maps $\{|F_{\lambda}(r)\rangle\}_{r \in \{0,1\}^{n(\lambda)}} \mapsto \{|r, 0^{n(\lambda)}\rangle\}_{r \in \{0,1\}^{n(\lambda)}}$. This way, the hamming weight of the second half of the string is 0, therefore attaining $\text{erg}_{\mathcal{D}} = \frac{n}{2}$.

To establish an upper bound on $\widehat{\text{erg}}_{\mathcal{D}}$, we consider a slightly relaxed problem, i.e. we allow all algorithms to access $F_{\lambda}(k, \cdot)$. Since we are only giving extra information to the algorithms, work extracted in this relaxed setting, will still serve as an upper bound on $\widehat{\text{erg}}_{\mathcal{D}}$.

In this relaxed setting, we want to ensure that

1. ρ is computationally indistinguishable from σ (see Equation (11)),
2. The unitary U_{λ} produced by any efficient algorithm $\mathcal{B}^{O_{\rho}, \text{H-desc}, F_{\lambda}(k, \cdot)}(1^{\lambda})$ can also be produced by an efficient algorithm $\mathcal{C}^{F_{\lambda}(k, \cdot)}(1^{\lambda})$ (where H-desc can be assumed to be hardcoded in $\mathcal{C}$ because it describes a local Hamiltonian).

The first point follows by applying a hybrid argument as follows: (i) replace the PRF with a random oracle (with negligible error using Definition 25) and (ii) in the random oracle model, use the indistinguishability of ρ_{RO} and σ (see Lemma 47) where $\rho_{\text{RO}} \propto (\sum_{r \in \{0,1\}^{n(\lambda)}} |f_{\lambda}(r)\rangle \langle f_{\lambda}(r)|)$ is the state in Equation (10). The second point can be proved by proceeding as we did in the proof of Claim 53.

One can now use Lemma 50 and Remark 51 (as we did in the proof of Claim 53) to conclude that there is a negligible function negl' such that $\mathcal{W}_{\rho, \text{H}, \mathcal{B}} \leq_{\infty} \text{negl}'$ (for every QPT algorithm $\mathcal{B}$). This establishes that $\widehat{\text{erg}}_{\mathcal{D}}$ is negligible (see Notation 44). □

5 Catalytic Computation

The definition of computational work extraction we considered above (see Definition 31), requires that the unitary output by the algorithm, must act strictly only on $n(\lambda)$ -qubits. This requirement, however, appears to be too restrictive. For instance, even the most basic cryptographic applications rely on keyed cryptographic primitives, where the key k is used explicitly. If one is limited to using only as many qubits as the input, one quickly runs into such issues as the following problem.

Problem 57. Given the key $k \in \{0, 1\}^{\lambda}$, implement a pseudo-random permutation π_k on $n(\lambda)$ -bit strings, using only $\lambda + n(\lambda)$ -qubits, i.e. *efficiently* implement $|k, x\rangle \mapsto |k, \pi_k(x)\rangle$.

The standard compilation of arbitrary classical circuits into reversible quantum circuits usually requires auxiliary space to store intermediate computations [Ben89]. The standard pseudorandom permutation oracle acts on $\lambda + 2n(\lambda)$ -qubits, λ -qubits to key the permutation to use, $n(\lambda)$ -qubits for input and $n(\lambda)$ -qubits for output,

$$O_{\pi} |k\rangle_k |x\rangle_i |0\rangle_o = |k\rangle_k |x\rangle_i |\pi_k(x)\rangle_o$$

Without auxiliary space, it is unclear whether standard cryptographic primitives can be implemented efficiently. One may be tempted to resolve this by allowing quantum algorithms to use auxiliary qubits (ancillas). While this approach is often taken when one considers polynomial-time machines, in the physical context here, this introduces a fundamental flaw—unconstrained fresh qubits may be exploited to extract near-maximal work even from passive states (see Definition 15), as illustrated by the following.

Example 58. Consider the Hamming-weight Hamiltonian $H_{\text{hw}(n)}$ (see Definition 13) acting on n qubits and consider the maximally mixed state $\sigma_n \propto \mathbb{I}_n$. Clearly, one should not be able to extract any work from σ_n using a unitary operation. However, if one allows for ancillas initialised to $|0^n\rangle \langle 0^n|$, then one can easily swap the system and ancilla registers (which is a unitary operation) to extract $n/2$ units of work.

The discussion so far, suggests that the physically relevant definition of computational work extraction must be some kind of generalisation of Definition 31. We propose the following generalisation: allow auxiliary qubits that must be restored exactly to the state they were initialised in. We call such auxiliary qubits, catalyst qubits and the corresponding model of computation, the *catalytic model of computation*. Clearly, the pathological case described in Example 58 no longer applies—and indeed, as we show later (see Section 6), the operational meaning of ergotropy remains meaningful.

Furthermore, in this catalytic model of computation, it is immediate that one can now solve Problem 57 by implementing the PRP as follows:

$$|k, x, 0\rangle \mapsto |k, x, \pi_k(x)\rangle \mapsto |k, 0, \pi_k(x)\rangle \mapsto |k, \pi_k(x), 0\rangle.$$

where the second step erases x by computing π_k^{-1} of the third register. Each classical logic gates gate is mapped to controlled gate that acts as $|x\rangle |b\rangle \mapsto |x\rangle |b \oplus \text{gate}(x)\rangle$.

Clearly, some catalytic models of computation allows for potentially more operations. There could be various notions of catalysts (e.g. whether they are restored exactly or approximately, whether they can be left slightly entangled, whether the catalytic condition is required for a fixed state or all states, etc.). A priori, it is unclear whether these additional operations allow one to map an input state $|\psi\rangle$ to a larger class of output states $|\phi\rangle$. We address this in both the computationally bounded and unbounded setting—starting with the latter.

5.1 Catalytic Unitaries

Consider a bipartite system with Hilbert space $\mathcal{H} = \mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$, where $\mathcal{H}_{\text{IO}}$ denotes the Hilbert space corresponding to an *input/output* (unconstrained) quantum space and $\mathcal{H}_{\text{catalyst}}$ denotes the *catalytic space*. We start by considering a model of catalytic computation where unitaries act on $\mathcal{H}$ in such a way that every state in the catalytic space $\mathcal{H}_{\text{catalyst}}$, is restored exactly after the computation.

Let $n, \ell \in \mathbb{N}$, $\mathcal{H}_{\text{IO}} = (\mathbb{C}^2)^{\otimes n}$ and $\mathcal{H}_{\text{catalyst}} = (\mathbb{C}^2)^{\otimes \ell}$.

Definition 59 (Catalytic Unitaries $\text{cat-}\mathcal{U}(n, \ell)$). Let the $(n + \ell)$ -qubit composite Hilbert space be denoted by $\mathcal{H} := \mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$. Then, the set of *catalytic unitaries* $\text{cat-}\mathcal{U}(n, \ell)$ is defined as

$$\text{cat-}\mathcal{U}(n, \ell) := \left\{ U \in \mathcal{U}(n + \ell) : \begin{array}{l} \forall |\phi\rangle \otimes |\eta\rangle \in \mathcal{H}, \quad \exists |\psi\rangle \in \mathcal{H}_{\text{IO}} \\ \text{s.t. } U |\phi\rangle \otimes |\eta\rangle = |\psi\rangle \otimes |\eta\rangle \end{array} \right\}.$$

Since the primary system of interest is the register IO, one would often want to consider only the reduced state on IO. For brevity, we introduce the following notation to represent the reduced state of the system after the application of a catalytic unitary.

Notation 60. Let $\mathcal{V} \subseteq \mathcal{U}(n + \ell)$ be set of unitaries on $\mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$. Then, by $C_{\mathcal{V}}$ we denote the set of *channels associated* with the action of these unitaries using the catalyst as

$$C_{\mathcal{V}} := \{C_U^\sigma(\cdot) : U \in \mathcal{V}, \sigma \in \text{Den}(\ell)\}.$$

where $C_U^\sigma(\cdot) := \text{tr}_{\text{catalyst}} U(\cdot \otimes \sigma) U^\dagger$ is the channel associated with (U, σ) .

Example 61. These may help to clarify the notation.

1. Note that for $\ell = 0$, $C_{\mathcal{U}(n)} = \{V(\cdot)V^\dagger : V \in \mathcal{U}(n)\}$.
2. Note also that $C_{\text{cat-}\mathcal{U}(n,\ell)}$ is simply $\{\text{tr}_{\text{catalyst}} U(\cdot \otimes \sigma)U^\dagger : U \in \text{cat-}\mathcal{U}(n,\ell), \sigma \in \text{Den}(\ell)\}$.

Clearly, it holds that $C_{\mathcal{U}(n)} = C_{\text{cat-}\mathcal{U}(n,\ell)}$ when $\ell = 0$ because $\mathcal{U}(n) = \text{cat-}\mathcal{U}(n,0)$. On the other hand, with $\ell > 0$, $C_{\text{cat-}\mathcal{U}(n,\ell)}$ may allow one to map $|\phi\rangle$ to a larger class of states $|\psi\rangle$, i.e. the containment

$$C_{\mathcal{U}(n)} \subseteq C_{\text{cat-}\mathcal{U}(n,\ell)}$$

could potentially be strict.

A little thought reveals that the two classes are equivalent, i.e. $C_{\text{cat-}\mathcal{U}(n,\ell)} = C_{\mathcal{U}(n)}$. Suppose $U \in \text{cat-}\mathcal{U}(n,\ell)$. For an orthonormal basis $\{|\phi_i\rangle\}$ of $\mathcal{H}_{\text{IO}}$, and for any $|\eta\rangle \in \mathcal{H}_{\text{catalyst}}$, the action of U must be $U|\phi_i\rangle \otimes |\eta\rangle = |\psi_i\rangle \otimes |\eta\rangle$. Since U must map orthonormal vectors to orthonormal vectors, we have that $\{|\psi_i\rangle\}_i$ must also be an orthonormal basis of $\mathcal{H}_{\text{IO}}$. Therefore, for $V := \sum_i |\psi_i\rangle \langle \phi_i| \in \mathcal{U}(n)$, we have that $C_U^\eta(\phi) = C_V(\phi)$ for all $|\phi\rangle \in \mathcal{H}_{\text{IO}}$.

Claim 62 ($C_{\mathcal{U}(n)} = C_{\text{cat-}\mathcal{U}(n,\ell)}$). For all $U \in \text{cat-}\mathcal{U}(n,\ell)$ there exists $V \in \mathcal{U}(n)$ such that, for all $|\phi\rangle_{\text{IO}} \otimes |\eta\rangle_{\text{catalyst}} \in \mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$ it holds that

$$C_U^\eta(\phi) = C_V(\phi).$$

In view of this, one might argue that perhaps the definition is too strict—requiring *all* catalysts to be restored *exactly* is not very physical. Indeed, no physical process happens with zero error. Further, using a completely unknown state in a register that, must be restored exactly, seems rather difficult.

We thus, address both of these arguments at once and consider a more relaxed notion of catalytic computation: we only require that for some fixed state, the catalytic property holds—and that too up to some error ε . In fact, we even allow the catalyst register to stay slightly entangled with the IO register, as long as it is close to being a tensor product. More formally, we consider the following notion.

Definition 63 (Approximate Catalytic Unitaries $\text{cat-}\mathcal{U}(n,\ell,\varepsilon)$). Let the $(n+\ell)$ -qubit composite Hilbert space be denoted by $\mathcal{H} := \mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$. Then, the set of *approximate catalytic unitaries* $\text{cat-}\mathcal{U}(n,\ell,\varepsilon)$ is defined as

$$\text{cat-}\mathcal{U}(n,\ell,\varepsilon) := \left\{ U \in \mathcal{U}(n+\ell) : \begin{array}{l} \exists |\eta\rangle \in \mathcal{H}_{\text{catalyst}} \forall |\phi\rangle \in \mathcal{H}_{\text{IO}}, \quad \exists |\psi\rangle \otimes |\xi\rangle \in \mathcal{H} \\ \text{s.t. } \text{TD}(U(\phi \otimes \eta)U^\dagger, \psi \otimes \xi) \leq \varepsilon \end{array} \right\}.$$

We now investigate the same question: does relaxing the catalyst restoration to being approximate, result in more computational power? To wit:

$$C_{\text{cat-}\mathcal{U}(n,\ell,\varepsilon)} \stackrel{?}{\subsetneq} C_{\mathcal{U}(n)}.$$

It turns out that, up to some error (that vanishes with ε) these operations are still the same. We prove this in Lemma 64 and Theorem 65 below.

Lemma 64. Let S be a finite-dimensional quantum system with Hilbert space $\mathcal{H}_S$, and let C be a catalyst register with Hilbert space $\mathcal{H}_C$. Let the catalyst be initially prepared in a fixed pure state $|\eta\rangle_C$, and let U_{SC} be a fixed unitary on $\mathcal{H}_S \otimes \mathcal{H}_C$. Define the isometry $W : \mathcal{H}_S \rightarrow \mathcal{H}_S \otimes \mathcal{H}_C$ by

$$W|\phi\rangle := U_{SC}(|\phi\rangle_S |\eta\rangle_C), \tag{19}$$

and let the induced channel on S be

$$\mathcal{N}_S(\rho) := \text{tr}_C [W\rho W^\dagger]. \tag{20}$$

Suppose that $0 \leq \varepsilon < 1$ and that, for every pure state $|\phi\rangle_S$, there exists a pure state $|\psi_\phi\rangle_S$ such that

$$\text{TD} \left(W |\phi\rangle \langle \phi| W^\dagger, |\psi_\phi\rangle \langle \psi_\phi|_S \otimes |\eta\rangle \langle \eta|_C \right) \leq \varepsilon. \quad (21)$$

Then there exists a fixed unitary V_S on $\mathcal{H}_S$ such that, for every $\rho \in \text{Den}(S)$,

$$\text{TD} \left(\mathcal{N}_S(\rho), V_S \rho V_S^\dagger \right) \leq \frac{3\varepsilon^2}{2}. \quad (22)$$

Proof. Define the linear operator $\mathcal{L} : \mathcal{H}_S \rightarrow \mathcal{H}_S$ by

$$\mathcal{L} := (\mathbb{I}_S \otimes \langle \eta |_C) W. \quad (23)$$

For every pure state $|\phi\rangle_S$, the two states appearing in Equation (21) are pure. Hence, using

$$\text{TD}(|a\rangle \langle a|, |b\rangle \langle b|) = \sqrt{1 - |\langle a | b \rangle|^2}, \quad (24)$$

we obtain

$$\begin{aligned} |(\langle \psi_\phi |_S \otimes \langle \eta |_C) W |\phi\rangle| &\geq \sqrt{1 - \varepsilon^2} \\ \implies |\langle \psi_\phi | \mathcal{L} |\phi\rangle| &\geq \sqrt{1 - \varepsilon^2}. \end{aligned} \quad (25)$$

Since $|\psi_\phi\rangle$ is normalized, Equation (25) implies

$$\|\mathcal{L} |\phi\rangle\| \geq \sqrt{1 - \varepsilon^2} \quad \text{for every normalized } |\phi\rangle \in \mathcal{H}_S. \quad (26)$$

Moreover, $\mathcal{L}$ is a contraction because it is obtained by projecting the isometry W onto the catalyst state $|\eta\rangle$. Therefore,

$$\sqrt{1 - \varepsilon^2} \leq \|\mathcal{L} |\phi\rangle\| \leq 1 \quad \text{for every normalized } |\phi\rangle \in \mathcal{H}_S. \quad (27)$$

Equivalently,

$$(1 - \varepsilon^2)\mathbb{I}_S \preceq \mathcal{L}^\dagger \mathcal{L} \preceq \mathbb{I}_S. \quad (28)$$

Let

$$\mathcal{L} = \sum_{j=1}^d s_j |u_j\rangle \langle v_j| \quad (29)$$

be a singular-value decomposition. By Equation (28),

$$\sqrt{1 - \varepsilon^2} \leq s_j \leq 1 \quad \text{for every } j \in [d]. \quad (30)$$

Define

$$V_S := \sum_{j=1}^d |u_j\rangle \langle v_j|. \quad (31)$$

Since $\varepsilon < 1$, all singular values of $\mathcal{L}$ are strictly positive, and hence V_S is unitary. Furthermore,

$$\begin{aligned} \|\mathcal{L} - V_S\|_\infty &= \max_j |s_j - 1| \\ &\leq 1 - \sqrt{1 - \varepsilon^2}. \end{aligned} \quad (32)$$

We now separate the component of W in which the catalyst is restored exactly from the component orthogonal to $|\eta\rangle_C$. Define $K : \mathcal{H}_S \rightarrow \mathcal{H}_S \otimes \mathcal{H}_C$ by

$$K := W - (\mathcal{L} \otimes |\eta\rangle_C). \quad (33)$$

By the definition of $\mathcal{L}$,

$$(\mathbb{I}_S \otimes \langle \eta |_C) K = 0. \quad (34)$$

Consequently, the two terms in Equation (33) have orthogonal catalyst components. Since $W^\dagger W = \mathbb{I}_S$, we therefore have

$$K^\dagger K = \mathbb{I}_S - \mathcal{L}^\dagger \mathcal{L} \preceq \varepsilon^2 \mathbb{I}_S, \quad (35)$$

where the final inequality follows from Equation (28).

Using Equations (33) and (34), the reduced system channel can be written as

$$\mathcal{N}_S(\rho) = \mathcal{L}\rho\mathcal{L}^\dagger + \text{tr}_C \left[K\rho K^\dagger \right]. \quad (36)$$

Indeed, the cross terms vanish after tracing out the catalyst because the image of K is orthogonal to the catalyst state $|\eta\rangle_C$.

Let

$$R_\rho := \text{tr}_C \left[K\rho K^\dagger \right]. \quad (37)$$

This is positive semidefinite and, by Equation (35),

$$\text{tr} R_\rho = \text{tr} \left[K^\dagger K \rho \right] \leq \varepsilon^2.$$

We first compare the first term in Equation (36) with the unitary channel generated by V_S . Using the Hölder inequality, $\|\rho\|_1 = 1$, $\|\mathcal{L}\|_\infty \leq 1$, and $\|V_S\|_\infty = 1$, we obtain

$$\begin{aligned} \frac{1}{2} \left\| \mathcal{L}\rho\mathcal{L}^\dagger - V_S\rho V_S^\dagger \right\|_1 &\leq \frac{1}{2} \left\| (\mathcal{L} - V_S)\rho\mathcal{L}^\dagger \right\|_1 + \frac{1}{2} \left\| V_S\rho(\mathcal{L}^\dagger - V_S^\dagger) \right\|_1 \\ &\leq \|\mathcal{L} - V_S\|_\infty \\ &\leq 1 - \sqrt{1 - \varepsilon^2}. \end{aligned} \quad (38)$$

Moreover, since $R_\rho \succeq 0$,

$$\frac{1}{2} \|R_\rho\|_1 = \frac{1}{2} \text{tr} R_\rho \leq \frac{\varepsilon^2}{2}. \quad (39)$$

Combining Equations (36), (38) and (39) with the triangle inequality yields

$$\text{TD} \left(\mathcal{N}_S(\rho), V_S\rho V_S^\dagger \right) \leq 1 - \sqrt{1 - \varepsilon^2} + \frac{\varepsilon^2}{2} \leq \frac{3}{2}\varepsilon^2.$$

□

By inspecting the definitions of $\text{cat-}\mathcal{U}(n, \ell, \varepsilon)$ and $\mathcal{U}(n)$ above, Lemma 64 can be recast more compactly as follows.

Recall that $n, \ell \in \mathbb{N}$, $\mathcal{H}_{\text{IO}} = (\mathbb{C}^2)^{\otimes n}$ and $\mathcal{H}_{\text{catalyst}} = (\mathbb{C}^2)^{\otimes \ell}$.

Theorem 65 ($C_{\text{cat-}\mathcal{U}(n, \ell, \varepsilon)} \approx_{\varepsilon^2} C_{\mathcal{U}(n)}$). *Let $0 \leq \varepsilon < 1$, and let $U \in \text{cat-}\mathcal{U}(n, \ell, \varepsilon)$. For every pure catalyst state $\eta = |\eta\rangle\langle\eta| \in \text{Den}(\ell)$, there exists a unitary $V_\eta \in \mathcal{U}(n)$ such that*

$$\max_{\rho \in \text{Den}(n)} \text{TD} \left(C_U^\eta(\rho), C_{V_\eta}(\rho) \right) \leq \frac{3\varepsilon^2}{2}. \quad (40)$$

Proof. Fix a pure catalyst state $\eta = |\eta\rangle\langle\eta|$. By the definition of $\text{cat-}\mathcal{U}(n, \ell, \varepsilon)$, for every pure state $|\phi\rangle \in \mathcal{H}_S$ there exists a pure state $|\psi_\phi\rangle \in \mathcal{H}_S$ such that

$$\text{TD} \left(U(|\phi\rangle\langle\phi| \otimes \eta) U^\dagger, |\psi_\phi\rangle\langle\psi_\phi| \otimes \eta \right) \leq \varepsilon. \quad (41)$$

Therefore, from Lemma 64, there exists a unitary $V_\eta \in \mathcal{U}(n)$ such that, for every $\rho \in \text{Den}(n)$,

$$\text{TD} \left(\text{tr}_{\text{catalyst}} \left[U(\rho \otimes \eta) U^\dagger \right], V_\eta \rho V_\eta^\dagger \right) \leq \frac{3\varepsilon^2}{2}. \quad (42)$$

Using the definitions

$$C_U^\eta(\rho) = \text{tr}_{\text{catalyst}} \left[U(\rho \otimes \eta) U^\dagger \right]$$

and

$$C_{V_\eta}(\rho) = V_\eta \rho V_\eta^\dagger,$$

we obtain

$$\max_{\rho \in \text{Den}(n)} \text{TD} (C_U^\eta(\rho), C_{V_\eta}(\rho)) \leq \frac{3\varepsilon^2}{2}.$$

□

As alluded to earlier, from Theorem 65 above, we essentially get $C_{\text{cat-}\mathcal{U}(n, \ell, \varepsilon)} \approx_{\varepsilon^2} C_{\mathcal{U}(n)}$, i.e. the catalytic model of computation does not offer an advantage in the computationally unbounded setting—even where the catalyst may be correlated with the system.

Could it be that in the computationally bounded setting, catalysts turn out to be advantageous?

To answer this question, we first define computationally efficient variants of catalytic unitaries discussed above.

5.2 Computational Catalytic setting

The most natural way to define a computational version of the catalytic model, seems to be one where the catalyst register is initialised in some fixed state $|0^\ell\rangle$ of ℓ qubits and must be restored exactly. Restricting to $|0^\ell\rangle$ ensures that one cannot inject “advice” into the model, making it potentially too powerful. Requiring that $|0^\ell\rangle$ is restored—but other states can be mapped arbitrarily—makes it easier to consider to use the catalyst in a meaningful way. Finally, the requirement that the catalyst be exactly restored, seems reasonable in the idealised setting (e.g. where fault-tolerance allows one to focus on logical qubits). One may justifiably argue that perhaps allowing some error could substantially increase the power of this model. We account for this later where we prove impossibilities (i.e. problems one cannot solve with few/no catalysts in the computational setting), we allow arbitrary errors—in fact, we prove space lower bounds.

It would be helpful to first define a class of catalytic unitaries, for a fixed $n, \ell \in \mathbb{N}$, that is required to restore a given catalyst state $|\eta\rangle$ on ℓ -qubits.

Definition 66 (Catalytic unitaries given a catalyst $\text{cat-}\mathcal{U}_{|\eta\rangle}(n, \ell)$). Let the $(n + \ell)$ -qubit composite Hilbert space be denoted by $\mathcal{H} := \mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$. Then, the set of *catalytic unitaries* $\text{cat-}\mathcal{U}_{|\eta\rangle}(n, \ell)$ given the catalyst $|\eta\rangle \in \mathcal{H}_{\text{catalyst}}$, is defined as

$$\text{cat-}\mathcal{U}_{|\eta\rangle}(n, \ell) := \left\{ U \in \mathcal{U}(n + \ell) : \begin{array}{l} \forall |\phi\rangle \in \mathcal{H}_{\text{IO}}, \quad \exists |\psi\rangle \in \mathcal{H}_{\text{IO}} \\ \text{s.t. } U(|\phi\rangle \otimes |\eta\rangle) = |\psi\rangle \otimes |\eta\rangle \end{array} \right\}.$$

To restrict to a computational notion of catalytic computation, we now define a class of PPT algorithms that generate the description of catalytic unitaries that are in $\text{cat-}\mathcal{U}_{|0^\ell\rangle}$.

Fix polynomials $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$. For $\lambda \in \mathbb{N}$, let $\mathcal{H}_{\text{IO}, \lambda} = (\mathbb{C}^2)^{\otimes n(\lambda)}$ and $\mathcal{H}_{\text{catalyst}, \lambda} = (\mathbb{C}^2)^{\otimes \ell(\lambda)}$ be the family of Hilbert spaces indexed by λ , corresponding to the quantum workspace and the catalyst respectively.

Definition 67 ($(n, \ell)\text{-}\widehat{\text{catGens}}$). A PPT algorithm $\mathcal{A}$ belongs to $(n, \ell)\text{-}\widehat{\text{catGens}}$, if for all $\lambda \in \mathbb{N}$, running $\mathcal{A}$ on input 1^λ produces a circuit description of a projector $\Pi_\lambda \in \text{Proj}(n(\lambda))$ and a catalytic unitary $U_\lambda \in \text{cat-}\mathcal{U}_{|0^{\ell(\lambda)}\rangle}(n(\lambda), \ell(\lambda))$, i.e. $\text{desc}(\Pi_\lambda), \text{desc}(U_\lambda) \leftarrow \mathcal{A}(1^\lambda)$.

Using this notion of computational catalytic computation, one can define a corresponding class of relational problems. We first setup some notation to describe “valid relations” based on the sizes of the IO register and the catalyst register. Informally, we require that both the input and output of the relation, to fit into the IO register.

Using the notation setup above, we denote the set of *valid relations* as

$$R := \left\{ \text{rel} = \{\text{rel}_\lambda\}_\lambda : \begin{array}{l} \exists m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N} \text{ s.t. } \forall \lambda \in \mathbb{N}, n(\lambda) \geq \max(m_1(\lambda), m_2(\lambda)) \\ \text{rel}_\lambda \subseteq \{0, 1\}^{m_1(\lambda)} \times \{0, 1\}^{m_2(\lambda)} \end{array} \right\}.$$

We can now define the relation $\widehat{\text{Fcat}}$ that is a set of relations one can solve with some computational catalytic algorithm, i.e. an algorithm in $\widehat{\text{catGens}}$.

Definition 68 ($(n, \ell)\text{-}\widehat{\text{Fcat}}$). We say a relation $\text{rel} \in R$ belongs to the complexity class $(n, \ell)\text{-}\widehat{\text{Fcat}}$ if there exists an algorithm $\mathcal{A} \in (n, \ell)\text{-}\widehat{\text{catGens}}$ such that with probability at least $2/3$ rds, $\mathcal{A}$ produces U_λ satisfying the following: For all $(x_\lambda, \cdot) \in \text{rel}_\lambda$,

- there exists y_λ such that $(x_\lambda, y_\lambda) \in \text{rel}_\lambda$ and
- some normalised state $|\varphi\rangle$ such that

$$U_\lambda \underbrace{|x_\lambda, 0^{n(\lambda)-|x_\lambda|}\rangle}_{\text{IO}} \underbrace{|0^{\ell(\lambda)}\rangle}_{\text{catalyst}} = \underbrace{|y_\lambda\rangle}_{\text{IO}} \underbrace{|\varphi\rangle}_{\text{catalyst}} \underbrace{|0^{\ell(\lambda)}\rangle}_{\text{catalyst}}.$$

Definition 69. To clarify the notation, consider the following problem—that we eventually use for one of our separations.

Example 70. Given $x \in \{0, 1\}^*$, produce

$$y := (x, f(g(x))).$$

Where $f, g : \{0, 1\}^* \rightarrow \{0, 1\}^*$ are arbitrary functions such that, $f = \{f_\lambda\}_\lambda$ where $f_\lambda : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ and similarly $g = \{g_\lambda\}_\lambda$ where $g_\lambda : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$. In our notation, this problem can be described as follows.

- $\text{rel} = \{\text{rel}_\lambda\}_\lambda$ where $\text{rel}_\lambda = \{(x, y) : y = (x, f(g(x))), \forall x \in \{0, 1\}^\lambda\}$,
- For this relation $m_1(\lambda) = \lambda, m_2(\lambda) = 2\lambda$.

Note that this relation is only valid when the size of $\mathcal{H}_{\text{IO}}, n : \mathbb{N} \rightarrow \mathbb{N}$, is such that $n(\lambda) \geq 2\lambda$.

Using this complexity class, one can concretely ask whether catalysts help in the computational setting:

$$(n, 0)\text{-}\widehat{\text{Fcat}} \stackrel{?}{\subsetneq} (n, \ell)\text{-}\widehat{\text{Fcat}}$$

for some $\ell > 0$?

In the following discussion, we work in the oracle setting and therefore first describe oracular versions of the definitions considered so far. These are fairly natural but we include them for completeness.

5.2.1 Computational catalytic computation with oracles

Let $c \in \mathbb{N}$ be a constant and oracles $\mathbf{O} = \{O_1, \dots, O_c\}$, where O_i encodes a function $f_i : \mathcal{Q} \rightarrow \mathcal{R}$ via

$$O_i |q, r\rangle = |q, r \oplus f_i(q)\rangle, \quad q \in \mathcal{Q}, r \in \mathcal{R}$$

A machine $\mathcal{A} \in \text{PPT}$ that outputs a circuit description of the unitaries can do the following actions when given access to $\mathbf{O}$:

1. At any step, $\mathcal{A}$ may make queries to $\mathbf{O}$;
2. The circuit description of the unitary, $\text{desc}(U_\lambda) \leftarrow \mathcal{A}^{\mathbf{O}}(1^\lambda)$, that is output by $\mathcal{A}$, can use the oracles $O_1, \dots, O_c$ as a gate. Therefore, the unitary circuit U_λ can query $\mathbf{O}$ at any point in its execution.

In the oracle setting, we allow the set of relations $R^{\mathbf{O}}$ to also depend on $\mathbf{O}$.

Remark 71 ($(n, \ell)\text{-}\widehat{\text{catGens}}^{\mathbf{O}}$ and $(n, \ell)\text{-}\widehat{\text{Fcat}}^{\mathbf{O}}$). We extend the definitions of $(n, \ell)\text{-}\widehat{\text{catGens}}$ and $(n, \ell)\text{-}\widehat{\text{Fcat}}$ naturally.

1. A PPT algorithm $\mathcal{A}^{\mathbf{O}}$ belongs to $(n, \ell)\text{-}\widehat{\text{catGens}}^{\mathbf{O}}$, if for all $\lambda \in \mathbb{N}$, running $\mathcal{A}^{\mathbf{O}}$ on input 1^λ produces a circuit description of a projector $\Pi_\lambda \in \text{Proj}(n(\lambda))$ and a catalytic unitary $U_\lambda \in \text{cat-}\mathcal{U}_{|0^{\ell(\lambda)}\rangle}(n(\lambda), \ell(\lambda))$ i.e. $(\text{desc}(\Pi_\lambda), \text{desc}(U_\lambda)) \leftarrow \mathcal{A}^{\mathbf{O}}(1^\lambda)$.
2. We say a relation $\text{rel} \in R^{\mathbf{O}}$ belongs to the complexity class $(n, \ell)\text{-}\widehat{\text{Fcat}}^{\mathbf{O}}$ if there exists an algorithm $\mathcal{A}^{\mathbf{O}} \in (n, \ell)\text{-}\widehat{\text{catGens}}^{\mathbf{O}}$ such that with probability at least $2/3$ it produces U_λ satisfying the following: For all $(x_\lambda, \cdot) \in \text{rel}_\lambda^{\mathbf{O}}$,
 - there exists y_λ such that $(x_\lambda, y_\lambda) \in \text{rel}_\lambda^{\mathbf{O}}$ and
 - some normalised state $|\varphi\rangle$ such that

$$U_\lambda \underbrace{|x_\lambda, 0^{n(\lambda)-|x_\lambda|}\rangle}_{\text{IO}} \underbrace{|0^{\ell(\lambda)}\rangle}_{\text{catalyst}} = \underbrace{|y_\lambda\rangle}_{\text{IO}} \underbrace{|\varphi\rangle}_{\text{catalyst}} |0^{\ell(\lambda)}\rangle.$$

5.3 Relational Separation $(2\lambda, c\lambda)\text{-}\widehat{\text{Fcat}} \subsetneq (2\lambda, \lambda)\text{-}\widehat{\text{Fcat}}$ (for $c < 1$)

With the notation in place, we now focus on establishing one of our main results. As we alluded to earlier (in Example 70), the problem we consider is as follows.

Let $\mathbf{O} := (O_f, O_g)$ be oracles corresponding to the *length preserving* random functions $f, g : \{0, 1\}^* \rightarrow \{0, 1\}^*$.

Problem 72. We denote by $\mathcal{P} := \{\mathcal{P}_\lambda\}_\lambda$ the relation where,

$$\mathcal{P}_\lambda := \left\{ (x, y) : x \in \{0, 1\}^\lambda, y = (x, f(g(x))) \right\}.$$

Remark 73. We observe that the relation above, satisfies the requirements in Section 5.2.1. Let

- $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$ be functions, $n(\lambda) = 2\lambda$, $\ell(\lambda) = \lambda$.
- For $\lambda \in \mathbb{N}$, let $\mathcal{H}_{\text{IO}, \lambda} = (\mathbb{C}^2)^{\otimes n(\lambda)}$ and $\mathcal{H}_{\text{catalyst}, \lambda} = (\mathbb{C}^2)^{\otimes \ell(\lambda)}$ be the family of Hilbert spaces indexed by λ .
- Let O_f, O_g be as above.

The relation $\mathcal{P}$ (see Problem 72) is a valid relation because for each $(x, y) \in \mathcal{P}_\lambda$, it holds that $\max(|x|, |y|) = 2\lambda = n(\lambda)$.

We start by establishing that with enough catalysts, one can solve the problem $\mathcal{P}$ in the computational catalytic setting.

Lemma 74. *Let $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$ be functions, $n(\lambda) = 2\lambda$, $\ell(\lambda) = \lambda$. Then, it holds that*

$$\mathcal{P} \in (n, \ell)\text{-}\widehat{\text{Fcat}}^{\mathcal{O}}.$$

Proof. Consider the family of unitaries $\mathbf{U} = \{U_\lambda\}_\lambda$ that we describe below, where each U_λ acts on 3λ . For clarity, we denote by register A, B, C the first, second and third λ -qubits the unitary acts on. We describe the action of each unitary $U_\lambda \in \mathbf{U}$ as follows.

$$\begin{aligned} |x\rangle_A |y\rangle_B |0^\lambda\rangle_C &\xrightarrow{O_{g,AC}} |x\rangle_A |y\rangle_B |g(x)\rangle_C \\ &\xrightarrow{O_{f,CB}} |x\rangle_A |y \oplus f(g(x))\rangle_B |g(x)\rangle_C \\ &\xrightarrow{O_{g,AC}} |x\rangle_A |y \oplus f(g(x))\rangle_B |0^\lambda\rangle_C. \end{aligned}$$

This unitary family $\mathbf{U}$ solves the relation rel . Since one can also efficiently describe this family using $\mathcal{O}(\lambda)$ -gates in its canonical representation—three gates corresponding to the oracles, and $\mathcal{O}(\lambda)$ gates to swap registers A, B using 2-qubit SWAP gates—this unitary family can be produced by an algorithm $\mathcal{A} \in (n, \ell)\text{-}\widehat{\text{catGens}}^{\mathcal{O}}$. $\square$

Proving exclusion from, even say $(2\lambda, 0)\text{-}\widehat{\text{Fcat}}$, turns out to be much more challenging. In fact, we will show that even with $2\lambda + c\lambda$ many qubits (where $c < 1$), one cannot solve $\mathcal{P}$ (on most inputs). We first setup some notation.

Let $\Pi_\lambda, \Pi_\lambda^\perp$ be projectors on $(n(\lambda) + \ell(\lambda))$ -qubits such that $\Pi + \Pi^\perp = \mathbb{I}$. Every $(n(\lambda) + \ell(\lambda))$ -qubit unitary U_λ with access to $\mathcal{O} = (O_f, O_g)$ can be written as

$$U_\lambda = V_{q(\lambda)} O_f V_{q(\lambda)-1} O_f \cdots V_2 O_f V_1$$

where

- V_i , for $i \in [q(\lambda)]$, contains gates from G (the universal gate set), and can potentially query the oracle O_g , and
- V_i is the unitary transformation right before the i -th query to oracle O_f . Further, one can write the unitary U (we suppress the security parameter for brevity)

$$\begin{aligned} U &= V_q O_f (\Pi + \Pi^\perp) V_{q-1} O_f (\Pi + \Pi^\perp) \cdots V_2 O_f (\Pi + \Pi^\perp) V_1 \\ &= \underbrace{V_q O_f V_{q-1} O_f \cdots O_f V_3 O_f V_2 O_f \Pi}_{A_1} \underbrace{V_1}_{Z_1} + \\ &\quad \underbrace{V_q O_f V_{q-1} O_f \cdots O_f V_3 O_f \Pi}_{A_2} \underbrace{V_2 O_f \Pi^\perp V_1}_{Z_2} + \\ &\quad \vdots \\ &\quad \underbrace{V_q O_f \Pi}_{A_{q-1}} \underbrace{V_{q-1} O_f \Pi^\perp \cdots O_f \Pi^\perp V_3 O_f \Pi^\perp V_2 O_f \Pi^\perp V_1}_{Z_{q-1}} + \\ &\quad \underbrace{\mathbb{I}}_{A_q} \underbrace{V_q O_f \Pi^\perp V_{q-1} O_f \Pi^\perp \cdots O_f \Pi^\perp V_3 O_f \Pi^\perp V_2 O_f \Pi^\perp V_1}_{Z_q}. \end{aligned}$$

Notation 75. One can break the unitary U into branches based on the step where f was queried when the state was parallel to Π .

$$U = \sum_{i=1}^{q-1} A_i \Pi Z_i + A_q \overset{\Pi}{Z}_q$$

We are now ready to state one of our main technical results, namely, that even with $c\lambda$ catalysts, for $c < 1$, the problem $\mathcal{P}$ cannot be solved (on most instances). This provides a sharp separation.

Lemma 76 ($\mathcal{P} \notin (2\lambda, c\lambda)\text{-}\widehat{\text{Fcat}}$ for $c < 1$). *Let $n, \ell, \ell' : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomials $n(\lambda) = 2\lambda$, $\ell(\lambda) = \lfloor c \cdot \lambda \rfloor$ for $c < 1$. Let $\mathcal{P}$ be the relation as above (see Problem 72). Given oracle access to length preserving random oracles $\mathbf{O} = (O_f, O_g)$, it holds that $\mathcal{P} \notin (n, \ell)\text{-}\widehat{\text{Fcat}}^{\mathbf{O}}$.*

Proof. We give the proof by showing that any circuit mapping $|x, f(g(x)), 0^\ell\rangle \mapsto |x, 0^\lambda\rangle$ has negligible success probability when $\ell(\lambda) = c\lambda$.

Suppose, for contradiction, that $\mathcal{P} \in (2\lambda, \ell)\text{-}\widehat{\text{Fcat}}^{\mathbf{O}}$. Since the relation requires the entire 2λ -qubit system register to contain $(x, f(g(x)))$, and the catalytic register must be restored exactly, the inverse of the corresponding unitary circuit satisfies

$$U |x, f(g(x)), 0^\ell\rangle = |x, 0^\lambda, 0^\ell\rangle \quad (43)$$

for every x , up to a phase that does not affect any success probability. Reversing the circuit replaces each elementary gate by its adjoint and preserves the oracle-query counts, since $O_f^\dagger = O_f$ and $O_g^\dagger = O_g$.

Fix the security parameter λ , write $\ell = \ell(\lambda)$, and suppress the security parameter in the circuit notation. Let $q - 1$ denote the number of queries to O_f , and let q_g denote the number of queries to O_g . The non-oracle gates of U are fixed independently of f, g, x . We first fix any oracle-independent randomness in the choice of the circuit; the bounds below also hold after averaging over this randomness.

Recall that $f, g : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ are independent uniformly random functions, and $x \in \{0, 1\}^\lambda$ is independently uniform. Define

$$\text{succ}_{\text{search}} := \mathbb{E}_{f, g, x} \left\| \Pi^{\text{search}} U |x, f(g(x)), 0^\ell\rangle \right\|^2, \quad (44)$$

where $\Pi^{\text{search}} := |x\rangle \langle x|_Q \otimes |0^\lambda\rangle \langle 0^\lambda|_R \otimes \mathbb{I}_C$. The registers are Q (λ -qubit query register), R (λ -qubit response register), and C (ℓ -qubit catalytic register). In particular, Π^{search} is independent of $f(g(x))$ when x is fixed.

The circuit in Equation (43) would have $\text{succ}_{\text{search}} = 1$. We will bound $\text{succ}_{\text{search}}$ for an arbitrary circuit U , without requiring that its catalytic register be restored.

If $q - 1 = 0$, the circuit is independent of f . Conditional on g, x , averaging over the uniform value $f(g(x))$ gives

$$\begin{aligned} \mathbb{E}_f \left\| \Pi^{\text{search}} U |x, f(g(x)), 0^\ell\rangle \right\|^2 &= \frac{1}{2^\lambda} \text{tr} \left[\Pi^{\text{search}} U \left(|x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes |0^\ell\rangle \langle 0^\ell|_C \right) U^\dagger \right] \\ &\leq \frac{\text{rank}(\Pi^{\text{search}})}{2^\lambda} = \frac{2^\ell}{2^\lambda}. \end{aligned} \quad (45)$$

This is negligible for the stated choice of ℓ . Hence assume $q \geq 2$.

one may condition on g, x when defining the branches below, but retain the averaging over them in all success probabilities. Define $\Pi := |g(x)\rangle\langle g(x)|_Q \otimes \mathbb{I}_{RC}$ and $\Pi^\perp := \mathbb{I} - \Pi$. Recalling Notation 75, write $U = V_q O_f V_{q-1} O_f \cdots V_2 O_f V_1$ and

$$\Pi^{\text{search}} U \left| x, f(g(x)), 0^\ell \right\rangle = \Pi^{\text{search}} \left(\sum_{i=1}^{q-1} A_i \Pi Z_i + Z_q \right) \left| x, f(g(x)), 0^\ell \right\rangle, \quad (46)$$

where

- $Z_1 := V_1$ and $Z_i := V_i O_f \Pi^\perp Z_{i-1}$ for $2 \leq i \leq q$;
- $A_{q-1} := V_q O_f$ and $A_i := A_{i+1} V_{i+1} O_f$ for $1 \leq i < q-1$.

Each A_i is unitary, and each Z_i is a contraction. Conditional on g, x and the restriction of f to all points other than $g(x)$, every Z_i is independent of $f(g(x))$, since every f -query within Z_i is preceded by $\Pi^\perp$.

With this notation in place, we proceed in two steps: (1) upper bound $\text{succ}_{\text{search}}$ in terms of the weights of the branches $\Pi Z_i \left| x, f(g(x)), 0^\ell \right\rangle$; and (2) construct an inverter whose success probability is bounded below by these weights.

Step (1).

The upper bound is computed as follows:

$$\begin{aligned} \left\| \Pi^{\text{search}} U \left| x, f(g(x)), 0^\ell \right\rangle \right\|^2 &= \left\| \Pi^{\text{search}} \left(\sum_{i=1}^{q-1} A_i \Pi Z_i + Z_q \right) \left| x, f(g(x)), 0^\ell \right\rangle \right\|^2 \\ &\leq q \left(\sum_{i=1}^{q-1} \left\| \Pi^{\text{search}} A_i \Pi Z_i \left| x, f(g(x)), 0^\ell \right\rangle \right\|^2 + \left\| \Pi^{\text{search}} Z_q \left| x, f(g(x)), 0^\ell \right\rangle \right\|^2 \right) \\ &\leq q \left(\sum_{i=1}^{q-1} \left\| \Pi Z_i \left| x, f(g(x)), 0^\ell \right\rangle \right\|^2 + \text{tr} \left[M \left| x, f(g(x)), 0^\ell \right\rangle \left\langle x, f(g(x)), 0^\ell \right| \right] \right), \end{aligned} \quad (47)$$

where $M := Z_q^\dagger \Pi^{\text{search}} Z_q$. The first inequality follows from Cauchy-Schwarz, while the second uses the unitarity of A_i and the fact that Π^{search} is a contraction.

From the definitions of Π^{search} and Z_q , one obtains $0 \preceq M \preceq \mathbb{I}$ and $\text{rank}(M) \leq 2^\ell$. Thus one may write $M = \sum_{j=1}^{2^\ell} c_j |m_j\rangle\langle m_j|$, where $0 \leq c_j \leq 1$ and the $|m_j\rangle$ are orthonormal, padding with zero eigenvalues if necessary. For fixed g, x and f away from $g(x)$, both c_j and $|m_j\rangle$ can be chosen independently of $f(g(x))$.

Let $\bar{f}$ denote the restriction of f to $\{0, 1\}^\lambda \setminus \{g(x)\}$. Since the vectors $\{|x, a, 0^\ell\rangle : a \in \{0, 1\}^\lambda\}$ are orthonormal, for each normalized $|m_j\rangle$ one obtains $\sum_a |\langle m_j | x, a, 0^\ell \rangle|^2 \leq 1$. Consequently,

$$\begin{aligned} \mathbb{E}_{f, g, x} \text{tr} \left[M \left| x, f(g(x)), 0^\ell \right\rangle \left\langle x, f(g(x)), 0^\ell \right| \right] &= \mathbb{E}_{g, x, \bar{f}} \left[\frac{1}{2^\lambda} \sum_{j=1}^{2^\ell} c_j \sum_{a \in \{0, 1\}^\lambda} |\langle m_j | x, a, 0^\ell \rangle|^2 \right] \\ &\leq \frac{1}{2^\lambda} \mathbb{E}_{g, x, \bar{f}} \sum_{j=1}^{2^\ell} c_j \\ &\leq \frac{2^\ell}{2^\lambda}. \end{aligned} \quad (48)$$

Averaging Equation (47) and using Equation (48), we obtain

$$\begin{aligned}
\frac{\text{succ}_{\text{search}}}{q} - \frac{2^\ell}{2^\lambda} &\leq \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \left\| \Pi Z_i |x, f(g(x)), 0^\ell \rangle \right\|^2 \\
&= \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \left\| \langle x, f(g(x)), 0^\ell | Z_i^\dagger \Pi \right\|^2 \\
&= \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \text{tr} \left[\left(|x\rangle \langle x|_Q \otimes |f(g(x))\rangle \langle f(g(x))|_R \otimes |0^\ell\rangle \langle 0^\ell|_C \right) Z_i^\dagger \Pi Z_i \right] \\
&= \mathbb{E}_{g,x,\bar{f}} \sum_{i=1}^{q-1} \frac{1}{2^\lambda} \text{tr} \left[\left(|x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes |0^\ell\rangle \langle 0^\ell|_C \right) Z_i^\dagger \Pi Z_i \right] \\
&= \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \frac{1}{2^\lambda} \text{tr} \left[\left(|x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes |0^\ell\rangle \langle 0^\ell|_C \right) Z_i^\dagger \Pi Z_i \right].
\end{aligned} \tag{49}$$

In the second-last equality, we averaged over $f(g(x))$, using its conditional uniformity and the independence of Z_i from that value. In the last equality, we restored the expectation over all of f , since the expression no longer depends on $f(g(x))$.

Step (2).

The inversion algorithm $\mathcal{I}$ receives $z = g(x)$ and has oracle access to O_g . It knows the circuit U , but is not given x or access to the original oracle O_f . It proceeds as follows:

1. Sample i uniformly from $\{1, \dots, q-1\}$ and sample an independent uniformly random function $f : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$.
2. Create the state $|z\rangle \langle z|_Q \otimes \mathbb{I}_{RC}/2^{\lambda+\ell}$ by preparing Q in $|z\rangle$ and choosing independent uniform computational-basis states for R and C .
3. Apply $V_i^\dagger$. For $j = i-1, i-2, \dots, 1$, apply $O_f^\dagger = O_f$, perform the projective test $\{\Pi^\perp, \Pi\}$, and, if the outcome is $\Pi^\perp$, apply $V_j^\dagger$. If any test returns Π , return $\perp$ and stop. Here $\Pi = |z\rangle \langle z|_Q \otimes \mathbb{I}_{RC}$.
4. Measure Q in the computational basis and return the resulting string.

The branch on which the algorithm does not abort has Kraus operator

$$V_1^\dagger \Pi^\perp O_f^\dagger V_2^\dagger \dots \Pi^\perp O_f^\dagger V_i^\dagger = Z_i^\dagger.$$

Thus the algorithm implements the trace-nonincreasing operation $\tau \mapsto Z_i^\dagger \tau Z_i$ on that branch. All aborts count as failures. Note that this is possible because $\mathcal{I}$ knows z . Each $V_j^\dagger$ is implemented by reversing the corresponding circuit. Since $O_g^\dagger = O_g$, the total number of queries to g is at most q_g . The independently sampled f is simulated internally.

Define $\text{succ}_{\text{inv}} := \Pr_{g,x,\mathcal{I}}[\mathcal{I}^g(g(x)) = x]$. Writing $\Pi'_\ell := \mathbb{I}_C - |0^\ell\rangle \langle 0^\ell|_C$, the success probability can

be computed as

$$\begin{aligned}
\text{succ}_{\text{inv}} &= \frac{1}{(q-1)2^{\lambda+\ell}} \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \text{tr} \left[\left(|x\rangle \langle x|_Q \otimes \mathbb{I}_{RC} \right) Z_i^\dagger \Pi Z_i \right] \\
&= \frac{1}{(q-1)2^{\lambda+\ell}} \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \text{tr} \left[\left(|x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes \left(|0^\ell\rangle \langle 0^\ell|_C + \Pi'_\ell \right) \right) Z_i^\dagger \Pi Z_i \right] \\
&\geq \frac{1}{(q-1)2^{\lambda+\ell}} \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \text{tr} \left[\left(|x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes |0^\ell\rangle \langle 0^\ell|_C \right) Z_i^\dagger \Pi Z_i \right] \\
&= \frac{1}{(q-1)2^\ell} \mathbb{E}_{f,g,x} \sum_{i=1}^{q-1} \left\| \Pi Z_i |x, f(g(x)), 0^\ell\rangle \right\|^2.
\end{aligned} \tag{51}$$

The first equality follows from the unnormalized state $Z_i^\dagger \Pi Z_i / 2^{\lambda+\ell}$ on the branch that does not abort. The inequality drops a nonnegative term, since $|x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes \Pi'_\ell$ and $Z_i^\dagger \Pi Z_i$ are positive semidefinite. The final equality is the same averaging identity used in Equation (50).

Combining Equations (49) to (51), we obtain

$$\begin{aligned}
2^\ell(q-1)\text{succ}_{\text{inv}} &\geq \frac{\text{succ}_{\text{search}}}{q} - \frac{2^\ell}{2^\lambda}, \\
\text{succ}_{\text{inv}} &\geq \frac{\text{succ}_{\text{search}}}{q(q-1)2^\ell} - \frac{1}{(q-1)2^\lambda}.
\end{aligned} \tag{52}$$

On the other hand, the random-function inversion bound obtained gives

$$\text{succ}_{\text{inv}} \leq \frac{(2q_g + 1)^2}{2^\lambda}. \tag{53}$$

Therefore,

$$\begin{aligned}
\text{succ}_{\text{search}} &\leq q 2^\ell(q-1)\text{succ}_{\text{inv}} + q \frac{2^\ell}{2^\lambda} \\
&\leq q \left(1 + (q-1)(2q_g + 1)^2 \right) 2^{\ell-\lambda}.
\end{aligned} \tag{54}$$

This bound also holds when $q = 1$, by Equation (45). For polynomially bounded q, q_g , it is negligible whenever $\lambda - \ell(\lambda) = \omega(\log \lambda)$. In particular, for $\ell(\lambda) = \lfloor c\lambda \rfloor$ with fixed $c < 1$, it is bounded by a polynomial in λ times $2^{-(1-c)\lambda}$, which is negligible. $\square$

Remark 77 (Queries made by $\mathcal{A}$ don't matter). The queries to $\mathbf{O}$ made PPT algorithm $\mathcal{A}^{\mathbf{O}}$ only affects the success probability negligibly. The algorithm $\mathcal{A}^{\mathbf{O}}$ can only query polynomially many points of the random oracle $\mathbf{O}$, while we need the output of the unitary to satisfy the relation on all inputs.

It immediately follows that one can solve a problem with λ many catalysts but not with 0 catalysts. We separate out the 0 catalyst case because of its implications in separating computational catalytic ergotropy and computational ergotropy.

Lemma 78 ($\mathcal{P} \notin (2\lambda, 0)\text{-Fcat}$). *Let $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomials $n(\lambda) = 2\lambda$ and $\ell(\lambda) = 0$. Let $\mathcal{P}$ be the relation as above (see Problem 72). Given oracle access to length preserving random oracles $\mathbf{O} = (O_f, O_g)$, it holds that $\mathcal{P} \notin (n, \ell)\text{-Fcat}^{\mathbf{O}}$.*

Proof. Set $\ell(\lambda) = 0$ in Lemma 76. For q, q_g polynomially bounded, we have that $\text{succ}_{\text{search}}$ is negligible. $\square$

Combining the results so far, we obtain our final result here.

Theorem 79 $((2\lambda, c\lambda)\text{-}\widehat{\text{Fcat}} \subsetneq (2\lambda, \lambda)\text{-}\widehat{\text{Fcat}})$. Let $n, \ell, \ell' : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomials $n(\lambda) = 2\lambda$, $\ell'(\lambda) = \lambda$ and $\ell(\lambda) = c \cdot \lambda$ for $c < 1$. In the random oracle model, it holds that

$$(n, \ell)\text{-}\widehat{\text{Fcat}}^O \subsetneq (n, \ell')\text{-}\widehat{\text{Fcat}}^O.$$

From Lemma 74 and Lemma 76, we know that $\mathcal{P} \in (n, \ell')\text{-}\widehat{\text{Fcat}}^O$ and $\mathcal{P} \notin (n, \ell)\text{-}\widehat{\text{Fcat}}^O$. As a direct consequence we have that $(n, \ell)\text{-}\widehat{\text{Fcat}}^O \subsetneq (n, \ell')\text{-}\widehat{\text{Fcat}}^O$.

We conclude with the following remark.

Remark 80. We emphasise that the proof of Lemma 76 shows something stronger—it establishes that even with $2\lambda + c\lambda$ -many qubits, the problem $\mathcal{P}$ remains intractable. Thus, even allowing for arbitrary errors in the definition of computational catalytic computation, preserves the separation.

5.4 Decision problem separation $(2\lambda, c\lambda)\text{-}\widehat{\text{cat}} \subsetneq (2\lambda, \lambda)\text{-}\widehat{\text{cat}}$ (for $c < 1$)

Separations in relational problems are often easier to establish than decision problems. In fact, we have reason to believe that in many cases, decision separations relative to a random oracle are impossible—such as the well known Aaronson Ambainis conjecture. Furthermore, decision problems turn out to be much more relevant for cryptographic applications. For instance, indistinguishability of encryptions may be viewed as a decision problem.

Thus, here we show how to establish a separation between computational catalytic computations using a decision problem. We start by defining the decision variant $\widehat{\text{cat}}$ of the relational problem class $\widehat{\text{Fcat}}$. Here, we directly introduce the oracle version.

Denote a (promise) decision problem $\mathcal{P}$ by the sets (YES, NO) where $\text{YES} \subseteq \{0, 1\}^*$ and $\text{NO} \subseteq \{0, 1\}^*$. When $\mathcal{P}$ depends on the oracle O , we denote it by $\mathcal{P}^O$. Fix $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$.

Definition 81 $((n, \ell)\text{-}\widehat{\text{cat}})$. We say a problem $\mathcal{P}^O$ is in $(n, \ell)\text{-}\widehat{\text{cat}}$ if there exists an algorithm $\mathcal{A}^O \in (n, \ell)\text{-}\widehat{\text{catGens}}^O$ such that

- Completeness: For all $x \in \text{YES}^O$ such that $|x| = n(\lambda)$,

$$\mathbb{E}_{\text{desc}(\Pi, U) \leftarrow \mathcal{A}^O(1^\lambda)} \left\| \Pi U \left| x, 0^{n(\lambda)-|x|}, 0^{\ell(\lambda)} \right\rangle \right\|^2 \geq 2/3$$

- Soundness: For all $y \in \text{NO}^O$ such that

$$\mathbb{E}_{\text{desc}(\Pi, U) \leftarrow \mathcal{A}^O(1^\lambda)} \left\| \Pi U \left| x, 0^{n(\lambda)-|x|}, 0^{\ell(\lambda)} \right\rangle \right\|^2 \leq 1/3$$

The decision problem we consider, is closely related to the relational problem we considered.

Problem 82. Let $O = (O_g, O_f)$ be the oracles corresponding to length preserving random functions f, g . The decision problem $\mathcal{P}^O$ is defined as follows:

$$\text{YES}^O := \{(x, f(g(x))) : x \in \{0, 1\}^*\}$$

and

$$\text{NO}^O := \{0, 1\}^* \setminus \text{YES}^O.$$

While part of the argument we used to establish Lemma 76 goes through, it fails when one tries to bound the contribution of the term $A_q \Pi Z_q |x, f(g(x))\rangle$. This has to do with the fact that for search, the projector Π^{search} is rank-1 while for a decision problem, we have no such guarantee on the corresponding projector Π^{dec} . We now prove the decision analogue of Lemma 76, where the catalyst register has $\ell(\lambda) = c \cdot \lambda$ qubits.

Lemma 83 ($(x, f(g(x))) \approx (x, r)$ with ℓ catalysts). *Let $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomials $n(\lambda) = 2\lambda$, $\ell(\lambda) = c \cdot \lambda$ for $c < 1$. Given oracle access to $\mathcal{O} = (O_g, O_f)$, it holds that for every $\mathcal{A} \in (n, \ell)\text{-cat}$ there is a negligible function negl such that*

$$\left| \mathbb{E}_{\substack{\text{desc}(\Pi_\lambda^{\text{dec}}, U_\lambda) \leftarrow \mathcal{A}^{\mathcal{O}} \\ \mathcal{O}}} \left\| \Pi_\lambda^{\text{dec}} (U_\lambda |x, f(g(x))) \right\| - \mathbb{E}_{\substack{\text{desc}(\Pi_\lambda^{\text{dec}}, U_\lambda) \leftarrow \mathcal{A}^{\mathcal{O}} \\ \mathcal{O}}} \left\| \Pi_\lambda^{\text{dec}} (U_\lambda |x, r) \right\| \right| \leq \text{negl}(\lambda).$$

Proof. Similar to Lemma 76, we show that if one can distinguish $(x, f(g(x)))$ from (x, r) with probability succ_{dec} then there is an inversion algorithm that inverts $g(x)$ with success probability at least

$$\text{succ}_{\text{inv}} \geq \frac{\text{succ}_{\text{dec}}^2}{16(q-1)^2 \times 2^\ell},$$

where $q-1 \geq 1$ is the number of queries to O_f .

Fix the security parameter λ , write $\ell = \ell(\lambda)$, and suppress the security parameter in the circuit notation. Let q_g denote the number of queries to O_g . The circuit U acts on registers Q, R, C containing λ, λ, ℓ qubits, respectively, with C initially in $|0^\ell\rangle$. We implicitly extend Π^{dec} by the identity on C .

The non-oracle gates of U and the projector Π^{dec} are fixed independently of f, g, x, r . Recall that the functions $f, g : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ are independent and uniformly random, and $x, r \in \{0, 1\}^\lambda$ are independent uniform strings, independent of f, g . All expectations below include f, g, x, r , as applicable.

If $q-1 = 0$, then conditional on g, x , the circuit is independent of the uniform value $f(g(x))$. The two acceptance probabilities are therefore equal. Hence assume $q \geq 2$.

Define

$$\text{succ}_{\text{dec}} := \left| \mathbb{E}_{f, g, x} \left\| \Pi^{\text{dec}} U |x, f(g(x)), 0^\ell\rangle \right\|^2 - \mathbb{E}_{f, g, x, r} \left\| \Pi^{\text{dec}} U |x, r, 0^\ell\rangle \right\|^2 \right|. \quad (55)$$

We may condition on g, x when defining the branches below, but retain the averaging over them in all success probabilities.

Recalling the notation from Notation 75, write $U = V_q O_f V_{q-1} O_f \cdots V_2 O_f V_1$ and

$$U = \sum_{i=1}^{q-1} A_i \Pi Z_i + Z_q, \quad (56)$$

where

- $\Pi := |g(x)\rangle \langle g(x)|_Q \otimes \mathbb{I}_{RC}$ and $\Pi^\perp := \mathbb{I} - \Pi$;
- $Z_1 := V_1$ and $Z_i := V_i O_f \Pi^\perp Z_{i-1}$ for $2 \leq i \leq q$;
- $A_{q-1} := V_q O_f$ and $A_i := A_{i+1} V_{i+1} O_f$ for $1 \leq i < q-1$.

Each A_i is unitary, and each Z_i is a contraction.

We proceed in two steps.

1. We prove that

$$\text{succ}_{\text{dec}} \leq 4 \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i \middle| x, f(g(x)), 0^\ell \right\|^2. \quad (57)$$

2. One can construct an inversion algorithm $\mathcal{I}$ whose success probability $\text{succ}_{\text{inv}} := \Pr_{g,x,\mathcal{I}}[\mathcal{I}^g(g(x)) = x]$ satisfies

$$\begin{aligned} (q-1)^2 2^\ell \text{succ}_{\text{inv}} &\geq (q-1) \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i \middle| x, f(g(x)), 0^\ell \right\|^2 \\ &\geq \left(\sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i \middle| x, f(g(x)), 0^\ell \right\|^2 \right)^2. \end{aligned} \quad (58)$$

Step 2 uses exactly the same inversion algorithm as in Lemma 76. Given $z = g(x)$, the inverter samples i uniformly from $\{1, \dots, q-1\}$, samples an independent random function f , prepares $|z\rangle \langle z|_Q \otimes \mathbb{I}_{RC} / 2^{\lambda+\ell}$, and implements the branch $Z_i^\dagger$ before measuring Q .

To implement this branch, it runs the gates defining Z_i in reverse order and implements each occurrence of $\Pi^\perp$ by the projective test $\{\Pi^\perp, \Pi\}$, aborting on outcome Π . The branch on which it does not abort has Kraus operator $Z_i^\dagger$. These tests are available because the inverter knows z . The inverter makes at most q_g queries to g , since $O_g^\dagger = O_g$. Its independently sampled f can be simulated internally: the inversion bound restricts the queries to g , not the inverter's other computation or workspace.

For completeness, define

$$\begin{aligned} P_x &:= |x\rangle \langle x|_Q \otimes \mathbb{I}_{RC}, \\ P_{x,0} &:= |x\rangle \langle x|_Q \otimes \mathbb{I}_R \otimes |0^\ell\rangle \langle 0^\ell|_C. \end{aligned} \quad (59)$$

The success probability, including the possibility of abort, satisfies

$$\begin{aligned} \text{succ}_{\text{inv}} &= \frac{1}{(q-1)2^{\lambda+\ell}} \sum_{i=1}^{q-1} \mathbb{E}_{f,g,x} \text{tr} \left(P_x Z_i^\dagger \Pi Z_i \right) \\ &\geq \frac{1}{(q-1)2^{\lambda+\ell}} \sum_{i=1}^{q-1} \mathbb{E}_{f,g,x} \text{tr} \left(P_{x,0} Z_i^\dagger \Pi Z_i \right) \\ &= \frac{1}{(q-1)2^\ell} \sum_{i=1}^{q-1} \mathbb{E}_{f,g,x} \left\| \Pi Z_i \middle| x, f(g(x)), 0^\ell \right\|^2. \end{aligned} \quad (60)$$

The inequality follows from $P_x \succeq P_{x,0}$ and $Z_i^\dagger \Pi Z_i \succeq 0$. The final equality follows by conditioning on g, x and the restriction of f to all points other than $g(x)$. Under this conditioning, Z_i is independent of the uniform value $f(g(x))$, because each f -query within Z_i is preceded by $\Pi^\perp$. Averaging over that value gives the factor $2^{-\lambda}$. Cauchy-Schwarz, together with $(\mathbb{E} X)^2 \leq \mathbb{E} X^2$, then proves Equation (58).

It remains to establish Step 1. For any vectors u, v with $\|u\| \leq 1$ and $\|u+v\| \leq 1$, the reverse triangle inequality gives

$$\begin{aligned} \left| \|u+v\|^2 - \|u\|^2 \right| &= (\|u+v\| + \|u\|) \left| \|u+v\| - \|u\| \right| \\ &\leq 2\|v\|. \end{aligned} \quad (61)$$

Consequently, if random vectors u_1, v_1, u_2, v_2 satisfy $\|u_j\| \leq 1$ and $\|u_j + v_j\| \leq 1$ almost surely for $j \in \{1, 2\}$, and $\mathbb{E} \|u_1\|^2 = \mathbb{E} \|u_2\|^2$, then

$$\begin{aligned} |\mathbb{E} (\|u_1 + v_1\|^2 - \|u_2 + v_2\|^2)| &\leq \mathbb{E} |\|u_1 + v_1\|^2 - \|u_1\|^2| \\ &\quad + \mathbb{E} |\|u_2 + v_2\|^2 - \|u_2\|^2| \\ &\leq 2 (\mathbb{E} \|v_1\| + \mathbb{E} \|v_2\|). \end{aligned} \quad (62)$$

This estimate does not require $\|v_j\| \leq 1$ or negligibility of either expectation.

We apply Equation (62) with

$$\begin{aligned} |v_1\rangle &:= \Pi^{\text{dec}} \sum_{i=1}^{q-1} A_i \Pi Z_i |x, f(g(x)), 0^\ell\rangle, \\ |u_1\rangle &:= \Pi^{\text{dec}} Z_q |x, f(g(x)), 0^\ell\rangle, \\ |v_2\rangle &:= \Pi^{\text{dec}} \sum_{i=1}^{q-1} A_i \Pi Z_i |x, r, 0^\ell\rangle, \\ |u_2\rangle &:= \Pi^{\text{dec}} Z_q |x, r, 0^\ell\rangle. \end{aligned} \quad (63)$$

Since Z_q is a contraction, $\|u_1\|, \|u_2\| \leq 1$. Moreover, $u_j + v_j$ is the projection of a unit vector, so $\|u_j + v_j\| \leq 1$ for $j \in \{1, 2\}$.

Conditional on g, x and f away from $g(x)$, the operator Z_q is fixed, while both $f(g(x))$ and r are uniform strings. Since Π^{dec} is independent of the oracles and the sampled strings, we obtain

$$\mathbb{E} \|u_1\|^2 = \mathbb{E} \|u_2\|^2. \quad (64)$$

The triangle inequality, together with the unitarity of A_i and the fact that Π^{dec} is a contraction, gives

$$\begin{aligned} \mathbb{E} \|v_1\| &= \mathbb{E} \left\| \Pi^{\text{dec}} \sum_{i=1}^{q-1} A_i \Pi Z_i |x, f(g(x)), 0^\ell\rangle \right\| \\ &\leq \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi^{\text{dec}} A_i \Pi Z_i |x, f(g(x)), 0^\ell\rangle \right\| \\ &\leq \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i |x, f(g(x)), 0^\ell\rangle \right\|. \end{aligned} \quad (65)$$

Similarly,

$$\begin{aligned} \mathbb{E} \|v_2\| &\leq \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi^{\text{dec}} A_i \Pi Z_i |x, r, 0^\ell\rangle \right\| \\ &\leq \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i |x, r, 0^\ell\rangle \right\| \\ &= \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i |x, f(g(x)), 0^\ell\rangle \right\|. \end{aligned} \quad (66)$$

Now using Equation (62), we conclude that

$$\begin{aligned}
\text{succ}_{\text{dec}} &= |\mathbb{E}(\|u_1 + v_1\|^2 - \|u_2 + v_2\|^2)| \\
&\leq 2(\mathbb{E}\|v_1\| + \mathbb{E}\|v_2\|) \\
&\leq 4 \sum_{i=1}^{q-1} \mathbb{E} \left\| \Pi Z_i \left| x, f(g(x)), 0^\ell \right\rangle \right\|.
\end{aligned} \tag{67}$$

This proves Step 1.

Combining Steps 1 and 2 gives

$$\text{succ}_{\text{inv}} \geq \frac{\text{succ}_{\text{dec}}^2}{16(q-1)^2 \times 2^\ell}. \tag{68}$$

On the other hand, the random-function inversion bound gives

$$\text{succ}_{\text{inv}} \leq \frac{(2q_g + 1)^2}{2^\lambda}. \tag{69}$$

Therefore,

$$\text{succ}_{\text{dec}} \leq 4(q-1)(2q_g + 1) 2^{-(\lambda-\ell)/2}. \tag{70}$$

This is negligible for polynomially bounded q, q_g whenever $\lambda - \ell(\lambda) = \omega(\log \lambda)$. In particular, this holds for $\ell(\lambda) = \lfloor c\lambda \rfloor$ with any fixed $0 \leq c < 1$. Note that this does not require the catalytic register to be restored. $\square$

As before, combining the lemma and using essentially the same algorithm that solved the relational problem, we obtain our main result for a decision problem.

Theorem 84 $((2\lambda, c\lambda)\text{-}\widehat{\text{cat}}^O \subsetneq (2\lambda, \lambda)\text{-}\widehat{\text{cat}}^O \text{ for } c < 1)$. *Let $n, \ell, \ell' : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomials $n(\lambda) = 2\lambda$, $\ell'(\lambda) = \lambda$ and $\ell(\lambda) = c \cdot \lambda$ for $c < 1$. In the random oracle model, it holds that*

$$(n, \ell)\text{-}\widehat{\text{cat}}^O \subsetneq (n, \ell')\text{-}\widehat{\text{cat}}^O.$$

Proof. Since from Lemma 83 it holds that (x, r) and $(x, f(g(x)))$ are indistinguishable with less than $\ell(\lambda) = \lambda$ catalysts, for the decision problem $\mathcal{P}$ (see Problem 82) it follows that: $\mathcal{P} \notin (2\lambda, c\lambda)\text{-}\widehat{\text{cat}}^O$ while $\mathcal{P} \in (2\lambda, \lambda)\text{-}\widehat{\text{cat}}^O$. $\square$

6 Catalytic Work Extraction

Since the definition of computational work extraction in Section 4.1 appears to be too restrictive (see Problem 57), one can relax the definition to allow for catalytic computations in the work extraction process. In this section we allow the use of catalysts for work extraction and make statements about catalytic ergotropy and its computational variant, separations between computational ergotropy and computational catalytic ergotropy and catalytic work potential of Hamiltonians.

We start by defining catalytic ergotropy as follows.

Definition 85 ($\text{cat-erg}_{\rho,H}(\ell)$, $\text{cat-erg}_{\rho,H}$). Given a quantum state (complete description) $\rho \in \text{Den}(n)$ and a Hamiltonian $H \in \text{Herm}(n)$, the catalytic ergotropy of the state ρ with respect to the Hamiltonian H using ℓ -catalysts is given by

$$\text{cat-erg}_{\rho,h}(\ell) := \max_{\substack{|\eta\rangle \in \mathcal{H}_{\text{catalyst}} \\ U \in \text{cat-}\mathcal{U}_{|\eta\rangle}(n,\ell)}} \text{tr} \left[H \otimes \mathbb{I}_{2^\ell} \left(\rho \otimes |\eta\rangle \langle \eta|^{\otimes \ell} - U \left(\rho \otimes |\eta\rangle \langle \eta|^{\otimes \ell} \right) U^\dagger \right) \right]$$

and with arbitrarily many catalysts is given by

$$\text{cat-erg}_{\rho,h} := \max_{\ell \geq 0} \text{cat-erg}_{\rho,h}(\ell).$$

We introduce the following notation to be consistent with Section 4.

Notation 86. Let $\boldsymbol{\rho} = \{\rho_\lambda\}_\lambda$ be a family of states and $\boldsymbol{H} = \{H_\lambda\}_\lambda$ be a family of Hamiltonians. We use these alternate notations to represent the catalytic ergotropy for families

- $\text{cat-erg}_{\boldsymbol{\rho},\boldsymbol{H}}(\lambda, \ell(\lambda)) := \text{cat-erg}_{\rho^\lambda, h^\lambda}(\ell(\lambda))$ and
- $\text{cat-erg}_{\boldsymbol{\rho},\boldsymbol{H}}(\lambda) := \text{cat-erg}_{\rho^\lambda, h^\lambda}$.

In order to define catalytic ergotropy, one has to extend the n -qubit Hamiltonian to define it on the catalytic space as well. It is crucial to ensure that the extended Hamiltonian does not trivially lead to higher work extraction.

As a consequence of Lemma 64, in the following theorem we show that the value of catalytic ergotropy coincides with the value of ergotropy without catalysts in the information theoretic setting.

Theorem 87 (Equivalence erg and cat-erg). *Let $n \in \mathbb{N}$ denote the size of the system. For state $\rho \in \text{Den}(n)$ and Hamiltonian $H \in \text{Herm}(n)$, the maximum extractable work remains unchanged even with the assistance of a catalytic system, i.e.*

$$\text{erg}_{\rho,H} = \text{cat-erg}_{\rho,H} \quad \forall \rho \in \text{Den}(n), \forall h \in \text{Ham}(n), \forall n.$$

Proof. Since for unitaries $U \in \text{cat-}\mathcal{U}_{|\eta\rangle}(n, \ell)$, it holds that $U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger = \sigma \otimes |\eta\rangle \langle \eta|$, one can write the resultant state as

$$U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger = \text{tr}_{\text{catalyst}} \left[U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right] \otimes \text{tr}_{\text{IO}} \left[U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right]$$

Using this in this in the definition of catalytic ergotropy, one can write

$$\begin{aligned} \text{cat-erg}_{\rho,h}(\ell) &= \max_{U \in \text{cat-}\mathcal{U}(n,\ell)} \text{tr} \left[H \otimes \mathbb{I}_{2^\ell} \left(\rho \otimes |\eta\rangle \langle \eta| - U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right) \right] \\ &= \max_{U \in \text{cat-}\mathcal{U}(n,\ell)} \text{tr} \left[(H\rho) \otimes (\mathbb{I}_{2^\ell} |\eta\rangle \langle \eta|) \right] \\ &\quad - \text{tr} \left[\left(H \cdot \text{tr}_{\text{catalyst}} \left[U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right] \right) \times \left(\mathbb{I}_{2^\ell} \cdot \text{tr}_{\text{IO}} \left[U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right] \right) \right] \\ &= \max_{U \in \text{cat-}\mathcal{U}(n,\ell)} \text{tr} \left[(H\rho) \times 1 \right] - \text{tr} \left[\left(H \cdot \text{tr}_{\text{catalyst}} \left[U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right] \right) \times 1 \right] \end{aligned}$$

Since for all $n, \ell \in \mathbb{N}$ we have $C_{\mathcal{U}(n)} = C_{\text{cat-}\mathcal{U}(n,\ell)}$ (from Claim 62 and theorem 65), it holds that for every $U \in \text{cat-}\mathcal{U}(n, \ell)$ there exists a unitary $V \in \mathcal{U}(n)$ such that,

$$\text{tr}[(H\rho)] - \text{tr} \left[\left(H \cdot \text{tr}_{\text{catalyst}} \left[U(\rho \otimes |\eta\rangle \langle \eta|)U^\dagger \right] \right) \right] = \text{tr}[(H\rho)] - \text{tr} \left[\left(H \cdot V\rho V^\dagger \right) \right].$$

As a consequence one it holds that $\text{erg}_{\rho,H} = \text{cat-erg}_{\rho,H}(\ell)$ for all ℓ , states and Hamiltonians. Therefore one obtains that $\text{erg}_{\rho,H} = \text{cat-erg}_{\rho,H}$. $\square$

6.1 Computational Catalytic Work Extraction

From the theorem above it holds that catalytic ergotropy, retains the meaning of the notion of ergotropy as defined in [Len78], while allowing one to use catalytic space.

Analogous to the definition of computational ergotropy, one can naturally extend the notion of catalytic ergotropy to the computationally bounded setting as follows.

Let $n, \ell : \mathbb{N} \rightarrow \mathbb{N}$ be functions denoting the size of the quantum system and the catalyst registers respectively. For $\lambda \in \mathbb{N}$, let $\rho = \{\rho_\lambda\}_\lambda$ and $H = \{H_\lambda\}_\lambda$ denote a family of states and Hamiltonians, such that $\rho_\lambda \in \text{Den}(n(\lambda))$, $H_\lambda \in \text{Herm}(n(\lambda))$.

Definition 88 (Work extracted by an algorithm). Let ρ, H be as above, and let $\mathcal{A}$ be a quantum oracle algorithm. We define the work extracted by the algorithm $\mathcal{A}$, using ℓ catalyst qubits, from the state ρ_λ with respect to the Hamiltonian H_λ as

$$\mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_\lambda \otimes \mathbb{I}_{2^{\ell(\lambda)}} \left(\rho_\lambda \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| - U_\lambda \left(\rho_\lambda \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| \right) U_\lambda^\dagger \right) \right] : \right. \\ \left. \begin{array}{l} \text{desc}(U_\lambda) \leftarrow \mathcal{A}^{O_{\rho, H-\text{desc}}(1^\lambda)}, \\ U_\lambda \in \text{cat-}\mathcal{U}_{|0^{\ell(\lambda)}\rangle}(n(\lambda), \ell(\lambda)) \end{array} \right]$$

where the expectation is over the internal randomness of the algorithms, and oracles.²

Similar to computational ergotropy, one can define computational catalytic ergotropy for distributions over states and Hamiltonians.

Let $\mathcal{D}$ denote a probability distribution over families of states $\{\rho_i\}_i$ and families of Hamiltonian $\{H_i\}_i$. We use $\rho, H \leftarrow \mathcal{D}$ to denote the result of sampling a pair of state and Hamiltonian families from $\mathcal{D}$.

Definition 89 (Computational Catalytic Ergotropy ($\text{cat-}\widehat{\text{erg}}_{\mathcal{D}, \ell}$)). Let $\text{ub}, \text{lb} : \mathbb{N} \rightarrow \mathbb{R}$ be functions, and $\mathcal{D}$ be a distribution over families of states and Hamiltonians (as described above). We say that the *computational catalytic ergotropy using ℓ catalyst qubits and for the distribution $\mathcal{D}$* , denoted by $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}, \ell}$, is (lb, ub) bounded if

$$\begin{aligned} \forall \mathcal{A} \in \text{PPT}, \text{ it holds that } \mathbb{E}_{\rho, H \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)} \right] &\leq_\infty \text{ub}, \text{ and} \\ \exists \mathcal{A} \in \text{PPT} \text{ s.t. } \mathbb{E}_{\rho, H \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)} \right] &\geq_\infty \text{lb}. \end{aligned}$$

We say that the *computational catalytic ergotropy for the distribution $\mathcal{D}$* , denoted by $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}}$, is (lb, ub) bounded if

$$\begin{aligned} \forall \mathcal{A} \in \mathcal{C}, \text{ and for all polynomials } \ell, \text{ it holds that } \mathbb{E}_{\rho, H \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)} \right] &\leq_\infty \text{ub}, \text{ and} \\ \text{there exists } \mathcal{A} \in \text{PPT}, \text{ and polynomial } \ell \text{ s.t. } \mathbb{E}_{\rho, H \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, H, \mathcal{A}}^{(\ell)} \right] &\geq_\infty \text{lb}. \end{aligned}$$

6.2 Separating $\widehat{\text{erg}}_{\mathcal{D}}$ and $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}}$ in the random oracle model

Clearly, since the catalytic model of computation allows for more operations in the computationally bounded setting, one can potentially extract more work using catalysts. We give the explicit distributions over families of states and Hamiltonians for which one attains maximal separation between ergotropy and catalytic ergotropy in the computationally bounded setting, relative to a random oracle.

Before we show the separation, we make the following remarks.

²We had to do this instead of directly using $(n, \ell)\text{-}\widehat{\text{cat}}$ because we wanted to explicitly give the algorithm access to O_ρ .

Remark 90 (Catalytic ergotropy relative to oracles). One can naturally extend the definitions above to the oracle setting.

1. The probability distribution $\mathcal{D}$ is over the family of states, Hamiltonians $\rho, \mathbf{H}$ and the oracle $\mathbf{O}$.
2. For a particular family of states, Hamiltonians and oracle, $(\rho, \mathbf{H}, \mathbf{O}) \leftarrow \mathcal{D}$, we define

$$\mathcal{W}_{\rho, \mathbf{H}, \mathbf{A}^{\mathbf{O}}}^{(\ell)}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_{\lambda} \otimes \mathbb{I}_{2^{\ell(\lambda)}} \left(\rho_{\lambda} \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| - U_{\lambda} \left(\rho_{\lambda} \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| \right) U_{\lambda}^{\dagger} \right) \right] \right],$$

$$\begin{aligned} &: \begin{aligned} &\text{desc}(U_{\lambda}) \leftarrow \mathcal{A}^{O_{\rho}, \mathbf{H}, \text{desc}, \mathbf{O}}(1^{\lambda}), \\ &U_{\lambda} \in \text{cat-}\mathcal{U}_{|0^{\ell(\lambda)}\rangle}(n(\lambda), \ell(\lambda)) \end{aligned} \end{aligned}$$

3. We say $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}}$, is (lb, ub) bounded if

$$\forall \mathcal{A} \in \text{PPT}, \text{ and for all polynomials } \ell, \text{ it holds that } \mathbb{E}_{\rho, \mathbf{H}, \mathbf{O} \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, \mathbf{H}, \mathbf{A}^{\mathbf{O}}}^{(\ell)} \right] \leq_{\infty} \text{ub}, \text{ and}$$

$$\text{there exists } \mathcal{A} \in \text{PPT}, \text{ and polynomial } \ell \text{ s.t. } \mathbb{E}_{\rho, \mathbf{H}, \mathbf{O} \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, \mathbf{H}, \mathbf{A}^{\mathbf{O}}}^{(\ell)} \right] \geq_{\infty} \text{lb}.$$

For the proof, we use a modified version of the Hamiltonian lemma (Lemma 50), that we state in the following remark.

Remark 91 ((n, ℓ) - $\widehat{\text{catGens}}$ indistinguishability).

- We say the family of states τ_1 and τ_2 are (n, ℓ) - $\widehat{\text{catGens}}$ indistinguishable if for any algorithm $\mathcal{A} \in (n, \ell)$ - $\widehat{\text{cat}}$, such that $\text{desc}(\Pi_{\lambda}, U_{\lambda}) \leftarrow \mathcal{A}(1^{\lambda})$, it holds that

$$\left| \Pr \left[\left[\Pi_{\lambda} U_{\lambda} \tau_{1, \lambda} U_{\lambda}^{\dagger} \right] = 1 \right] - \Pr \left[\left[\Pi_{\lambda} U_{\lambda} \tau_{2, \lambda} U_{\lambda}^{\dagger} \right] = 1 \right] \right| \leq \text{negl}$$

for some negligible function negl .

- One can naturally extend the Lemma 50 to (n, ℓ) - $\widehat{\text{catGens}}$ indistinguishable states as follows: For every family of extensive k -local extensive Hamiltonians, we have that

$$\left| \text{tr} \left[H_{\lambda} U_{\lambda} \tau_{1, \lambda} U_{\lambda}^{\dagger} \right] - \text{tr} \left[H_{\lambda} U_{\lambda} \tau_{2, \lambda} U_{\lambda}^{\dagger} \right] \right| \leq \text{negl}(\lambda)$$

where the unitary U_{λ} is produced by an algorithm $\mathcal{A} \in (n, \ell)$ - $\widehat{\text{cat}}$, i.e. $\text{desc}(\Pi_{\lambda}, U_{\lambda}) \leftarrow \mathcal{A}(1^{\lambda})$.

With the notations in place, we define the distribution $\mathcal{D}$ over $\rho, \mathbf{H}, \mathbf{O}$, for which we show the separation as follows.

Let $\mathbf{O} = (O_f, O_g)$ be the oracles corresponding to the length preserving functions f, g that are drawn uniformly at random from the set of all length preserving functions. Corresponding to each $\mathbf{O}$ we define the following families of states and Hamiltonians, $\rho = \{\rho_{\lambda}\}_{\lambda}$ and $\mathbf{H} = \{H_{\lambda}\}_{\lambda}$ as follows.

$$\rho_{\lambda} := \sum_{x \in \{0,1\}^{\lambda}} |x, f(g(x))\rangle \langle x, f(g(x))| \text{ and}$$

$$H_{\lambda} := \mathbb{I}_{2^{\lambda}} \otimes \text{hw}_{\lambda}$$

We denote the family of maximally mixed states $\sigma = \{\sigma_\lambda\}_\lambda$, where

$$\sigma_\lambda = \sum_{x,r \in \{0,1\}^\lambda} |x, r\rangle \langle x, r|.$$

The high level intuition the separation follows from Lemma 78. Since one can not distinguish between $|x, f(g(x))\rangle$ and $|x, r\rangle$ in the absence of catalysts, one can not map the state ρ_λ to a low energy state of the Hamiltonian. Whereas, with sufficient catalysts, one can map $|x, f(g(x))\rangle \mapsto |x, 0\rangle$, therefore minimising the energy.

Theorem 92 ($\widehat{\text{erg}}_{\mathcal{D}}$ is strictly lesser than $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}}$). *In the random oracle model, for the family of states, Hamiltonians and the oracle, ρ, H, O sampled from $\mathcal{D}$ as defined above (see Definition 46), it holds that*

$$\nexists \text{ non-negl } s.t. \widehat{\text{erg}}_{\mathcal{D}} \geq_{\infty} \text{ non-negl, while } \text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \geq_{\infty} \frac{n}{2}$$

where non-negl is any non-negligible function.

Proof. We prove this statement by breaking it into two smaller claims.

Claim. For any PPT algorithm $\mathcal{A}^{O_\rho, H\text{-desc}, O}$ such that, $\text{desc}(U_\lambda) \leftarrow \mathcal{A}^{O_\rho, H\text{-desc}, O}(1^\lambda)$, where $U_\lambda \in \mathcal{U}(n)$, it holds that $\mathbb{E}_{\mathcal{D}} [\mathcal{W}_{\rho, H, \mathcal{A}}(\lambda)] \leq \text{negl}$.

Proof. Notice that a PPT algorithm that has access to O can efficiently simulate $\mathcal{A}^{O_\rho, H\text{-desc}, O}(1^\lambda)$. This is because the given access to O_f, O_g the algorithm can efficiently prepare the (classical) state ρ_λ upon receiving input 1^λ . Therefore, any unitary U_λ whose circuit description is output by $\mathcal{A}^{O_\rho, H\text{-desc}, O}(1^\lambda)$ belongs to $\text{PPT-cat-}\mathcal{U}(2\lambda, 0)$.

From Lemma 83 we know that, for all $U \in \text{PPT-cat-}\mathcal{U}(2\lambda, 0)$ and for any projector Π^{dec} that does not depend on f, g , there exists a negligible function negl' such that

$$\mathbb{E}_{\mathcal{D}} \left[\text{tr} \left[\Pi^{\text{dec}} U_\lambda \rho_\lambda U_\lambda^\dagger \right] \right] \approx_{\text{negl}'} \mathbb{E}_{\mathcal{D}} \left[\text{tr} \left[\Pi^{\text{dec}} U_\lambda \sigma_\lambda U_\lambda^\dagger \right] \right].$$

Since the Hamiltonian H_λ is extensive, and can be decomposed into projectors that are independent of f, g , one can show (by arguments similar to Lemma 50) that

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} [\text{tr} [H_\lambda \rho_\lambda] - \text{tr} [H_\lambda U_\lambda \rho_\lambda U_\lambda^\dagger]] - \mathbb{E}_{\mathcal{D}} [\text{tr} [H_\lambda \sigma_\lambda] - \text{tr} [H_\lambda U_\lambda \sigma_\lambda U_\lambda^\dagger]] &= \text{negl}(\lambda) \\ \implies \mathbb{E}_{\mathcal{D}} [\mathcal{W}_{\rho, H, \mathcal{A}^O}(\lambda)] - \mathbb{E}_{\mathcal{D}} [\mathcal{W}_{\sigma, H, \mathcal{A}^O}(\lambda)] &\leq \text{negl}(\lambda) \quad \forall \mathcal{A} \in \text{PPT}. \end{aligned}$$

Since σ is the family of maximally mixed states, on average one can extract no work from the state σ_λ using unitary operations. Therefore, as a consequence, we have that $\widehat{\text{erg}}_{\mathcal{D}} \leq_{\infty} \text{negl}$. $\square$

Claim. $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \geq_{\infty} \frac{n}{2}$.

Proof. From Lemma 74 we know that there exists an algorithm $\mathcal{A}^O \in (n, \ell)\text{-}\widehat{\text{catGens}}^O$ that produces a unitary U_λ such that

$$U_\lambda \left(\rho_\lambda \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| \right) U_\lambda^\dagger = \underbrace{\sum_{x \in \{0,1\}^\lambda} |x, 0^{n(\lambda)}\rangle \langle x, 0^{n(\lambda)}| \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}|}_{\rho'_\lambda}$$

For such a unitary U_λ it holds that

$$\mathbb{E} [\text{tr} [H_\lambda \rho_\lambda] - \text{tr} [H_\lambda \rho'_\lambda]] = \frac{n(\lambda)}{2}.$$

Therefore, there exists an oracle PPT algorithm $\mathcal{A}^O$ such that $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}^O} \geq \infty \frac{n}{2}$. □

□

6.3 Catalytic work potential

From the above theorem, one can notice that there exists distributions over families of states and Hamiltonians for which one can extract more work through catalytic processes than without catalysts. We call such distributions as having *catalytic work potential* as formally define it as follows.

Definition 93 (Catalytic work potential). Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be a function denoting the size of the system. We say a distribution $\mathcal{D}$ over families of states ρ , Hamiltonians $\mathbf{H}$ (and oracles $\mathbf{O}$) have computational *catalytic-work potential* with gap Δ if either of the following hold:

- $\widehat{\text{erg}}_{\mathcal{D}} \leq \infty$ ub but $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \geq \infty$ lb, for some non-negligible function $\Delta := \text{lb} - \text{ub}$.
- $\widehat{\text{erg}}_{\mathcal{D}}$ is negligible (see Notation 44) but $\text{cat-}\widehat{\text{erg}}_{\mathcal{D}} \geq \infty \Delta$, for some non-negligible function Δ .

We say the *catalytic work potential* is *near-maximal* if Δ is at least the energy difference between the average energy and the ground energy, i.e. $\Delta \geq \infty \bar{E} - E^{\min}$ where

- $E = (E_\lambda)_\lambda$ with $\bar{E}_\lambda = \text{tr} [H_\lambda] / 2^{n(\lambda)}$ while
- $E^{\min} = (E_\lambda^{\min})$ where $E_\lambda^{\min}$ denotes the smallest eigenvalue of the Hamiltonian (H_λ)

One can extend this notion and say that a *family of Hamiltonians has catalytic work potential*, if there exists a distribution over family of states from which one can extract more work (with respect to the Hamiltonian) by using catalysts.

Theorem 94. Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be the polynomial $n(\lambda) = 2\lambda$. Let $\mathbf{H} = \{H_\lambda\}_\lambda$ be a family of efficient extensive Hamiltonians acting on $n(\lambda)$ qubits.

- Denote its average energy by $\bar{E}_\lambda := \text{tr} [H_\lambda] / 2^{n(\lambda)}$.
- There is polynomial $\ell : \mathbb{N} \rightarrow \mathbb{N}$ and an algorithm $\mathcal{A} \in (n, \ell)\text{-}\widehat{\text{cat}}$ such that

$$\sigma_\lambda := \mathbb{E}_{\text{desc}(\Pi_\lambda, U_\lambda) \leftarrow \mathcal{A}(1^\lambda)} U \left(\mathbb{I}_{2^\lambda} \otimes |0\rangle \langle 0|^\lambda \otimes |0\rangle \langle 0|^{\ell(\lambda)} \right) U^\dagger$$

and its corresponding energy is $E_\lambda = \text{tr}[H_\lambda \sigma_\lambda] = \text{non-negl}(\lambda)$.

Then $\{H_n\}_n$ has catalytic work-potential of $\Delta = E - \bar{E}$, relative to a random oracle.

Proof. (Proof sketch) Consider the following distribution over family of states ρ and oracles $\mathcal{O}$. Let $\mathcal{O} = (O_f, O_g)$ be the random oracle, and the family corresponding to it be $\rho = \{\rho_\lambda\}_\lambda$ where $\rho_\lambda = \sum_{x \in \{0,1\}^\lambda} |x, f(g(x))\rangle \langle x, f(g(x))|$. From Lemma 83, no computationally bounded algorithm can distinguish the state from maximally mixed state. Therefore it holds that the computational ergotropy for the distribution is negligible.

In the presence of catalysts, one can map the state $\rho_\lambda = \sum_{x \in \{0,1\}^{n(\lambda)}} |x, f(g(x))\rangle \langle x, f(g(x))|$ to $\rho'_\lambda = \sum_{x \in \{0,1\}^\lambda} |x, 0^{\ell(\lambda)}\rangle \langle x, 0^{\ell(\lambda)}| = \mathbb{I} \otimes |0^\lambda\rangle \langle 0^\lambda| \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}|$

Since one can map the state $\rho'_\lambda \mapsto \sigma_\lambda$ efficiently (by premise), it holds that the family of Hamiltonians $\mathbf{H} = H_{\lambda\lambda}$ has catalytic work potential of $\Delta = E - \bar{E}$. $\square$

As a consequence of the above theorem, we have the following.

Corollary 95. *The Hamming weight Hamiltonian has catalytic work-potential with near-maximal gap.*

7 Pseudoergotropy

Ergotropy is a resource, in some ways analogous to a battery. One can think of the battery as a quantum state ρ which follows dynamics corresponding to the Hamiltonian H and one can extract work from the battery by evolving through unitary evolutions. To ensure that only the intended user is able to use the battery and extract work from it, we define the notion of catalytic pseudoergotropy (used interchangeably with pseudoergotropy). Informally, we say that a distribution over families of state-Hamiltonian pair is has pseudoergotropy if the states are efficiently constructible and have low ergotropy, but are indistinguishable from a distribution over families state-Hamiltonian pairs with high ergotropy. This notion is analogous to pseudorandom numbers. To an adversary that does not know the secret key, the the high ergotropy state is indistinguishable from the low ergotropy state, therefore it can extract no work from the high ergotropic state. This is the case where the adversary has stolen the battery, but the adversary can still not extract any work out of it.

Symmetrically, one can also think about it this way, for computational purposes, the low ergotropy state can mimic the high ergotropy state and can be used as a resource. This is similar to how we think about Pseudorandomness, a state with low randomness (the pseudorandom state) which mimics a truly random state, to the entity that does not know the secret key.

In this section, we formalise the notion of pseudoergotropy and give explicit distributions that have catalytic pseudoergotropy. The construction we give in this section is also a counterexample to the Watanabe and Takagi's conjecture.

To begin with, we first set up some notations and definitions.

A keyed distribution $\mathcal{D}$ is a probability distribution over families of states, Hamiltonians and keys $(\rho, \mathbf{H}, \mathbf{k})$.

Let $n, \kappa : \mathbb{N} \rightarrow \mathbb{N}$ be polynomials. Consider the keyed distributions $\mathcal{D}$ and $\mathcal{D}'$ over $(\rho, \mathbf{H}, \mathbf{k})$ and $(\sigma, \mathbf{H}, \mathbf{k})$ respectively, where the families of states, Hamiltonians are defined on n -qubits and keys are of size κ .

Remark 96 (Catalytic ergotropy for keyed distributions). We extend cat-erg for keyed distributions as follows: For a particular family of states, Hamiltonians and oracle, $(\rho, \mathbf{H}, \mathbf{k}) \leftarrow \mathcal{D}$, we define the work

extracted by an algorithm $\mathcal{A}$ as

$$\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}^k}^{(\ell)}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_\lambda \otimes \mathbb{I}_{2^{\ell(\lambda)}} \left(\rho_\lambda \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| - U_\lambda \left(\rho_\lambda \otimes |0^{\ell(\lambda)}\rangle \langle 0^{\ell(\lambda)}| \right) U_\lambda^\dagger \right) \right] \right. \\ \left. : \begin{array}{l} \text{desc}(U_\lambda) \leftarrow \mathcal{A}^{O_\rho, \mathbf{H}\text{-desc}, \mathbf{k}}(1^\lambda), \\ U_\lambda \in \text{cat-}\mathcal{U}_{|0^{\ell(\lambda)}\rangle}(n(\lambda), \ell(\lambda)) \end{array} \right]$$

i.e. the algorithm knows the key $\mathbf{k}$ corresponding to the family of states ρ sampled from $\mathcal{D}$. The definition of $\text{cat-}\widehat{\text{erg}}$ is then defined using $\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}^k}^{(\ell)}$ above, by proceeding as in Remark 90.

With the notations set up, we formally define Catalytic pseudoergotropy (used interchangeably with pseudoergotropy) as follows.

Definition 97 (Catalytic pseudoergotropy). Let $c, d : \mathbb{N} \rightarrow \mathbb{N}$ be functions. We say the distribution $\mathcal{D}$ has catalytic pseudoergotropy (c, d) with respect to $\mathcal{D}'$ if the following conditions hold.

- Ergotropy conditions:

$$\begin{aligned} \text{cat-}\widehat{\text{erg}}_{\mathcal{D}} &\leq_\infty c \\ \text{cat-}\widehat{\text{erg}}_{\mathcal{D}'} &\geq_\infty d. \end{aligned}$$

- Indistinguishability condition. For all QPT algorithms $\mathcal{A}$, there is a negligible function negl such that

$$\left| \Pr_{(\rho, \mathbf{H}, \mathbf{k}) \leftarrow \mathcal{D}} \left[1 \leftarrow \mathcal{A}^{O_\rho}(1^\lambda) \right] - \Pr_{(\sigma, \mathbf{H}, \mathbf{k}) \leftarrow \mathcal{D}'} \left[1 \leftarrow \mathcal{A}^{O_\sigma}(1^\lambda) \right] \right| \leq \text{negl}(\lambda).$$

- Efficiency condition: There exists an efficient sampling procedure for the distribution $\mathcal{D}$.

7.1 Construction for Catalytic Pseudoergotropy

In this section, we explicitly describe the distributions $\mathcal{D}, \mathcal{D}'$ for which one can attain $(0, \text{non-negl})$ -catalytic pseudoergotropy. We start by setting up the following notations.

Consider a PRP family (F, Flnv) and let $n, m : \mathbb{N} \rightarrow \mathbb{N}$ be s.t. $m > n$. We consider $\mathbf{k}_f, \mathbf{k}_g, \mathbf{k}_h \in \{0, 1\}^m$, and define the following functions $f = \{f_\lambda\}_\lambda, g = \{g_\lambda\}_\lambda, h = \{h_\lambda\}_\lambda$ where

- $f_\lambda : \{0, 1\}^{m(\lambda)} \rightarrow \{0, 1\}^{m(\lambda)}$ as $f(x) = F(\mathbf{k}_f, x)$,
- $g_\lambda : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{m(\lambda)}$ as $g(x) = F(\mathbf{k}_g, x \| 0^{m-n})$, and
- $h_\lambda : \{0, 1\}^{m(\lambda)} \rightarrow \{0, 1\}^{n(\lambda)}$ as $h(x) = F(\mathbf{k}_h, x)_n$ ³
(i.e. the n -bit prefix of the outcome).

Definition 98. Let $n, m : \mathbb{N} \rightarrow \mathbb{N}$ be polynomials denoting the size of the quantum system and size of the key respectively. We define a keyed distribution $\mathcal{D}'$ over families of states, Hamiltonians and keys $(\rho, \mathbf{H}, \mathbf{k})$ as follows.

- The family of keys $\mathbf{k} = (\mathbf{k}_f, \mathbf{k}_g, \mathbf{k}_h)$ are sampled uniformly from $\{0, 1\}^m$

³One note in the above lemma is that, the adversary $\mathcal{A}$ is only given access to the forward direction oracles, $f(\cdot)$ or $g(h(\cdot))$ and not the inverses as well. This observation will be crucial for us in catalytic ergotropy.

- The family of states $\rho = \{\rho_\lambda\}_\lambda$ acting on $2n(\lambda)$ -qubits where

$$\rho_\lambda := \frac{1}{2^{m(\lambda)}} \sum_{x \in \{0,1\}^{m(\lambda)}} |g_\lambda(h_\lambda(x))\rangle \langle g_\lambda(h_\lambda(x))|.$$

- The family of Hamiltonians $\mathbf{H} = \{H_\lambda\}_\lambda$ (independent of $\mathbf{k}$) where $H_\lambda = \mathbb{I}_{2^{n(\lambda)}} \otimes H_{\text{hw}(m(\lambda)-n(\lambda))}$ is the padded Hamming weight Hamiltonian (see Equation (12)).

The keyed distribution $\mathcal{D}$ is defined exactly as above, except, the family of states $\sigma = \{\sigma_\lambda\}_\lambda$ is defined as

$$\sigma_\lambda := \frac{1}{2^{m(\lambda)}} \sum_{x \in \{0,1\}^{m(\lambda)}} |f_\lambda(x)\rangle \langle f_\lambda(x)| = \frac{\mathbb{I}_{2^{m(\lambda)}}}{2^{m(\lambda)}}.$$

With the distributions set up we now show the main result of the section.

Theorem 99 ($\mathcal{D}$ is $(0, \frac{n}{4})$ -catalytic pseudoergotropy w.r.t. $\mathcal{D}'$). *Let the function $m = 2n$. Assuming the existence of quantum secure PRPs, the keyed distribution $\mathcal{D}$ has $(0, \frac{n}{4})$ -catalytic pseudoergotropy w.r.t. the family $\mathcal{D}'$.*

Notice that the only difference between the distributions $\mathcal{D}$ and $\mathcal{D}'$ is in the families of states used. The family in distribution $\mathcal{D}$ is maximally mixed, whereas in $\mathcal{D}'$ it is less mixed. In the presence of sufficient catalysts, one can convert the less mixed state to the ground of the Hamiltonian, by inverting the keyed permutation.

We prove theorem 99 formally through the following lemmas.

Lemma 100. *For all polynomial functions $t : \mathbb{N} \rightarrow \mathbb{N}$, it holds that*

$$\left| \Pr_{(\rho, \mathbf{H}, \mathbf{k}) \leftarrow \mathcal{D}} \left[1 \leftarrow \mathcal{A}^{\rho}(1^\lambda) \right] - \Pr_{(\sigma, \mathbf{H}, \mathbf{k}) \leftarrow \mathcal{D}'} \left[1 \leftarrow \mathcal{A}^{\sigma}(1^\lambda) \right] \right| \leq \text{negl}(\lambda).$$

Proof. This immediately follows from Lemma 27. □

Lemma 101. *It holds that $\text{cat-erg}_{\mathcal{D}} \leq_\infty 0$ for every Hamiltonian*

Proof. Since the family of states in $\mathcal{D}$ is the family of maximally mixed states, it is invariant under unitary operations. Therefore, $\text{cat-erg}_{\mathcal{D}} \leq_\infty 0$. □

Lemma 102. *The distribution $\mathcal{D}$ is efficiently sampleable.*

Proof. Sample a family of keys $\mathbf{k} = (\mathbf{k}_f, \mathbf{k}_g, \mathbf{k}_h)$ uniformly at random. The family of states is just the family of maximally mixed states. □

Lemma 103. $\text{cat-erg}_{\mathcal{D}'} \geq_\infty \frac{(m-n) \cdot n}{2m}$.

Proof. We compute $\mathbb{E}_{(\rho, \mathbf{H}, \mathbf{k}) \leftarrow \mathcal{D}} [\text{tr} [H_\lambda \rho_\lambda]]$ as follows

- $g_\lambda(h_\lambda(x))$ has at least $m(\lambda) - n(\lambda)$ zeroes. This is because $g_\lambda(\cdot)$ appends $0^{m(\lambda)-n(\lambda)}$ before applying the permutation.

- In expectation over $x \in \{0, 1\}^{m(\lambda)}$ and the key k , the remaining qubits are indistinguishable from a uniformly distribution over 0 and 1. Therefore, there are $\frac{n(\lambda)}{2}$ qubits that are 1.
- Of these, only an $\frac{m(\lambda)-n(\lambda)}{m(\lambda)}$ fraction aligns with the Hamming weight part of the Hamiltonian H_λ . Therefore, we have

$$\mathbb{E}_{(\rho, H, k) \leftarrow \mathcal{D}} \text{tr} [H_\lambda \rho_\lambda] = \frac{m(\lambda) - n(\lambda)}{2} \times \frac{n(\lambda)}{m(\lambda)}.$$

Consider the oracle algorithm $\mathcal{A}$, such that $U_\lambda \leftarrow \mathcal{A}^{O_{\rho, H, \text{desc}, k}}(1^\lambda)$, i.e. on input 1^λ it outputs the description of a catalytic unitary, that does the following:

- The catalytic unitary U_λ applies the inverse permutation $\text{FInv}(k_{g, \lambda}, \cdot)$ on the state ρ_λ . The state thus obtained is

$$\begin{aligned} \rho'_\lambda &= \frac{1}{2^{m(\lambda)}} \sum_{x \in \{0, 1\}^{m(\lambda)}} |\text{FInv}(k_{g, \lambda}, g_\lambda(h_\lambda(x)))\rangle \langle \text{FInv}(k_{g, \lambda}, g_\lambda(h_\lambda(x)))|_{\text{IO}} \otimes |0^{m(\lambda)}\rangle \langle 0^{m(\lambda)}|_{\text{catalyst}} \\ &= \frac{1}{2^{m(\lambda)}} \sum_{x \in \{0, 1\}^{m(\lambda)}} |h_\lambda(x) \parallel 0^{m(\lambda)-n(\lambda)}\rangle \langle h_\lambda(x) \parallel 0^{m(\lambda)-n(\lambda)}|_{\text{IO}} \otimes |0^{\otimes m(\lambda)}\rangle \langle 0^{\otimes m(\lambda)}|_{\text{catalyst}}. \end{aligned}$$

- Therefore, it follows that

$$\mathbb{E}_{(\rho, H, k) \leftarrow \mathcal{D}} \text{tr} [H_\lambda \rho'_\lambda] = 0.$$

One can compute the catalytic work extracted by the algorithm $\mathcal{A}$ as,

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} [\mathcal{W}_{\rho, H, \mathcal{A}^k}^{(\ell)}(\lambda)] &= \mathbb{E}_{\mathcal{D}} [\text{tr} [H_\lambda \rho_\lambda] - \text{tr} [H_\lambda \rho'_\lambda]] \\ &= \frac{m(\lambda) - n(\lambda)}{2} \times \frac{n(\lambda)}{m(\lambda)}. \end{aligned}$$

Setting $m = 2n$ we get that $\text{cat-erg}_{\mathcal{D}} \geq_\infty (0, \frac{n}{4})$. □

Remark 104. The same state ensembles also exhibit a separation in nonequilibrium free energy. Fix an inverse temperature $\beta > 0$ independent of λ , and equip the system with the Hamiltonian $\tilde{H}_\lambda = 0$. Its Gibbs state is $\tau_\lambda = \sigma_\lambda$. Define $F_{\beta, \lambda}(\omega) = \text{tr}[\tilde{H}_\lambda \omega] - \beta^{-1} S(\omega)$, where the entropy uses natural logarithms.

Since h_λ is balanced and g_λ is injective, ρ_λ is uniform on $2^{n(\lambda)}$ computational-basis states. Hence $S(\rho_\lambda) = n(\lambda) \ln 2$ and $S(\sigma_\lambda) = m(\lambda) \ln 2$. For every key, it follows that $F_{\beta, \lambda}(\rho_\lambda) - F_{\beta, \lambda}(\tau_\lambda) = \beta^{-1}(m(\lambda) - n(\lambda)) \ln 2$, whereas the reference state has zero free-energy excess. Taking $m(\lambda) = 2n(\lambda)$ makes this gap linear in the number of qubits. Computational indistinguishability given polynomially many copies is unchanged.

Thus, assuming quantum-secure pseudorandom permutations exist, these ensembles provide a counterexample to Watanabe and Takagi's conjecture that pseudo-nonequilibrium states do not exist [WT26].

8 Classical vs Quantum Computational Ergotropy

In this section we explore the notion of work extraction using classical operations instead of unitaries. Clearly allowing the set of operations to reversible classical computation (permutations), restricts the set of operations one can perform to extract work. In this section, assuming $\text{BPP} \subsetneq \text{BQP}$, we show that, there exists

families of classical states and Hamiltonians from which one can extract non-negligible work catalytically using efficient unitary operations, while one can not do the same using permutations.

We proceed by setting up the notation for classical work extraction as follows.

Definition 105 (Catalytic permutations $\text{cat-Perm}(n, \ell, \varepsilon)$). Let the $(n + \ell)$ -qubit composite Hilbert space be denoted by $\mathcal{H} := \mathcal{H}_{\text{IO}} \otimes \mathcal{H}_{\text{catalyst}}$. Then, the set of *approximate catalytic permutations* $\text{cat-Perm}(n, \ell, \varepsilon)$ is defined as

$$\text{cat-Perm}(n, \ell, \varepsilon) := \left\{ \pi \in \text{Perm}(n + \ell) : \begin{array}{l} \exists |\eta\rangle \in \text{Cl}(\mathcal{H}_{\text{catalyst}}) \ \forall |\phi\rangle \in \text{Cl}(\mathcal{H}_{\text{IO}}), \ \exists |\psi\rangle \otimes |\xi\rangle \in \text{Cl}(\mathcal{H}) \\ \text{s.t. } \text{TD}(\pi(\phi \otimes \eta) \pi^{-1}, \psi \otimes \xi) \leq \varepsilon \end{array} \right\}$$

where $\text{Cl}(\mathcal{H}')$ represents all the computational basis states in the Hilbert space $\mathcal{H}'$.

Since the notion of using reversible classical computation is too restrictive, we study classical work extraction in the catalytic setting with approximate restoration. More formally, we define the work extracted by a classical algorithm as follows.

Definition 106 (Classically extracted work $(\mathcal{WC}_{\rho, \mathbf{H}, \mathcal{A}})$). Let $\rho, \mathbf{H}$ be families of classical states and Hamiltonians. For an algorithm $\mathcal{A}$ we define the *work extracted by an ε -approximate ℓ -catalytic permutation* (i.e. reversible classical circuit) as follows.

$$\mathcal{WC}_{\rho, \mathbf{H}, \mathcal{A}}^{(\ell, \varepsilon)}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_\lambda \left(\rho_\lambda \otimes |0^\ell\rangle \langle 0^\ell| - \pi_\lambda \left(\rho_\lambda \otimes |0^\ell\rangle \langle 0^\ell| \right) \pi_\lambda^{-1} \right) \right] : \begin{array}{l} \text{desc}(\pi_\lambda) \leftarrow \mathcal{A}^{O_{\rho, \mathbf{H}}\text{-desc}}(1^\lambda), \\ \pi_\lambda \in \text{cat-Perm}(n(\lambda), \ell(\lambda), \varepsilon) \end{array} \right].$$

Similarly for an algorithm $\mathcal{A}$ one can define the *work extracted by an ε -approximate ℓ -catalytic unitary* as follows.

$$\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}^{(\ell, \varepsilon)}(\lambda) := \mathbb{E} \left[\text{tr} \left[H_\lambda \left(\rho_\lambda \otimes |0^\ell\rangle \langle 0^\ell| - U_\lambda \left(\rho_\lambda \otimes |0^\ell\rangle \langle 0^\ell| \right) U_\lambda^\dagger \right) \right] : \begin{array}{l} \text{desc}(U_\lambda) \leftarrow \mathcal{A}^{O_{\rho, \mathbf{H}}\text{-desc}}(1^\lambda), \\ U_\lambda \in \text{cat-}\mathcal{U}(n, \ell, \varepsilon) \end{array} \right].$$

Naturally, one can extend the notion of work extracted by an algorithm to define the notion of classical catalytic ergotropy. For classical catalytic ergotropy, we allow at most negligible imperfections when returning the catalyst.

Definition 107 ($\widehat{\text{cat-ergC}}_{\mathcal{D}}(\ell)$). We say $\widehat{\text{cat-ergC}}_{\mathcal{D}}(\ell)$ is (lb, ub)-bounded if

$$\begin{aligned} \forall \mathcal{A} \in \text{PPT} \ \exists \text{negl} \ \text{s.t.}, \ \mathbb{E}_{\rho, \mathbf{H} \leftarrow \mathcal{D}} \left[\mathcal{WC}_{\rho, \mathbf{H}, \mathcal{A}}^{(\ell, \text{negl})} \right] &\leq_\infty \text{ub}, \text{ and} \\ \exists \mathcal{A} \in \text{PPT}, \text{negl}' \ \text{s.t.} \ \mathbb{E}_{\rho, \mathbf{H} \leftarrow \mathcal{D}} \left[\mathcal{WC}_{\rho, \mathbf{H}, \mathcal{A}}^{(\ell, \text{negl}')} \right] &\geq_\infty \text{lb} \end{aligned}$$

where ub, lb are functions and negl, negl' are negligible functions.

Analogously, we define quantum catalytic ergotropy with approximate restoration of catalysts as follows.

Definition 108 ($\widehat{\text{cat-ergQ}}_{\mathcal{D}}(\ell)$). We say $\widehat{\text{cat-ergQ}}_{\mathcal{D}}(\ell)$ is (lb, ub)-bounded if

$$\begin{aligned} \forall \mathcal{A} \in \text{PPT} \ \exists \text{negl} \ \text{s.t.}, \ \mathbb{E}_{\rho, \mathbf{H} \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}^{(\ell, \text{negl})} \right] &\leq_\infty \text{ub}, \text{ and} \\ \exists \mathcal{A} \in \text{PPT}, \text{negl}' \ \text{s.t.} \ \mathbb{E}_{\rho, \mathbf{H} \leftarrow \mathcal{D}} \left[\mathcal{W}_{\rho, \mathbf{H}, \mathcal{A}}^{(\ell, \text{negl}')} \right] &\geq_\infty \text{lb} \end{aligned}$$

where ub, lb are functions and negl, negl' are negligible functions.

In subsequent discussions, we set up some notation and lemmas, to prove a separation between the classical catalytic ergotropy, and quantum catalytic ergotropy in Section 8.3.

8.1 Pseudo-deterministic to catalytic circuits

A pseudo-deterministic circuit is one that for each input x produces a fixed answer y with $1 - \text{negl}$ probability. Here we show that any pseudo-deterministic circuit can be turned into a catalytic circuit with negligible restoration error on the catalyst.

Consider a relational problem rel that has only one solution for each input, i.e. for each x , there is a unique y such that $(x, y) \in \text{rel}$. This is essentially function evaluation.

For this exposition, we restrict ourselves to such relations. We say a QPT algorithm $\mathcal{A}$ pseudo-deterministically solves rel if there is a negligible function negl such that $\Pr[y \leftarrow \mathcal{A}(x)] = 1 - \text{negl}(|x|)$ for all inputs x . We assume that $\mathcal{A}$ defers all measurements to the end of the computation.

For the claim below, let $m, n, s : \mathbb{N} \rightarrow \mathbb{N}$ be functions denoting the size of the input, output and the workspace respectively.

Claim 109 (Pseudo-Deterministic to Catalytic). Let rel, n, m, s be as above. Suppose $\mathcal{A}$ is a pseudo-deterministic QPT algorithm that, on an $n(\lambda)$ -bit input x , uses an $s(\lambda)$ -qubit workspace and produces an $m(\lambda)$ -bit output y . Then, one can construct an algorithm $\mathcal{B}$ that uses an $n(\lambda) + m(\lambda)$ qubit IO register and $\ell(\lambda)$ qubit catalyst register where $\ell := n + m + s$. Furthermore, there is a negligible function negl' such that on input x , $\mathcal{B}$ applies a unitary U_λ satisfying

$$\left\| U_\lambda |x, 0\rangle_{\text{IO}} |0\rangle_{\text{catalyst}} - |x, y\rangle_{\text{IO}} |0\rangle_{\text{catalyst}} \right\| \leq \text{negl}'(\lambda)$$

for $\lambda > \lambda'$ for some $\lambda' \in \mathbb{N}$.

Proof. The proof is elementary. Suppose the action of $\mathcal{A}$ on input x is given by the unitary V_λ that (from the fact that $\mathcal{A}$ is a pseudo-deterministic QPT algorithm implies) acts as

$$V_\lambda |x\rangle |0^{m+s}\rangle = \sqrt{1 - a_{xy}^2} |x, y\rangle \otimes |\varphi_{xy}\rangle + a_{xy} |\Psi_{\text{bad}}\rangle$$

followed by a measurement, where $0 \leq a_{xy} \leq 1$ is at most a negligible function negl , and $|\varphi_{xy}\rangle, |\Psi\rangle$ are normalised vectors.

We define a unitary U_λ that acts on registers I (n -qubits), O (m -qubits), C_1 (n -qubits), $C_1 \cdot O$ (m -qubit) and C_2 (s -qubit) as follows (see also Figure 3)

$$U_\lambda = \text{CNOT}_{I, C_1} (V_\lambda)_{C_1 C_2} \text{CNOT}_{C_1, O} (V_\lambda^\dagger)_{C_1 C_2} \text{SWAP}_{IO, C_1}$$

where CNOT_{AB} is a controlled-NOT gate, with A as control and B as the target.

By direct calculation (and using the fact that $a_{xy} \leq \text{negl}$), one can verify that there is a negligible function negl' such that

$$\left\| U_\lambda |x, 0\rangle_{\text{IO}} |0, 0\rangle_{C_1 C_2} - |x, y\rangle_{\text{IO}} \otimes |0, 0\rangle_{C_1 C_2} \right\| \leq \text{negl}'(\lambda).$$

□

8.2 Classical circuit to classical Hamiltonian

The Cook-Levin theorem gives a conditions on the transcript of a Turing machine to ensure that a Turing machine arrived at the through valid intermediate computation. In this part, we adapt the same method for circuits, and show the procedure to construct an extensive Hamiltonian corresponding to a poly-sized classical circuit such that:

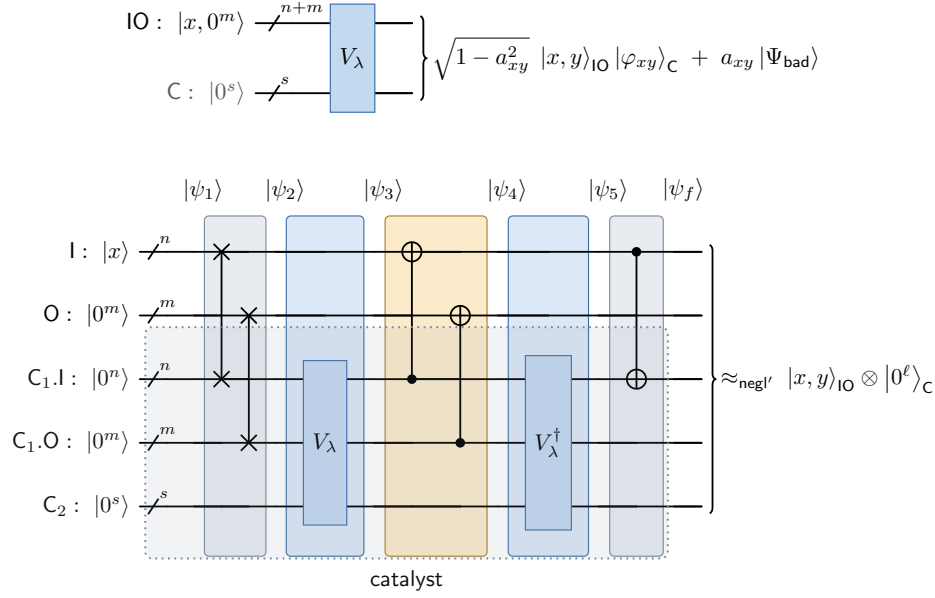


Figure 3: Pseudo-deterministic to catalytic

- given an input where the circuit accepts, one can efficiently construct the (classical) ground state of the Hamiltonian, and the converse,
- given a ground state of the Hamiltonian, one can efficiently construct an input where the circuit accepts.

We set up the notation and describe the circuit-Hamiltonian conversion procedure below.

Circuit. Consider a polynomially-sized classical circuit C specified by w wires and by gates $G_1, \dots, G_g$ where the gate G_ℓ acts on W_ℓ wires (where W_ℓ is a set of wires). For simplicity, assume that each gate act on 2 input wires and produce 2 output wires. Denote the output of the circuit (decision bit) by the state of the first wire (after all the gates have been applied).

Denote by $t_0, t_1, \dots, t_\ell, \dots, t_g$ the state of the wires (transcript) starting with the input t_0 , where t_ℓ is the transcript after gate G_ℓ is applied.

Hamiltonian. Given a circuit as described above, we define a Hamiltonian H acting on gw wires/bits (where w is the number wires and g is the number of gates in C , as described above), where the bits are labelled as $(i, j) \in \{1, 2 \dots g\} \times \{1, 2 \dots w\}$ as follows

- Let pred_ℓ be the predicate that applies G_ℓ on wires $\{(\ell - 1, i) : i \in W_\ell\}$ and outputs 0 (no penalty) if the result matches the wires $\{(\ell, i) : i \in W_\ell\}$ (i.e. it applies G_ℓ on $t_{\ell-1}$ on the $\ell - 1$ th block of wires and checks if it matches t_ℓ on the ℓ th block of wires; see) but 1 (penalty) otherwise.
- Let pred_{dec} be the predicate that outputs the *opposite* of the decision bit (i.e. state of wire $(g, 1)$).

Define the 4-local Hamiltonian H as

$$H := \sum_{\ell=1}^g \text{pred}_\ell + \text{pred}_{\text{dec}}.$$

Claim 110. Let C and H be as above.

- Given an input t_0 on which C accepts, one can efficiently construct the ground state $t_0, t_1, \dots, t_g$ of H
- Given a ground state $t_0, t_1, \dots, t_g$ of H , the string t_0 is an accepting input of C .

Proof Sketch (elementary). The Hamiltonian H acts on a sequence of ‘transcripts’ $t_0, t_1, \dots, t_g$ and applies an energy penalty for every inconsistency. It also applies a penalty if the decision bit is 0.

Clearly, if $t_0, t_1, \dots, t_g$ is in fact the transcript produced by running C such that the decision bit in t_g is 1, then clearly $t_0, t_1, \dots, t_g$ is the ground state of H .

Conversely, if the sequence $t_0, t_1, \dots, t_g$ is a ground state of H , it means that on input t_0 the circuit C produces t_g as the final transcript and the decision bit (i.e. the first bit of t_g) is 1. Thus, t_0 is an accepting input of C . $\square$

8.3 Putting everything together—the separation

With all the parts in place, we now establish the final separation.

(Amplified, pseudo-deterministic) Public-Coin Proof of Quantumness. A *public-coin proof of quantumness* is specified by a pair of algorithms $(\mathcal{P}, \mathcal{V})$ where

- The prover $\mathcal{P}$ is a QPT algorithm that takes as input c (challenge) and produces an output w (witness)
- The verifier $\mathcal{V}$ is a PPT algorithm that
 - on input 1^λ samples c from some distribution over the set of challenges C according to distribution $\mathcal{D}_C(1^\lambda)$ and sends it to $\mathcal{P}$,
 - on input w from $\mathcal{P}$, computes a deterministic function to compute $b(w, c) \in \{\text{accept}, \text{reject}\}$ and outputs b .

A public-coin proof of quantumness protocol must satisfy the following conditions.

- (completeness) there is a QPT prover $\mathcal{P}$ that can make the verifier accept with probability at least $1 - \text{negl}$ and
- (soundness) no PPT prover $\mathcal{P}^*$ can make the verifier accept with probability more than negl .
- We also require an additional ‘pseudo-deterministic’ property: for each c there is a unique w such that $b(w, c) = \text{accept}$.

We will work in the common reference string model. This is like the random oracle model but now, every party gets access to a string, sampled from some fixed distribution—as opposed to getting access to an oracle for a function sampled from the uniform distribution over all functions.

Theorem 111. Suppose $(\mathcal{P}, \mathcal{V})$ specifies a public-coin proof of quantumness as above. Suppose every party has access to the common reference string $\{c_\lambda\}_\lambda$ sampled from the distribution $\mathcal{D}_C$. Then, for each $\{c_\lambda\}_\lambda$, there is a family of 4-local extensive classical Hamiltonians $\mathbf{H} := \{H_\lambda\}_\lambda$ (normalisation is as in Definition 29) and corresponding family of states $\rho := \{\rho_\lambda\}_\lambda$ such that

$$\nexists \text{ non-negl } \text{ s.t. } \widehat{\text{cat-ergC}}_{\mathcal{D}}(\ell) \geq_\infty \text{ non-negl} \quad (71)$$

$$\exists \text{ negl}' \text{ s.t. } \widehat{\text{cat-ergQ}}_{\mathcal{D}}(\ell) \geq_\infty 1 - \text{negl}' \quad (72)$$

where $\mathcal{D}$ is the distribution over $\rho, \mathbf{H}$, specified implicitly by $\mathcal{D}_c$, ℓ is some polynomial and non-negl is a non-negligible function while negl' is some negligible function.

Proof Sketch. Since the verifier is a PPT algorithm, one can construct a reversible classical circuit corresponding to it. Let $\mathbf{H} = \{H_\lambda\}_\lambda$, where the reversible circuit for the verifier is converted into H_λ (using Claim 110). We define $\rho = \{\rho_\lambda\}_\lambda$, where ρ_λ is to be the transcript $t_0, t_1, \dots, t_g$ where t_0 is the all zero string, and this is propagated correctly through the verifier's circuit (see Section 8.2).

Notice that

- $\mathbb{E}_{\mathcal{D}_c} \text{tr}(H_\lambda \rho_\lambda) = 1 - \text{negl}(\lambda)$ because no penalties until the very last step where the decision bit is false—and so 1 unit penalty (with $1 - \text{negl}$ probability from the guarantee of the PoQ)
- ρ_λ can be converted to an all zero string catalytically (since we used a reversible circuit for the verifier, thus each element in the transcript can be reversed)

One can show that $\exists \text{ negl}'$ s.t. $\widehat{\text{cat-ergQ}}_{\mathcal{D}}(\ell) \geq_\infty 1 - \text{negl}'$ Equation (72)), through the following arguments. The PoQ has almost perfect completeness and has the extra pseudo-deterministic property; and these imply there is a pseudo-deterministic quantum algorithm V_λ that on input c produces a w such that $b(w, c) = \text{accept}$ with $1 - \text{negl}$ probability.

Using Claim 109 one can run V_λ as catalytic unitary U'_λ on an all zero string and produce the response w . Concretely, the PPT algorithm $\mathcal{A}^{c, \text{H-desc}, O_\rho}$ produces the description of U_λ that

- first undoes the verifier's reversible computations to obtain an all zero state starting from $t_0, t_1, \dots, t_g$, and
- then applies the catalytic unitary U'_λ that has the input c hard-coded.

One can show that $\nexists \text{ non-negl}$ s.t. $\widehat{\text{cat-ergC}}_{\mathcal{D}}(\ell) \geq_\infty \text{non-negl}$ (i.e. Equation (71)) through the following arguments. Since $\mathbb{E}_{\mathcal{D}_c} \text{tr}(H_\lambda \rho_\lambda)$ is already the 'first excited' state, the only way to lower the energy is to move to the ground state. Using Claim 110 it follows that if the PPT algorithm is able to produce the ground state, this would mean that from the ground state, one can also efficiently extract w (an accepting input to the verifier's circuit). However, the PoQ has almost perfect soundness and therefore no PPT algorithm on input c , can produce w such that $b(w, c) = \text{accept}$ with more than negl' probability. $\square$

As a consequence of the above separation, we make the following remark.

Remark 112 (Catalysts are useful computationally). (i) information-theoretically, quantum ergotropy and classical ergotropy for this family is identical.

(ii) computationally, quantum ergotropy and classical ergotropy for this family is identical.

(iii) Classical catalytic ergotropy is negligible.

(iv) Quantum catalytic ergotropy is constant.

This is another feature of thermodynamics that seems to only appear in the computational theory—and it is again aided by catalysts.

A Deferred Proofs

Lemma 113 (Restatement of Lemma 27; [ABV23]). *Consider a PRP family (F, Flnv) and let $n, m \in \mathbb{N}$ be s.t. $m > n$. We consider $k_f, k_g, k_h \in \{0, 1\}^m$, and define the following functions: $f : \{0, 1\}^m \rightarrow \{0, 1\}^m$ as $f(x) = F(k_f, x)$, $g : \{0, 1\}^n \rightarrow \{0, 1\}^m$ as $g(x) = F(k_g, x \| 0^{m-n})$, and $h : \{0, 1\}^m \rightarrow \{0, 1\}^n$ as $h(x) = F(k_h, x)_n$ (i.e. the n -bit prefix of the outcome). Then for any polynomial function $m = m(n)$ and for any polynomial-time quantum algorithm $\mathcal{A}$ with binary output it holds that*

$$\left| \Pr_{k_g, k_h} [\mathcal{A}^{g(h(\cdot))}(1^n) = 1] - \Pr_{k_f} [\mathcal{A}^{f(\cdot)}(1^n) = 1] \right| = \text{negl}(n).$$

Proof. (Proof sketch) The proof proceeds through a sequence of hybrids. Consider an adversary $\mathcal{A}$ and define the random variable $H_0 = H_0(n) = \mathcal{A}^{g(h(\cdot))}(1^n)$, where g, h are defined based on uniformly random keys k_g, k_h .

We then let π_g, π_h be uniformly random permutations on $\{0, 1\}^m$. In H_1 we change h to be defined using π_h instead of $F(k_h, \cdot)$. It holds that $\|H_0 - H_1\|_1 = \text{negl}(n)$ because of PRP pseudorandomness, where $\|\cdot\|_1$ denotes L_1 norm. Similarly, in H_2 we also change g to be defined using π_g instead of $F(k_g, \cdot)$. Again we have $\|H_1 - H_2\| = \text{negl}(n)$ for the same reason.

The next few hybrids do not rely on computational assumptions but rather on query lower bounds to random permutations and functions. Let Q be an upper bound on the number of queries made by $\mathcal{A}$ to its oracle. The next bounds only rely on the assumption that $Q = \text{poly}(n)$.

In hybrid H_3 we change π_h to be a random function rather than a permutation. It holds that $\|H_2 - H_3\| = \text{negl}(n)$ using the query lower bound for distinguishing random functions from random permutations. Hybrid H_4 does the same for π_g , and by the same theorem $\|H_3 - H_4\| = \text{negl}(n)$.

In H_5 we make a more radical change. Let $r : \{0, 1\}^m \rightarrow \{0, 1\}^m$ be a random function, we let $H_5 = \mathcal{A}^{r(\cdot)}(1^n)$. We show that $\|H_4 - H_5\| = \text{negl}(n)$ using Zhandry's small domain theorem (SDT). Fix the function $g(x) = \pi_g(x \| 0^{m-n})$. This function defines a multi-set $S_g \subseteq \{0, 1\}^m$ of size 2^n (namely, it possibly has repetitions). Let $D_{1,g}$ be the uniform distribution over S_g . Let D_2 be the uniform distribution over $\{0, 1\}^m$. Using SDT, an algorithm that distinguishes H_4 from H_5 with some advantage can be transformed into an algorithm that distinguishes $D_{1,g}$ from D_2 by getting polynomially samples from either distribution. Since the reduction is uniform, this distinguisher has advantage in expectation over g as well. However, when g is chosen at random, polynomially many samples from $D_{1,g}$ are statistically indistinguishable from polynomially many samples from D_2 , which completes the argument.

Next, we define $H_6 = \mathcal{A}^{\pi_f(\cdot)}(1^n)$, where π_f is a random permutation. Here again we use the function to permutation indistinguishability to argue that $\|H_5 - H_6\| = \text{negl}(n)$. Our last hybrid again relies on a cryptographic assumption. In H_7 we replace π_f with f and use PRP pseudorandomness to deduce that $\|H_6 - H_7\| = \text{negl}(n)$. We can conclude that $\|H_0 - H_7\| = \text{negl}(n)$ and the lemma follows. $\square$

Lemma 114 (Restatement of lemma 47). *Let $\mathcal{O}$ be the length doubling random oracle, and $q_1, q_2 : \mathbb{N} \rightarrow \mathbb{N}$ be polynomials. For every q_1 -query quantum algorithm $\mathcal{B}$ with oracle access to $\mathcal{O}$, there exists a q_2 -query quantum algorithm $\mathcal{A}$ with oracle access to $\mathcal{O}_\lambda$ corresponding to the function f_λ such that,*

$$\begin{aligned} & \left| \Pr [\mathcal{B}^{\mathcal{O}}(1^\lambda, \rho_\lambda) = 1] - \Pr [\mathcal{B}^{\mathcal{O}}(1^\lambda, \sigma_\lambda) = 1] \right| \\ &= \left| \Pr [\mathcal{A}^{\mathcal{O}_\lambda}(1^\lambda, \rho_\lambda) = 1] - \Pr [\mathcal{A}^{\mathcal{O}_\lambda}(1^\lambda, \sigma_\lambda) = 1] \right| \end{aligned} \tag{73}$$

$$\leq \text{negl}(\lambda). \tag{74}$$

Proof. We first prove that there exists an algorithm $\mathcal{A}$ such that the output distributions of $\mathcal{B}^O(1^\lambda, \tau_\lambda)$ and $\mathcal{A}^{O_\lambda}(1^\lambda, \tau_\lambda)$ are identical for any τ_λ that depends only on O_λ . Observe that each f_λ is sampled uniformly and independently from $\mathcal{F}_\lambda$.

Now, for every algorithm $\mathcal{B}^O$, one can construct such an algorithm $\mathcal{A}^{O_\lambda}$ as follows: $\mathcal{A}$ runs $\mathcal{B}$ identically except when $\mathcal{B}$ makes queries. When $\mathcal{B}$ makes a query to O at $\{0, 1\}^{n(\lambda)}$, then $\mathcal{A}$ uses O_λ , and when $\mathcal{B}$ makes any other query, it simulates a random oracle independently. This establishes Equation (73).

To establish Equation (74), we use a hybrid argument and the One-Way to Hiding (O2H) Lemma. We define a sequence of hybrid experiments $\text{Exp}_0, \text{Exp}_1, \text{Exp}_2$ as follows

- Exp_0 : The challenger samples $O_\lambda \leftarrow \mathcal{F}_\lambda$ and $r \leftarrow \{0, 1\}^{n(\lambda)}$. It outputs $\mathcal{A}^{O_\lambda}(O_\lambda(r))$.
- Exp_1 : The challenger samples $O_\lambda \leftarrow \mathcal{F}_\lambda$, $r \leftarrow \{0, 1\}^{n(\lambda)}$ and $u \leftarrow \{0, 1\}^{2n(\lambda)}$. It defines O'_λ such that $O'_\lambda(r) = u$ and $O'_\lambda(x) = O_\lambda(x)$ for all $x \neq r$. It outputs $\mathcal{A}^{O'_\lambda}(u)$.
- Exp_2 : The challenger samples $O_\lambda \leftarrow \mathcal{F}_\lambda$, $r \leftarrow \{0, 1\}^{n(\lambda)}$ and $u \leftarrow \{0, 1\}^{2n(\lambda)}$. It outputs $\mathcal{A}^{O_\lambda}(u)$.

Claim 115. $\Pr_{O_\lambda, r}[1 \leftarrow \text{Exp}_0] = \Pr_{O_\lambda, r, u}[1 \leftarrow \text{Exp}_1]$.

Proof of Claim 48. Evaluating a random oracle at any point yields a uniformly distributed string. Since at every point $x \neq r$, $O_\lambda(x) = O'_\lambda(x)$, the joint distribution $(O_\lambda, O_\lambda(r))$ in Exp_0 is statistically identical to the joint distribution (O'_λ, u) in Exp_1 . Therefore,

$$\Pr_{O_\lambda, r}[1 \leftarrow \text{Exp}_0] = \Pr_{O_\lambda, r, u}[1 \leftarrow \text{Exp}_1].$$

□

$$|\Pr_{O_\lambda, r, u}[1 \leftarrow \text{Exp}_1] - \Pr_{O_\lambda, u}[1 \leftarrow \text{Exp}_2]| \leq \text{negl}(n).$$

Proof of Claim 49. The inputs to $\mathcal{A}$ is identical in both Exp_1 and Exp_2 , the only difference in the experiments is the oracle the $\mathcal{A}$ has access to differs in a single point— r . Intuitively, for the outputs of Exp_1 and Exp_2 to be different, the algorithm $\mathcal{A}$ should query the point r . But since $\mathcal{A}$ is QPT and can query in superposition, we do the following.

Proof. We construct an algorithm $\mathcal{B}^{O_\lambda}$ that on input u , simulates $\mathcal{A}^{O_\lambda}(u)$ upto the i -th query, where $i \leftarrow \{1, 2, \dots, q\}$, measures the query register and returns the measurement output T . From Lemma 23, we know that such an algorithm $\mathcal{B}^{O_\lambda}$ will return r with probability P_{guess} such that,

$$\frac{|\Pr_{O_\lambda, r, u}[1 \leftarrow \text{Exp}_1] - \Pr_{O_\lambda, u}[1 \leftarrow \text{Exp}_2]|}{2q} \leq \sqrt{P_{\text{guess}}}$$

Notice that in Exp_2 , the variable r is sampled uniformly at random from $\{0, 1\}^n$ and is independent of both the classical input u and the oracle O_λ . Since u, r are completely uncorrelated, from the perspective of algorithm $\mathcal{B}^{O_\lambda}$, upon receiving input u , the probability with which it returns the random string r is $\frac{1}{2^n} = \text{negl}(n)$. Taking a union bound over $\text{poly}(n)$ many trials, we get $P_{\text{guess}} \leq \text{negl}(n)$. Therefore,

$$\frac{|\Pr_{O_\lambda, r, u}[1 \leftarrow \text{Exp}_1] - \Pr_{O_\lambda, u}[1 \leftarrow \text{Exp}_2]|}{2q} \leq \text{negl}(n)$$

Since $\mathcal{A}$ is a QPT algorithm, $q = \text{poly}(n)$, therefore we obtain $|\Pr_{O_\lambda, r, u}[1 \leftarrow \text{Exp}_1] - \Pr_{O_\lambda, u}[1 \leftarrow \text{Exp}_2]| \leq \text{negl}(n)$, proving the claim. □

□

Now using the hybrid argument, we conclude $|\Pr_{O_\lambda, r}[1 \leftarrow \text{Exp}_0] - \Pr_{O_\lambda, u}[1 \leftarrow \text{Exp}_2]| \leq \text{negl}(n)$ proving the theorem statement. □

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